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Fischer–Tropsch synthesis to olefins boosted by MFI zeolite nanosheets

Chengtao Wang[1,2,6], Wei Fang[1,6], Zhiqiang Liu[3,6], Liang Wang[1]✉, Zuwei Liao[1], Yongrong Yang[1], Hangjie Li[2], Lu Liu[1], Hang Zhou[1], Xuedi Qin[2], Shaodan Xu[4], Xuefeng Chu[5], Yeqing Wang[2], Anmin Zheng[3]✉ and Feng-Shou Xiao[1]✉

Catalytic reactions are severely restricted by the strong adsorption of product molecules on the catalyst surface, where promoting desorption of the product and hindering its re-adsorption benefit the formation of free sites on the catalyst surface for continuous substrate conversion[1,2]. A solution to this issue is constructing a robust nanochannel for the rapid escape of products. We demonstrate here that MFI zeolite crystals with a short *b*-axis of 90–110 nm and a finely controllable microporous environment can effectively boost the Fischer–Tropsch synthesis to olefins by shipping the olefin molecules. The ferric carbide catalyst (Na-FeC$_x$) physically mixed with a zeolite promoter exhibited a CO conversion of 82.5% with an olefin selectivity of 72.0% at the low temperature of 260 °C. By contrast, Na-FeC$_x$ alone without the zeolite promoter is poorly active under equivalent conditions, and shows the significantly improved olefin productivity achieved through the zeolite promoter. These results show that the well-designed zeolite, as a promising promoter, significantly boosts Fischer–Tropsch synthesis to olefins by accelerating escape of the product from the catalyst surface.

Olefins have been used extensively as carbon-based building blocks for polymers and platform chemicals in the production of valuable oxygenates[3]. Current olefin production depends strongly on petroleum, and recently much attention has been focused on exploring alternative routes[4–8]. Among them, a promising blueprint has been to use the feedstock of syngas, a mixture of hydrogen and carbon monoxide (H_2 + CO) derived from biomass, coal and natural gas[6–15]. A classical syngas-to-olefin transformation is well known as the Fischer–Tropsch synthesis to olefins (FTO), which is normally performed at 320–400 °C using iron-based catalysts[15–22] because of their advantages of adaptability for the syngas with wide flexibility in H_2/CO ratios. A typical feature of the FTO is a wide product distribution, from C_1 up to even C_{20+} molecules following the Anderson–Schulz–Flory distribution, that usually contains abundant and undesirable methane (CH_4)[16,17]. In studying the FTO, much attention has been focused on innovating this process for the selective production of valuable olefins, such as ethene, propene or 1-hexene, at lower reaction temperatures although this has been a great challenge. For example, the Fe-based catalysts usually exhibit poor CO conversions at temperatures lower than 300 °C (ref. [6]), and higher temperatures are favourable for CO conversion[18] with a wide product distribution. In addition to the classical FTO route, the route based on an oxide–zeolite-based catalyst (OX-ZEO) was developed by combining a metal oxide and a zeolite (for example, ZnCrO$_x$ and SAPO-34) for a tandem conversion of syngas into methanol/ketene and further transformation to C_2–C_4 olefins[11,12]. Superior selectivity was obtained but the reactions were still performed at relatively high temperatures (~400 °C).

Here we report an efficient methodology to boost the activity and optimize the product selectivity for the low-temperature FTO (LT-FTO) over an Fe-based catalyst, which is achieved through mixing a finely designed zeolite as a promoter for olefin desorption from the catalyst surface (Fig. 1). In a proof-of-concept experiment, the sodium-doped ferric carbide catalyst (Na-FeC$_x$, with the Fe_5C_2 phase dominant, synthesized following the strategy reported previously; Supplementary Figs. 1 and 2) was used because it is one of the most classical catalysts for the FTO[6,9,23]. At 260 °C (syngas with a CO/H_2/Ar ratio of 45/45/10, a flow rate of 2,400 ml per h per g FeC$_x$, at 2.0 MPa pressure), this Na-FeC$_x$ catalyst showed a low activity with a CO conversion of 1.9% (Fig. 2a; Supplementary Fig. 3). An enhanced CO conversion of 25.6% was obtained by increasing the temperature to 300 °C, giving the C_1–C_{20+} products with a total olefin selectivity of 72.4% (Fig. 2a; Supplementary Table 1), which is similar to the results reported previously[16–18]. Interestingly, when the aluminosilicate MFI zeolite with a *b*-axis thickness of 90–110 nm (s-ZSM-5, with a Si/Al ratio of 26.2; Supplementary Fig. 4 and Supplementary Table 2) was mixed (a granule mixture with Na-FeC$_x$; Supplementary Fig. 5), the Na-FeC$_x$/s-ZSM-5 catalyst gave a CO conversion of 82.5% with an olefin selectivity of 72.0% at 260 °C (Fig. 2a,b); this gives an olefin productivity of 15.3 mmol per g FeC$_x$ per h, which is significantly higher than that of Na-FeC$_x$ under equivalent test conditions (Fig. 1). These data support the so-called LT-FTO, where the zeolite without any catalytic activity for CO transformation was shown to be a promoter, boosting the catalysis.

To explore the effect of the zeolite in LT-FTO, we studied the catalytic performance of Na-FeC$_x$ combined with different zeolites, which were shown to exhibit a higher CO conversion than that of bare Na-FeC$_x$ (Supplementary Fig. 3 and Supplementary Table 1). For example, the normal ZSM-5 zeolite (n-ZSM-5, with a Si/Al

[1]Key Lab of Biomass Chemical Engineering of Ministry of Education, College of Chemical and Biological Engineering, Zhejiang University, Hangzhou, China. [2]Key Laboratory of Applied Chemistry of Zhejiang Province, Department of Chemistry, Zhejiang University, Hangzhou, China. [3]National Center for Magnetic Resonance in Wuhan, State Key Laboratory of Magnetic Resonance and Atomic and Molecular Physics and Mathematics, Wuhan Institute of Physics and Mathematics, Innovation Academy for Precision Measurement Science and Technology, Chinese Academy of Sciences, Wuhan, China. [4]College of Materials and Environmental Engineering, Hangzhou Dianzi University, Hangzhou, China. [5]Key Laboratory of Architectural Cold Climate Energy Management, Jilin Jianzhu University, Changchun, China. [6]These authors contributed equally: Chengtao Wang, Wei Fang, Zhiqiang Liu. ✉e-mail: liangwang@zju.edu.cn; zhenganm@wipm.ac.cn; fsxiao@zju.edu.cn

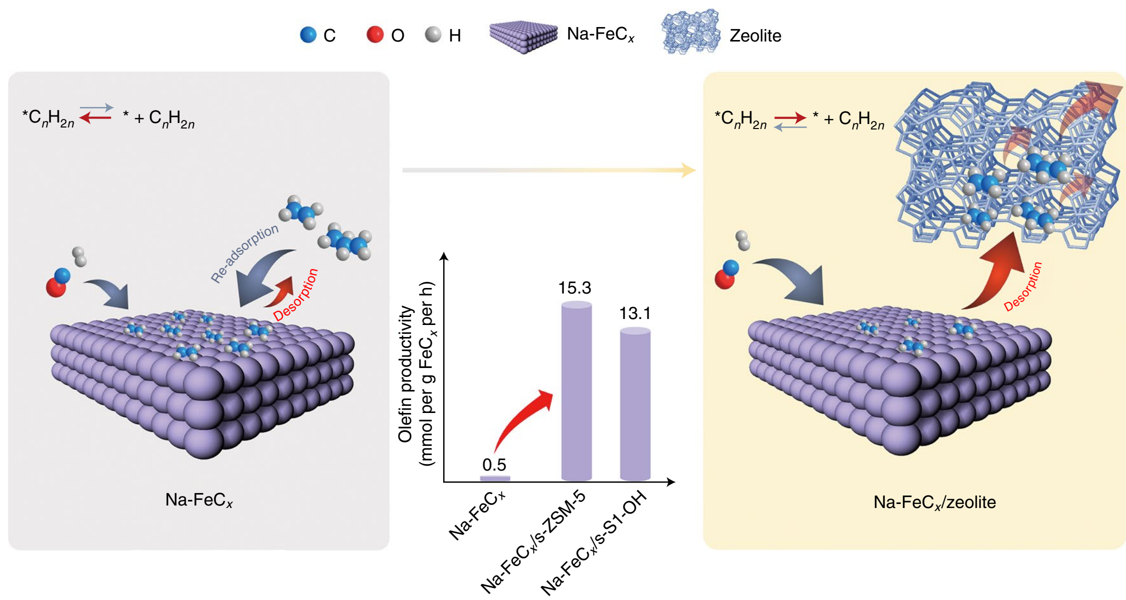

Fig. 1 | Schematic showing the strategy to boost the FTO via shifting the chemical equilibrium on the catalyst surface. In the model without the zeolite (left), the olefin products adsorbed on the Na-FeC$_x$ catalyst surface, as denoted by *, hinders CO hydrogenation to give a low olefin productivity. In the model with the zeolite (right), the olefin products rapidly escape from the zeolite micropores to form a free Na-FeC$_x$ surface for continuous CO hydrogenation. The olefin productivities over different catalysts are shown in the middle.

ratio of 30.5; Supplementary Fig. 6) combined with Na-FeC$_x$ (that is, Na-FeC$_x$/*n*-ZSM-5, with a zeolite/Na-FeC$_x$ mass ratio of 3) exhibited a CO conversion of 45.1% with C_2–C_{10} alkanes as the dominant products (an olefin/alkane molar ratio of only 0.1). The SSZ-13 zeolite (Supplementary Fig. 7) and the beta zeolite (Supplementary Fig. 8) mixed with the Na-FeC$_x$ improved the CO conversion to 39.8% and 32.5% with desired olefin/alkane (o/a) ratios of 4.2 and 4.1, respectively. The mordenite (or MOR) (Supplementary Fig. 9), SAPO-34 (Supplementary Fig. 10) and ZSM-22 zeolites (Supplementary Fig. 11), respectively, with Na-FeC$_x$ had a CO conversion of 62.2–74.1% and o/a ratios of 2.5–5.7. Among these zeolite-containing catalysts, the Na-FeC$_x$/s-ZSM-5 showed the highest CO conversion of 82.5% and an o/a ratio of 3.5 (Supplementary Figs. 12 and 13). By decreasing the zeolite/Na-FeC$_x$ ratio in the Na-FeC$_x$/s-ZSM-5 catalysts (with the zeolite/Na-FeC$_x$ mass ratio of 2), the o/a ratio could reach 4.2 with CO conversion at 66.8% (Supplementary Fig. 14 and Supplementary Table 3).

Regarding the product distribution, the Na-FeC$_x$-catalysed FTO reaction at 300 °C (that is, Na-FeC$_x$*) exhibited a wide carbon distribution of C_1–C_{20+} (Fig. 2b). By contrast, the LT-FTO over Na-FeC$_x$/s-ZSM-5 gave predominantly C_1–C_{10} molecules with limited C_{10+} products (Fig. 2b), indicating the significant advantage of LT-FTO for product selectivity compared with the conventional FTO. The LT-FTO narrows the product distribution to give predominantly C_2–C_{10} olefins (Supplementary Figs. 15 and 16), which include both the desired lower olefins (C_2–C_4) that are basic units for the modern chemical industry[9–12] and longer olefins that are valuable platform chemicals[18]. For example, 1-hexene and 1-octene are used as building units for high-performance polymers, and 1-pentene and 1-heptene can be converted into C_6/C_8 oxygenates via hydroformylation. In addition, this unique product distribution is also different from that of the OX-ZEO process, which predominantly produces lower (C_2–C_4) olefins[11,12,24–26].

The Na-FeC$_x$/s-ZSM-5 catalyst was evaluated using a continuous LT-FTO test. After activation for about 50 h, the CO conversion was constant in the steady state with an average value of about 68.1% (Fig. 2c). Such conversion was controlled to be lower than the aforementioned best result (82.5%) by adjusting the zeolite/Na-FeC$_x$ ratio, which is beneficial for evaluating the catalyst durability. Even after reaction for 600 h, the CO conversion was well maintained at 67.6%, with individual Na-FeC$_x$ and zeolite remaining stable (Supplementary Figs. 17 and 18). The selectivity for C_2–C_{10} olefins was also constant at 67.8–71.0% with an o/a ratio of 4.2–5.2. In these C_2–C_{10} olefin products, C_2–C_4 olefins and C_5–C_{10} olefins were about 45.3% and 54.7%, respectively (Supplementary Fig. 19). Notably, the selectivity for undesirable methane was lower than 3.8% (Fig. 2c). These data confirm the good durability of the Na-FeC$_x$/s-ZSM-5 catalyst in the LT-FTO reaction process. Normally, the carbon dioxide (CO_2) fraction is higher than 45% in the products, which is a general phenomenon in syngas-to-olefin reactions[6–8,15]. Owing to the combined advantages compared with classical FTO, which include being energy saving and using simpler equipment at a relatively low reaction temperature, as well as the high activity, optimized selectivity and good catalyst durability, we have compiled a process package based on this LT-FTO process to meet the demands for potential application (Supplementary Fig. 20).

Mixed catalysts that contain individual metal oxide and zeolite components have previously been studied in syngas conversion (for example, in OX-ZEO catalysts), where the zeolite catalyses the C–C coupling steps in cascade processes such as methanol-to-olefins reaction[11,12,24–27] and aromatization[28–30]. We emphasized that the zeolite in Na-FeC$_x$/s-ZSM-5 is conceptually different from the zeolite in the earlier composite catalysts, because the Na-FeC$_x$ can directly catalyse the syngas-to-olefin reaction[6,18]. Although the Na-FeC$_x$/s-ZSM-5 catalyst indeed controlled the product selectivity so that C_1–C_{10} products were dominant compared with the C_1–C_{20+} products that dominate on bare Na-FeC$_x$, this feature should not be assigned to acid-catalysed reactions, such as the cracking of large olefins formed on the Na-FeC$_x$. This possibility can be excluded in explaining the unique selectivity of LT-FTO, because the cracking of large olefins would produce isomerized olefins that cannot meet the high α-olefin selectivity in the Na-FeC$_x$/s-ZSM-5-catalysed LT-FTO, as confirmed by the model cracking reactions of 1-butene, 1-octene and 1-hexene over s-ZSM-5 at 260 °C (Supplementary Fig. 21a and Supplementary Tables 4–6). Therefore, the final α-olefin products in LT-FTO would result from the FTO step on Na-FeC$_x$ rather

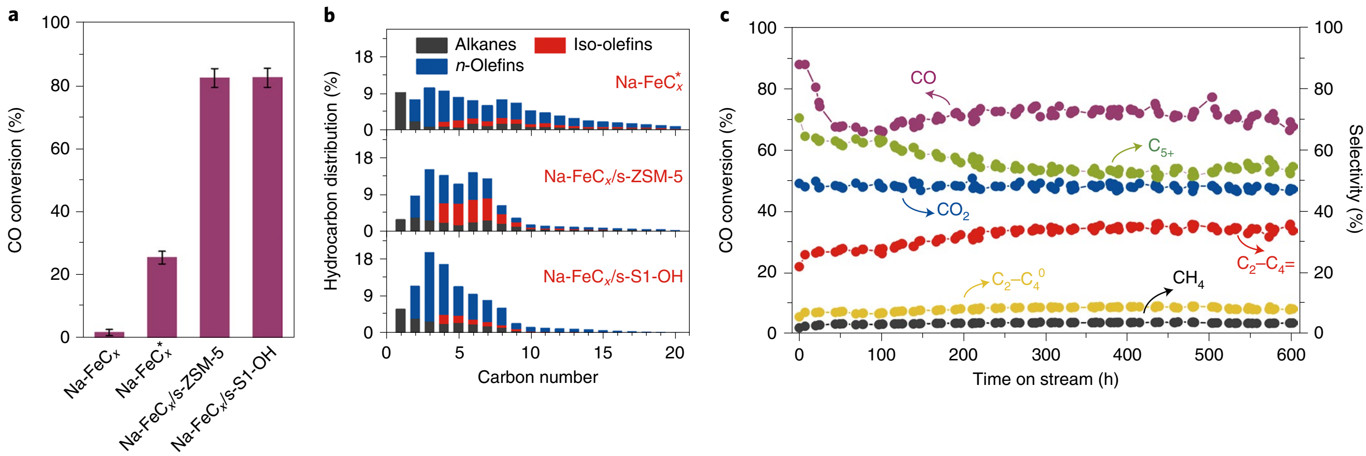

Fig. 2 | Catalytic performance of the Na-FeC$_x$/zeolite catalysts in the LT-FTO. a,b, CO conversion (**a**) and hydrocarbon distribution (**b**) for Na-FeC$_x$, Na-FeC$_x$/s-ZSM-5 and Na-FeC$_x$/s-S1-OH in FTO. Reaction conditions: zeolite:Na-FeC$_x$ weight ratio, 3; H_2/CO ratio, 1; gas flow rate, 2,400 ml per h per g FeC$_x$; reactor pressure, 2.0 MPa, temperature, 260 °C; granule mixing of the individual zeolite and Na-FeC$_x$ components. Na-FeC$_x$ marked with * denotes the reaction at 300 °C. For Na-FeC$_x$/s-S1-OH the H_2/CO molar ratio in the feed gas was 3. CO_2 was not included in the product distribution. **c**, Catalytic data for the Na-FeC$_x$/s-ZSM-5 catalyst in a continuous LT-FTO test. Reaction conditions: s-ZSM-5:Na-FeC$_x$, 2; H_2/CO, 1; 2,400 ml per h per g FeC$_x$; 2.0 MPa; 260 °C; granule mixing of the individual zeolite and Na-FeC$_x$ components. CO_2 was not included in calculating the selectivity for hydrocarbon products. C_2–C_4^0, lower alkanes; C_2–$C_4^=$, lower olefins; C_{5+}, C_5 products or higher.

than from acid-catalysed steps. For the production of α-olefins, s-ZSM-5 can serve as a promoter rather than a tandem acid catalyst. This feature is different from the role of zeolites in previous metal oxide/carbide and zeolite composite catalysts, where those zeolites act as acid catalysts for aromatization, isomerization, cracking and C–C coupling reactions, such as in the OX-ZEO system, where the final products are all produced via acid-catalysed reactions (Supplementary Fig. 21b)[11,12,24–26,31–37].

To confirm this concept and completely avoid the influence of acid sites on the aluminosilicate zeolite (for example, s-ZSM-5), we rationally designed a siliceous zeolite with a controlled *b*-axis thickness of 90–110 nm and artificially introduced silanol groups on the framework using an organosilane-assisted method[38,39], which is denoted as s-S1-OH (Supplementary Figs. 22–26). The Na-FeC$_x$/s-S1-OH catalyst showed a remarkably enhanced CO conversion of 66.5% compared with 1.9% on the bare Na-FeC$_x$ under equivalent test conditions (syngas with a CO/H_2/Ar ratio of 45/45/10, 2,400 ml per h per g FeC$_x$, 2.0 MPa; Supplementary Figs. 27 and 28 and Supplementary Table 7), exhibiting a olefin productivity of 13.1 mmol per g FeC$_x$ per h. By adjusting the H_2/CO ratio to 3, the CO conversion could even reach 82.6% in the LT-FTO (Fig. 2a; Supplementary Fig. 29 and Supplementary Table 8). Importantly, Na-FeC$_x$/s-S1-OH still showed a narrow distribution that was dominated by C_1–C_{10} molecules with an o/a ratio of 3.8, where the methane and C_{10+} product selectivities are 6.0% and 6.3% (Supplementary Table 8), respectively. Considering that the siliceous zeolite has no active sites for reactions in the LT-FTO process (Supplementary Fig. 30), the promoter role of the zeolite is confirmed in the Na-FeC$_x$-catalysed reactions. Moreover, the Na-FeC$_x$/s-S1-OH catalyst showed that 81.7% of the C_{4+} olefin products are α-olefins, which is much higher than the 52.1% seen over the Na-FeC$_x$/s-ZSM-5 catalyst (Fig. 2b), which is due to the s-S1-OH zeolite having no Brønsted acid sites that might cause undesirable α-olefin isomerization (Supplementary Fig. 21a). Although ferric-based catalysts for the FTO have been investigated for many years, it remains a challenge to simultaneously achieve both a high CO conversion and a high olefin selectivity at a given temperatures (for example, 260 °C). These data confirm the good performance of the Na-FeC$_x$/zeolite catalyst[10].

The influence of the mixing method for Na-FeC$_x$ and the zeolite in the LT-FTO process was also studied (Fig. 3a; Supplementary Fig. 31 and Supplementary Tables 9 and 10). Compared with granule mixing in the aforementioned cases, a powder mixture—prepared by mixing Na-FeC$_x$ and s-ZSM-5 powdered samples together and then made into granules for catalytic tests—exhibited a much lower CO conversion (21.5%) with a large amount of alkane by-products (o/a ratio 1.3). The selectivity for undesirable methane reached 18.8%, much higher than the 3.0% seen over the Na-FeC$_x$/s-ZSM-5 granule-mixing catalyst. This phenomenon could be due to massive sodium migration from Na-FeC$_x$ to the zeolite in the powder-mixing catalyst, as confirmed via Na 1*s* X-ray photoelectron spectroscopy characterization (Supplementary Figs. 32 and 33), which resulted in Na-FeC$_x$ having an insufficient sodium doping level to give a performance similar to the classical Fischer–Tropsch synthesis to alkanes. In addition, the nanometre-scale proximity in the powder-mixing catalyst might enhance hydrogen spillover in the catalyst bed that also benefits deep hydrogenation[39–41]. Using the dual-bed approach, s-ZSM-5 was packed below the Na-FeC$_x$ bed, separated by a layer of inert quartz sand, resulting in a poor CO conversion. In Na-FeC$_x$/s-ZSM-5-catalysed LT-FTO, raising the gas flow rate from 2,400 to 9,600 ml per h per g FeC$_x$ efficiently decreased the CO conversion but showed a similar product selectivity (Supplementary Fig. 34 and Supplementary Table 11). A similar phenomenon was also observed using the Na-FeC$_x$/s-S1-OH catalyst (Supplementary Fig. 35 and Supplementary Table 12).

To understand the promotion effect of the zeolite, we explored the essential factors for controlling the FTO reaction. As shown in Fig. 3b, the Na-FeC$_x$ catalyst gave a CO conversion of 29.9% in the FTO at 300 °C. When ethene was co-introduced with the feed gas (at 16.7 vol%), about 88% of the activity was lost, showing a CO conversion of 3.5% after the introduction of ethene for 5 h. Supplementary Figure 36 shows the dependence of the CO conversion on the ethene concentration for this FTO reaction, and gives lower activity with more ethene in the feed gas. This result can be explained by the known feature of the ferric carbide catalyst, whose surface strongly adsorbs the olefin molecules, causing difficult-to-desorb species to block the active sites[23,42,43]. When ethene was removed from the feed gas, the CO conversion could be regenerated to 28.8% (Fig. 3b).

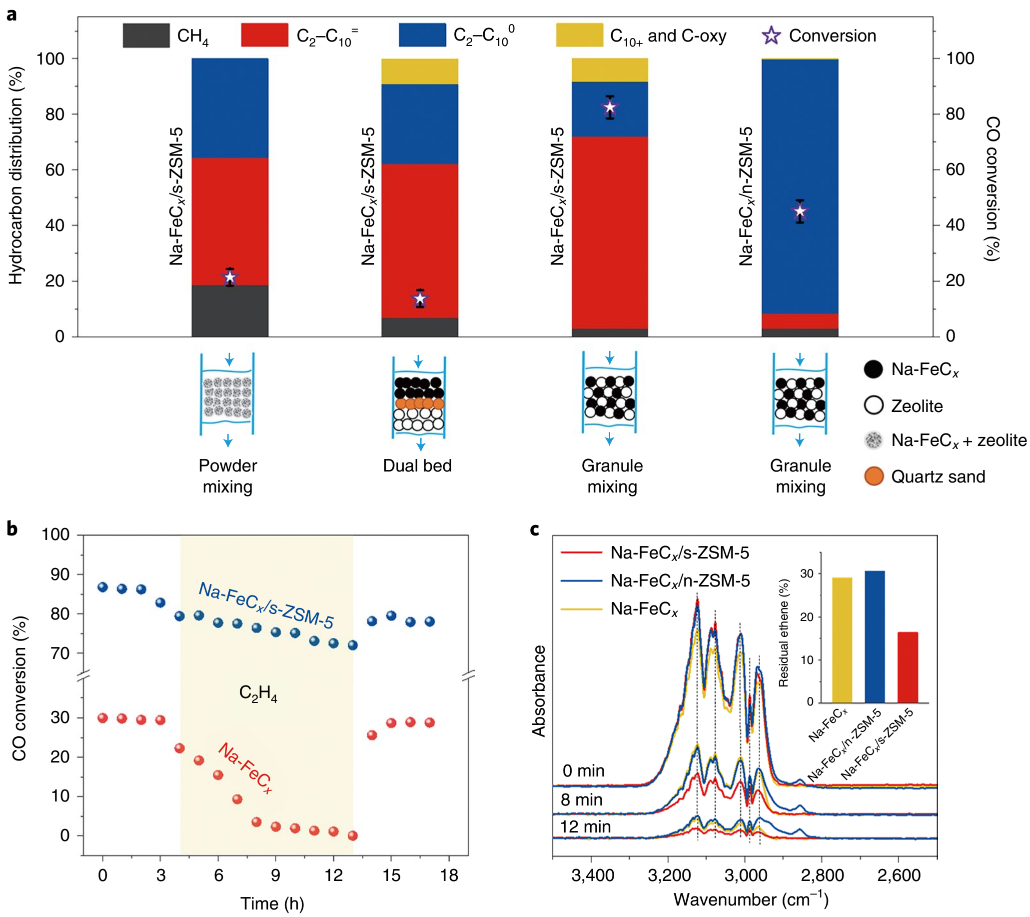

Fig. 3 | Catalytic data for the Na-FeC$_x$/zeolite catalysts in LT-FTO and ethene desorption DRIFT spectra over Na-FeC$_x$/zeolite catalysts. a, Data showing the performance of catalysts, using different mixing methods, in the LT-FTO. Reaction conditions: zeolite:Na-FeC$_x$ weight ratio, 3; H_2/CO ratio, 1; flow rate, 2,400 ml per h per g FeC$_x$; pressure, 2.0 MPa; temperature, 260 °C. CO_2 was not included in the product distribution. C_2–$C_{10}{}^0$, C_2–C_{10} alkanes; C_2–$C_{10}{}^=$, C_2–C_{10} olefins; C_{10+}, C_{10} products or higher; C-oxy, oxygenated carbon compounds. Error bounds for CO conversions were ±2%. **b**, Data showing the influence of ethene (C_2H_4) on the LT-FTO using the Na-FeC$_x$/s-ZSM-5 and Na-FeC$_x$ catalysts. Reaction conditions: s-ZSM-5:Na-FeC$_x$, 3; ethene/H_2/CO, 0.6/2.0/1.0; 2,880 ml per h per g FeC$_x$; 2.0 MPa, granule mixing of the individual s-ZSM-5 and Na-FeC$_x$ components for Na-FeC$_x$/s-ZSM-5; reaction temperature, 260 °C for Na-FeC$_x$/s-ZSM-5 and 300 °C for Na-FeC$_x$. **c**, Ethene desorption DRIFT spectra over the Na-FeC$_x$/s-ZSM-5, Na-FeC$_x$/n-ZSM-5 and Na-FeC$_x$ catalysts. The inset shows the residual percentage of ethene according to the peak-intensity changes (signal at ~2,986 cm^{-1}) over the different catalysts for 0–12 min (data calculated from Supplementary Table 13).

These results help us to understand that the poor activity of the Na-FeC$_x$ catalyst in the LT-FTO might be due to the difficulty of olefin product desorption at lower temperatures (Supplementary Fig. 37). By contrast, the equivalent introduction of ethene causes only a slight loss of activity over the Na-FeC$_x$/s-ZSM-5 and Na-FeC$_x$/s-S1-OH catalysts, at even lower temperature, confirming the improved olefin tolerance of Na-FeC$_x$ when mixed with s-ZSM-5 or s-S1-OH zeolites (Fig. 3b; Supplementary Fig. 38). It is possible that these rationally designed zeolites help desorption of the olefin products once they have been formed on the Na-FeC$_x$ surface (Supplementary Fig. 37), thus improving the activity.

This hypothesis was supported by diffuse reflectance infrared Fourier transform (DRIFT) spectra characterizing the olefin desorption on various catalysts including Na-FeC$_x$, Na-FeC$_x$/n-ZSM-5 and Na-FeC$_x$/s-ZSM-5. As shown in Fig. 3c, after pre-adsorption treatment of ethene as a model, all samples showed the bands which were assigned to the symmetrical and asymmetrical stretching vibration of =CH_2 (ν_s=CH_2 signals at 2,966, 2,986, 3,011 and 3,124 cm^{-1}; ν_{as}=CH_2 signal at 3,075 cm^{-1})[44–46]. Introducing an argon (Ar) flow to the samples, the signals were reduced because of the ethene desorption. According to the peak-intensity changes as a function of time, the ethene desorption rates followed the order of Na-FeC$_x$/s-ZSM-5 > Na-FeC$_x$ > Na-FeC$_x$/n-ZSM-5 (Supplementary Figs. 39–42 and Supplementary Table 13), which is confirmed in the equivalent infrared spectroscopy tests using propene as a probing molecule (Supplementary Figs. 43–47 and Supplementary Table 14). These data demonstrate that the s-ZSM-5 zeolite indeed accelerates the desorption of olefins once they are formed on the Na-FeC$_x$ surface via C–C coupling in the FTO, which is favourable for the continuous hydrogenation of CO. In addition, the rapid olefin desorption also inhibits the growth of carbon chains to optimize the selectivity, hindering the formation of long-chain hydrocarbons. The s-S1-OH zeolite displayed the same behaviour as s-ZSM-5 in accelerating the olefin desorption, as shown in the DRIFT spectra of the samples (Supplementary Figs. 48 and 49). By contrast, the n-ZSM-5 zeolite delayed the olefin desorption on the Na-FeC$_x$/n-ZSM-5 catalyst. The long retention time of olefins in the Na-FeC$_x$/n-ZSM-5 catalyst led to olefins being hydrogenated (Supplementary Fig. 50), which could explain the large amount of alkane products rather than olefins in the LT-FTO over the Na-FeC$_x$/n-ZSM-5 catalyst (Fig. 3a).

In previous studies using zeolite and sodium-modified oxide/carbide mixtures, sodium migration may occur to influence the

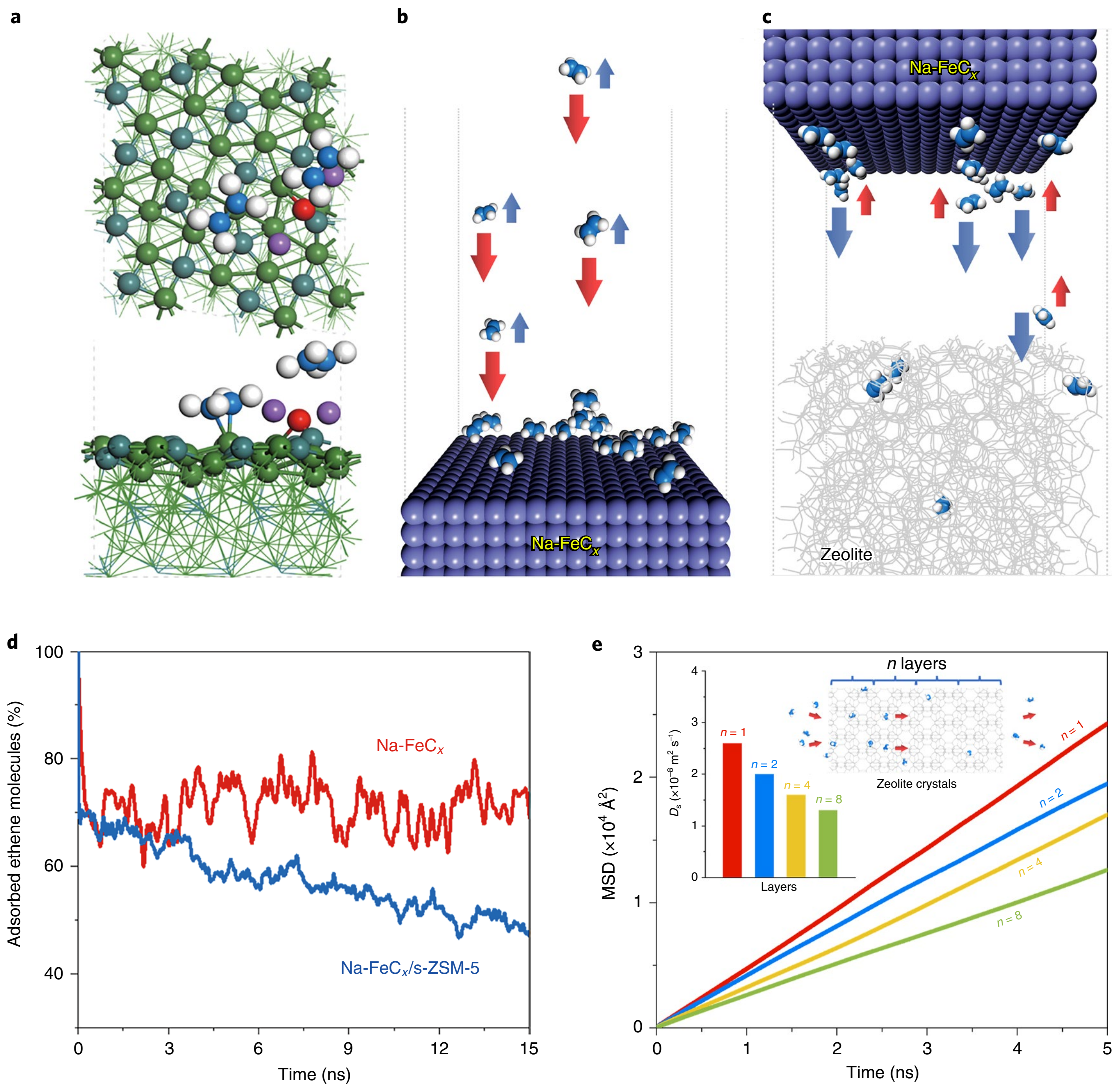

Fig. 4 | Computational models, DFT calculations, MD simulations and the MSD of ethene. **a**, Top and side view of the adsorption model showing ethene molecules on the Na-FeC$_x$ surface (C_2H_4: C atoms in blue and H atoms in white; Na-FeC$_x$: O atoms in red, Na atoms in purple, Fe atoms in green and C atoms in dark cyan). **b**,**c**, Schematics showing the adsorption–desorption equilibrium of ethene molecules on Na-FeC$_x$ (**b**) and Na-FeC$_x$/s-ZSM-5 (**c**). **d**, Percentage of ethene molecules adsorbed on the Na-FeC$_x$ surface during the desorption process over Na-FeC$_x$ and Na-FeC$_x$/s-ZSM-5. **e**, MSD of ethene molecules diffusing in s-ZSM-5 zeolites with different numbers of unit layers (n). Insets: the model (right) and D_s values (left) in the MFI zeolite micropores with one, two, four and eight unit layers.

intrinsic catalytic activity of the individual oxide/carbide compounds[31,33,36,47,48]. Notably, this hypothesis can be excluded to explain the high CO conversion on the Na-FeC$_x$/zeolite catalysts in the LT-FTO (Supplementary Figs. 51 and 52 and Supplementary Tables 10 and 15). Na-FeC$_x$ recycled from the used Na-FeC$_x$/s-ZSM-5 catalyst still exhibited a comparable performance to that of fresh Na-FeC$_x$ (Supplementary Table 15). Therefore, the fast desorption of olefins from the catalyst surface should dominate the accelerated reaction. Olefin desorption from the composite catalysts should proceed via efficient olefin transfer from the Na-FeC$_x$ surface to the zeolite micropores and then rapidly escape from the zeolite micropores. To understand this process, we performed theoretical simulations to provide a deeper insight into the adsorption and diffusion characteristics of ethene on the Na-FeC$_x$ surface as a model (Fig. 4a; Supplementary Fig. 53). In practice, the free olefins may either escape to form the final product or re-adsorb on the Na-FeC$_x$ surface to participate an adsorption–desorption equilibrium ($^*C_2H_4 \leftrightarrows {}^* + C_2H_4$), which hinders the escape of ethene (Fig. 4b; Supplementary Fig. 54). Figure 4c shows the model containing the Na-FeC$_x$ surface and adjacent zeolite framework with a free region between them (Supplementary Fig. 55), for considering the separated Na-FeC$_x$ and s-ZSM-5 components in the practical catalytic tests. Using this model, the ethene molecules on the Na-FeC$_x$ surface desorb into the free region, where these free olefins can be rapidly adsorbed by the zeolite to shift the sorption equilibrium on Na-FeC$_x$ and accelerate the desorption of more ethene molecules (Supplementary Fig. 56). The ethene desorption processes were qualitatively predicted on the Na-FeC$_x$ surface as a function of the desorption time using molecular dynamics (MD) simulations at 260 °C (Fig. 4d). It was observed that the number of ethene molecules on the Na-FeC$_x$ surface maintained a quantitative equilibrium of about 71% for the model without the zeolite. By contrast, this number clearly decreased to about 48% for the model with the zeolite, confirming the shifted equilibrium between desorption and re-adsorption in the presence of the microporous ZSM-5 zeolite. In this case, the zeolite did not directly interact with the olefin molecules on the Na-FeC_x surface, but captured the olefins in the free region to shift the equilibrium (Supplementary Figs. 57 and 58).

With regard to ethene molecules in the zeolite crystals, density functional theory (DFT) calculations showed adsorption energies of −0.42, −0.78 and −0.59 eV within the micropores for S-1, ZSM-5 and S1-OH, respectively (Supplementary Fig. 59), which confirmed the enhanced adsorption of ethene by the acid sites or silanol groups within the zeolite. This result has been illustrated by the well-known H–π interaction between a proton and a C=C bond[49], confirming the importance of the zeolite microporous environment for olefin adsorption. We also investigated the ethene molecule diffusing along the straight channels of the MFI zeolite structure with different unit layers (Supplementary Fig. 60). Interaction energy profiles showed that the diffusion barriers for ethene are strongly related to the adsorption sites, and the diffusion efficiency in one zeolite crystal should be inversely proportional to the number of adsorption sites and the zeolite channel distance, which can be adjusted via the MFI zeolite crystal morphology (for example, s-ZSM-5 versus n-ZSM-5 have a similar composition but a different morphology). Further studies on olefin diffusion in the zeolite micropores were performed, and from the slope of the mean square displacement (MSD) values of the ethene diffusion coefficient (D_s) with various zeolite layers were quantitatively determined (Fig. 4e; Supplementary Fig. 61)[50]. The D_s values were 2.6×10^{-8}, 2.0×10^{-8}, 1.6×10^{-8} and $1.3\times10^{-8}\,m^2\,s^{-1}$ in the s-ZSM-5 MFI zeolite with one, two, four and eight unit layers, respectively. These results suggest that thinner zeolite crystals with shorter diffusion distances have a significant advantage for the more rapid shipping of olefin products, resulting in short retention times of the olefin molecules in the zeolite crystals; this will be beneficial for the fast formation of free micropores for continuously adsorbing more olefin products. To further address this issue, we theoretically simulated the olefin product desorption from the Na-FeC_x surface using a zeolite with residual olefins in the micropores. In this case, the number of ethene molecules on the Na-FeC_x surface maintained a quantitative equilibrium of about 69%, which is much higher than 48% for the case of using zeolite with free micropores (Supplementary Fig. 62). This outcome is reasonable due to the lack of sufficient free micropores on the MFI zeolite because they are blocked by residual olefin molecules, confirming the importance of rapid olefin shipping in zeolite crystals.

We have demonstrated the LT-FTO process with boosted activity and optimized selectivity on a classical ferric carbide catalyst combined with a zeolite promoter via appropriate mixing. Multiple experimental and theoretical studies revealed that the zeolite, with a reasonably controlled microporous environment for adsorbing olefins and a short *b*-axis thickness for the shipping of olefins, can accelerate the escape of product molecules. This feature benefits the formation of a free catalyst surface for CO hydrogenation. We believe that the designed zeolites can act as a powerful promoter for syngas conversion. In addition to the equilibrium change, other interpretations of the zeolite promotion effect should be considered in future work for deeply understanding the synergism between metal carbide and zeolite, which could open an alternative route to the design and preparation of more efficient heterogeneous catalysts for syngas conversion.

Online content

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Methods

Synthesis of s-ZSM-5 zeolite. The ZSM-5 zeolite with a sheet morphology and a *b*-axis thickness of 90–110 nm (denoted as s-ZSM-5, where 's' stands for sheet) was synthesized using a urea-assisted method. As a typical run, tetrapropylammonium hydroxide (TPAOH; 4.0 g, 25% aqueous solution) was dissolved in deionized water (5.1 g) with stirring for 0.5 h, and then tetraethylorthosilicate (TEOS; 2.8 g) was added to the solution with stirring. After the addition of aluminium isopropoxide (0.07 g) and urea (0.65 g), the mixture was stirred continuously at room temperature for another 12 h. The resulting gel, with the molar composition of SiO_2/Al_2O_3/TPAOH/H_2O/urea of 1/0.0128/0.3684/33.8346/0.812, respectively, was transferred into an autoclave for crystallization at 180 °C for two days. The as-synthesized products were collected by filtration, washed with water, dried in air and calcined at 550 °C for 4 h to remove the organic template. Finally, the s-ZSM-5 zeolite with a Si/Al ratio of 26.2 was obtained. For this work, unless specified otherwise, the s-ZSM-5 zeolite is in the protonated form. For comparison, s-ZSM-5 zeolites with Si/Al ratios of 16.5, 84.3 and 141.2 were synthesized via similar procedures using 0.14, 0.035 and 0.02 g of aluminium isopropoxide in the starting gels, respectively. In this work, unless specified otherwise, the s-ZSM-5 zeolite has the Si/Al ratio of 26.2.

Synthesis of s-S1-OH zeolite. The S-1 zeolite with a sheet morphology, artificially introduced silanol groups and a *b*-axis thickness of 90–115 nm (denoted as s-S1-OH, where s stands for sheet) was synthesized using a urea-assisted method in the presence of organosilane in the starting gel. As a typical run, TPAOH (4.0 g, 25% aqueous solution) was dissolved in deionized water (5.1 g) with stirring for 0.5 h, and then TEOS (2.52 g) and diethoxydimethylsilane (DEMS; 0.199 g) was added to the solution with stirring (where the molar ratio of organosilane to the total amount of silica was 1:10). After the addition of urea (0.65 g), the mixture was stirred continuously at room temperature for another 12 h. The resulting gel was transferred into an autoclave for further crystallization at 180 °C for four days. The as-synthesized products were collected by filtration, washed with water, dried in air and calcined at 550 °C for 4 h for removal of the organic template and transformation of the methyl groups into silanol groups. For comparison, s-S1-OH zeolite samples with different molar ratios of organosilane to the total amount of silica of 1/50 and 1/20 were synthesized via similar procedures using 2.74 g of TEOS and 0.04 g of DEMS for the 1/50 sample, and 2.66 g of TEOS and 0.1 g of DEMS for the 1/20 sample. In this work, unless specified otherwise, the s-S1-OH zeolite has a molar ratio of organosilane to the total amount of silica of 1/10.

Procedures for the synthesis of ferric carbide catalysts and other zeolites are provided in the Supplementary Information.

Catalytic tests. The LT-FTO reactions were performed using a stainless-steel fixed-bed flow reactor with an inner diameter of 10 mm. Typically, the Fe-based catalyst (0.5 g) was mixed with the zeolite (1.5 g) (where the catalyst granule size was 20–40 mesh unless stated otherwise), and was positioned in the middle of the reactor. Additional quartz sand was added to the reactor to fix the position of the catalyst bed. For each reaction, the catalyst was pre-reduced at 400 °C for 6 h in a pure H_2 flow (flow rate, 10 ml min^{-1}) at atmospheric pressure. After reduction, the reactor was cooled to 260 °C. Then, the feed gas (H_2/CO/Ar with a molar ratio of 45/45/10) with a flow rate of 20 ml min^{-1} was introduced into the reactor, and the reactor was pressurized gradually to 2 MPa. For comparison with different zeolites, the the amount of Na-FeC_x catalyst (0.5 g) was constant in the reactor, and the different zeolites with various amounts were changed. The GHSV (gas hourly space velocity) was calculated on the basis of the weight of the Fe-based catalyst, that is 2,400 ml per h per g FeC_x, and the zeolite promoter was not included.

The gaseous products from the reactor were analysed using two online gas chromatographs. One is a Fu Li-9790 gas chromatograph equipped with a TDX-01 column (3 m × 3 mm) and a thermal conductivity detector; the other is a chromatograph equipped with a capillary PLOT-Al_2O_3 column (50 m × 0.53 mm × 0.25 μm) and a flame ionization detector (FID). The liquid-phase products collected in the cold trap were analysed using an offline Shimadzu GC-14C gas chromatograph equipped with an HP-5 column (100 m × 0.32 mm × 0.25 μm) and an FID. Argon was used as an internal standard for calculating the CO conversion and product selectivities (CO_2 and CH_4) in the thermal-conductivity-detector analysis. Methane was used as an internal standard for calculating the product selectivities in the online FID analysis. The offline products were analysed using *n*-hexadecane as an internal standard. The conversions and product selectivities were calculated on the basis of the amount of total carbon atoms. At the beginning of each reaction, the ferric oxide would change into carbide, so the CO conversion and selectivity data were unstable during this activation period. For characterizing the performance of each catalyst, the catalytic data were recorded after reaction for more than 40 h to obtain stable CO conversion and product selectivity values, which was to ensure that the reaction had reached steady operation.

Data availability

The data that support the findings of this study are presented in the Letter and Supplementary Information, and are available from the corresponding authors upon reasonable request. Source data are provided with this paper.

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Author contributions

C.W. and W.F. performed the catalyst preparation, characterization and catalytic tests. Z. Liu and A.Z. performed the theoretical calculations and wrote the corresponding part. H.L., L.L., H.Z., X.Q., S.X. and Y.W. participated in the catalyst characterization. Z. Liao and Y.Y. provided helpful discussion and compiled the process package. X.C. performed the X-ray photoelectron spectroscopy characterization. L.W. and F.-S.X. designed the study, analysed the data and wrote the paper. All authors discussed the results and commented on the manuscript.

Competing interests

This work has been protected by a Chinese patent with the application number 202110715110.6. The authors C.W., W.F., L.W. and F.-S.X. were involved in this patent. The other authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Liang Wang, Anmin Zheng or Feng-Shou Xiao.

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Supplementary information

Fischer–Tropsch synthesis to olefins boosted by MFI zeolite nanosheets

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Supplementary Information

Fischer-Tropsch synthesis to olefins boosted by MFI zeolite nanosheets

Chengtao Wang[1,2,#], Wei Fang[1,#], Zhiqiang Liu[3,#], Liang Wang[1,*], Zuwei Liao[1], Yongrong Yang[1], Hangjie Li[2], Lu Liu[1], Hang Zhou[1], Xuedi Qin[2], Shaodan Xu[4], Xuefeng Chu[5], Yeqing Wang[2], Anmin Zheng[3,*], Feng-Shou Xiao[1,*]

[1]Key Lab of Biomass Chemical Engineering of Ministry of Education, College of Chemical and Biological Engineering, Zhejiang University, Hangzhou 310027, China.

[2]Key Laboratory of Applied Chemistry of Zhejiang Province, Department of Chemistry, Zhejiang University, Hangzhou 310028, China.

[3]National Center for Magnetic Resonance in Wuhan, State Key Laboratory of Magnetic Resonance and Atomic and Molecular Physics and Mathematics, Wuhan Institute of Physics and Mathematics, Innovation Academy for Precision Measurement Science and Technology, Chinese Academy of Sciences, Wuhan 430071, China.

[4]College of Materials and Environmental Engineering, Hangzhou Dianzi University, Hangzhou 310018, China.

[5]Key Laboratory of Architectural Cold Climate Energy Management, Jilin Jianzhu University, Changchun 130118, China.

*Correspondence to: liangwang@zju.edu.cn; zhenganm@wipm.ac.cn; fsxiao@zju.edu.cn.

[#]These authors contributed equally.

Materials

Tetraethylorthosilicate (TEOS, >99%), diethoxydimethylsilane (DEMS, 99%), aluminum isopropoxide (99.99%), 1-hexene (99%, AR), 1-octene (98%, AR), and SiO_2 (AR) were obtained from Aladdin Chemical Reagent Company. $FeCl_3{\cdot}6H_2O$, $FeCl_2{\cdot}4H_2O$, HCl, and NaOH were obtained from Sinopharm Chemical Reagent Co. Ltd. Tetrapropylammonium hydroxide (TPAOH, 25 *wt*%) was supplied by Jiangsu Kente Chemical Technology Co. Ltd. Tetrapropylammonium hydroxide (TPAOH, 40 *wt*%) was supplied by Shanghai Cairui Chemical Technology Co. Ltd. Boehmite (AlOOH) was supplied by Changling Catalyst Co. Ltd. Beta, MOR, and SAPO-34 zeolites were purchased from Tianjin Nankai Catalyst Co. Ltd. SSZ-13 and ZSM-22 zeolites were purchased from Zhejiang Taide catalyst Co. Ltd. H_2, CO/Ar, H_2/CO/Ar, H_2/Ar, C_2H_4, C_3H_6, and C_4H_8 were supplied by Hangzhou Jingong special gas Co. Ltd. Tungsten trioxide (WO_3, 99.8%) was obtained from Shanghai Macklin Biochemical Co. Ltd.

Catalyst preparation

Synthesis of Na-Fe_3O_4 and Na-FeC_x catalysts. Typically, 30.0 g of $FeCl_3{\cdot}6H_2O$ and 11.9 g of $FeCl_2{\cdot}4H_2O$ were dissolved in a mixture of deionized water (200 mL) and aqueous HCl solution (5.0 mL, 12.0 mol L^{-1}). Then NaOH aqueous solution (2.0 mol L^{-1}) was added dropwise into the solution under vigorous stirring until the pH value reaching to 10. After stirring for another 1 h, the solid phase was separated from the solution *via* centrifugation. After washing with a large amount of deionized water for six times and drying at 100 °C overnight, the Na-Fe_3O_4 sample was finally obtained. The Na-FeC_x was prepared by treating the Na-Fe_3O_4 sample with syngas (molar ratio of CO to H_2 at 1:1) at 300 °C for 24 h with a gas flow rate 50 mL/min, which was denoted as the Na-FeC_x. The Na loading of the Na-FeC_x was 5.35% by ICP analysis. Different Na loading of the Na-FeC_x was prepared by washing with water for different times, denoted as nNa-FeC_x, where the n stands for the Na loadings. The Na-FeC_x means that the Na loading is 5.35 wt% unless otherwise statement.

Synthesis of *n*-ZSM-5 zeolite. The normal ZSM-5 zeolite (denoted as *n*-ZSM-5) was prepared by a conventional hydrothermal method. In a typical run, 3.5 g of TEOS was added into a mixture containing 11.28 g of deionized water and 3.3 g of tetrapropylammonium hydroxide (TPAOH, 40 wt %), followed by adding 0.03 g of AlOOH and stirring at room temperature for 12 h. Then, the mixture was transfer into an autoclave and heated at 180 °C for 3 days. After removal of the organic template by calcination at 550 °C for 4 h, the *n*-ZSM-5 zeolite with Si/Al ratio at 30.5

was finally obtained.

Synthesis of *s*-S1 zeolite. The S-1 zeolite with sheet morphology and *b*-axis thickness at 80-100 nm (denoted as *s*-S1, where *s* stands for sheet) was synthesized from a urea-assisted method. As a typical run, 4.0 g of TPAOH (25% aqueous solution) was mixed with 5.1 g of deionized water under stirring for 0.5 h, then 2.8 g of TEOS was added to the solution under stirring. After adding 0.65 g of urea, the mixture was continuously stirred at room temperature for another 12 h. The resulting gel was transferred into an autoclave for further crystallization at 180 °C for 2 days. The as-synthesized products were collected by filtrating, washing with water, drying in air, and calcining at 550 °C for 4 h to remove the organic template.

Synthesis of *n*-S1 zeolite. The normal S-1 zeolite (*n*-S1) was prepared by a conventional hydrothermal method. In a typical run, 3.5 g of TEOS was added into a mixture containing 11.28 g of deionized water and 3.3 g of TPAOH (40 wt %), followed by stirring at room temperature for 12 h. Then, the mixture was transfer into an autoclave and heated at 180 °C for 3 days. After removal of the organic template by calcination at 550 °C for 4 h, the *n*-S1 zeolite was finally obtained.

Synthesis of *n*-S1-OH zeolite. The S-1 zeolite with silanol groups (*n*-S1-OH) was prepared by solvent-free crystallization. As a typical run, 64 mL of ethanol was added to 80 mL of a water solution containing 6 mL of aqueous ammonia with stirring. After addition of 6.0 g of TEOS and 0.474 g of DEMS, the solution was stirred at room temperature for another 12 h, followed by distillation under vacuum to remove the water and ethanol and drying at 100 °C overnight. Then the solid powder of amorphous silica modified with methyl groups (SiO_2-Me) was obtained. The S-1-OH was synthesized by solvent-free crystallization with SiO_2-Me in the presence of TPAOH. After grinding of 1.6 g of TPAOH and 2 g of SiO_2-Me at room temperature for 0.5 h, the mixture was transferred into an autoclave for crystallization at 180 °C for 3 days to give the methyl group-modified S-1. The as-synthesized products were collected by filtrating, washing with water, drying in air, and calcining at 550 °C for 4 h for removal of the organic template and transformation of the methyl groups into silanol groups. Finally, the *n*-S1-OH zeolite was obtained.

Catalyst characterization

Powder X-ray diffraction (XRD) patterns were collected on a Rigaku D/MAX 2550 diffractometer with CuKα radiation (λ = 1.5418 Å). Nitrogen sorption isotherms were measured using a Micromeritics ASAP 2020M system. Transmission electron microscopy (TEM) images, scanning transmission electron microscopy (STEM) images, and Energy dispersive spectrometer (EDS) spectra were recorded on a JEM-2100F electron microscope (JEOL, Japan) with an acceleration voltage of 200 kV. Scanning electron microscopy (SEM) experiments were performed with Hitachi SU-8010 and SU-1510 electron microscopes. NH_3 temperature-programmed desorption (NH_3-TPD) was performed on a catalyst analyzer BELCAT II instrument. ^{29}Si MAS NMR was performed on a Bruker AVANCE-III 500 MHz spectrometer operating at Larmor frequencies of 500.57 and 99.44 MHz for the ^{1}H and ^{29}Si nucleus, respectively. ^{29}Si MAS NMR spectra with high power proton decoupling were recorded using a π/2 pulse of 3.9 us and a recycle delay of 60 s on a 7 mm MAS probe with a spinning rate of 4 kHz. A contact time of 4 ms and a recycle delay of 3 s were used for the ^{29}Si-^{1}H CP MAS measurement. The chemical shift of ^{29}Si was externally referenced to kaolinite (-91.5 ppm). IR spectra was performed with Perkin-Elmer Frontier system. Metal loadings and Si/Al ratios of the catalysts were determined by inductively coupled plasma analysis (ICP, Perkin-Elmer 3300DV). Gas chromatograph-mass spectrometer (GC-MS) analysis was performed with a Shimadzu GCMS-QP2020-NX system. X-ray photoelectron spectroscopy (XPS) measurements were carried out using a Thermo Fisher Scientific ESCALAB 250Xi with photoelectron spectroscopy system using a monochromatic Al Ka (1486.6 eV) X-ray source. The instrument work function was calibrated to give an Au4$f_{7/2}$ metallic gold binding energy of 83.96 eV. The spectrometer dispersion was adjusted to give a binding energy of 368.21 eV for metallic Ag3$d_{5/2}$ and 932.62 eV for metallic Cu2$p_{3/2}$. The XPS spectra were collected in constant analyzer energy (CAE) mode. Instrument base pressure was 2.0´10 -10 mbar and high-resolution spectra were collected using a spot size of 500 μm, with pass energy of 30 eV. The charge neutralization being monitored using the C1s (284.8 eV) signal for adventitious carbon. The granule-mixed catalysts were grinded into mixed powder before the Na1s XPS characterization.

Temperature-programmed desorption of C_2H_4 (C_2H_4-TPD) was performed on a catalyst analyzer BELCAT II instrument. Typically, the sample was pretreated with He flow at 400 °C for 1 h. Then, the He flow was switched off and a C_2H_4/He (20 vol% C_2H_4) gas mixture was induced for C_2H_4 adsorption. After adsorption at 40 °C for 1 h, the gas feed was switched to He for a swept treatment at 40 °C for 0.5 h. C_2H_4-TPD profiles were collected in flowing He by raising the temperature from 40 to 700 °C with ramping rate at 10 °C/min. The C_3H_6-TPD tests were performed from the same procedures except for using C_3H_6/He (10 vol% C_3H_6) as the feed gas for pre-adsorption.

C_2H_4(C_3H_6)-desorption DRIFT spectra were recorded using a Nicolet is50 FT-IR spectrometer equipped with an MCT/A detector and ZnSe windows and a high-temperature reaction chamber. As a typical run, the catalyst was localized in the sample cell and pre-treated in pure Ar at 400 °C for 1 h (flow rate at 30 mL/min), and then the temperature was decreased to 30 °C. After maintaining the gas flow for 0.5 h, the background data were collected. Then the pure C_2H_4 was introduced into the sample with a flow rate of 30 mL/min for 0.5 h to ensure the saturated adsorption. Then, the gas flow was switched to pure Ar with a flow rate of 10 mL/min to start the desorption step. After desorption for 4 min, the IR spectra were recorded at intervals of two minutes. The C_3H_6-desorption DRIFT spectra were recorded from the same procedures except for using C_3H_6/He (50 vol% C_3H_6) as the feed gas for pre-adsorption.

C_3H_6 hydrogenation in pulsed experiment

C_3H_6 hydrogenation was performed using a chemisorption instrument (iChem700, PhysiChem Instruments Ltd.) connected to a mass spectrometer (SRD200M, TILON GRP TECHNOLOGY LIMITED). As a typical run, 0.1 g of Na-FeC_x (40-60 mesh) was mixed with 0.3 g of *s*-ZSM-5 (40-60 mesh) or 0.3 g of *n*-ZSM-5 (40-60 mesh) in the reactor tube. The temperature of catalyst bed was raised to 260 °C in a pure Ar flow (200 mL/min). After that, the gas was switched to a flowing hydrogen gas (10%H_2/Ar, flow rate at 200 mL/min). Then, a slight amount of C_3H_6 (1 mL of mixed gas containing 10% C_3H_6 in Ar) was pulsed into the system, the signals of C_3H_6 and C_3H_8 were collected by a mass detector. The pulsed experiment was repeated for multiple times (C_3H_6 was pulsed every 4 min) in each test.

Computational Models

Model A (Fig. 4a, Supplementary Fig. 53) simulating the Na-FeC_x catalyst: The Na_2O/Fe_5C_2 was constructed to represent the Na-FeC_x catalyst[1]. The optimized Fe_5C_2 unit cell was a monoclinic crystal and had C2/c crystallographic symmetry, whose lattice constants are a = 11.588 Å, b = 4.579 Å, c = 5.059 Å and β = 97.75°, showing an excellent agreement with experiments, in which a, b, and c equal 11.562, 4.573 and 5.060 Å respectively, and β equals 97.74°(2). To rationalize the experimental observations of this work, we used the 2 × 2 supercell Fe_5C_2 with Na_2O. In particular, the Na-FeC_x surface contains 96 Fe, 48 C, 1 O and 2 Na atoms, in which the bottom

40 Fe and 16 C were fixed, and the rest atoms were relaxed. These models were used for calculating the adsorption energy between ethene and Na-FeC_x.

Model B simulating the Na-FeC_x/*s*-ZSM-5 catalyst: The simplified Na-FeC_x (Fig. 4b, Supplementary Fig. 54) and Na-FeC_x/*s*-ZSM-5 surface model (Fig. 4c, Supplementary Fig. 56) were constructed to investigate the diffusion of ethene. The initial framework structure of pure silicon MFI was taken from the International Zeolite Associations database[3] and Si(12)-O(26)-Al(12) site was chosen as the locations of Brønsted acidic proton. For surface model, the 2 × 1 × 3, 2 × 2 × 3, 2 × 4 × 3 and 2 × 8× 3 supercells were selected to represent ZSM-5 with 1-, 2-, 4-, and 8-unit layers. Then, the *s*-ZSM-5 surface was obtained by disrupting the periodicity of the bulk zeolite and 40 Å vacuum layer were added also along [010] (Supplementary Fig. 61). It should be noted that, the hydroxyl groups were added to the unsaturated atoms over the external surface. Due to the strong interacting between the Fe and ethene, only the Fe atoms have been built to simplify the models of Na-FeC_x surface. The distance between two Fe atoms is about 2.6 Å which could be subtle tuned to match the zeolites periodicity. Six layers of Na-FeC_x were built while two left layers are C (with the similar force field parameter of C from ethene) to get the single-sided Na-FeC_x model. The loading number ethene is 20 and the distance between Na-FeC_x and *s*-ZSM-5 is about 30 Å. The lattice parameters are 40.18 × 158.95 × 39.43 Å fort both Na-FeC_x and Na-FeC_x/*s*-ZSM-5.

DFT simulation

All geometry optimizations were performed according to the plane-wave-based periodic DFT method as implemented in the Vienna Ab Initio Simulation Package[4,5]. The electron-ion interaction was described with the projector augmented wave method[6,7]. The electron exchange and correlation energies were treated within the generalized gradient approximation in the Perdew-Burke-Ernzerhof functional (GGA-PBE)[8]. The energy cutoff of the plane-wave basis was set up to 400 eV. Electron smearing was used *via* the Methfessel-Paxton technique to speed up the convergence of the metallic systems with a width of $\sigma = 0.2$ eV and the 1 × 1 × 1 gamma-point sampling was used for geometry optimization.

The adsorption energy was calculated according to $E_{ads} = E_x/_{slab} - [E_{slab} + E_x]$, where $E_x/_{slab}$ was the total energy of the slab with adsorbates in its equilibrium geometry, E_{slab} was the total energy of the bare slab, and E_x was the total energy of the free adsorbates in the gas phase. Therefore, the more negative for the value of E_{ads}, the stronger for the adsorption.

Molecular dynamics (MD) simulation

MD simulations were performed in the canonical ensemble (NVT), where the number of particles (N), volume (V), and temperature (T) were kept constant. The simulated temperature was 260 °C and controlled by a Nosé-Hoover thermostat with a coupling time constant of 0.1 ps. The leapfrog Verlet algorithm was used to integrate the Newton's equations of motion with a time step of 0.5 fs. The TraPPE-UA[9] and TraPPE-zeo[10-12] force field was used for hydrocarbon and zeolites (Supplementary Table 16), respectively. All the Lennard–Jones cross-interaction parameters were determined by the Lorentz–Berthelot mixing rules, while the parameters for Fe_5C_2 and C_2H_4 were got by fitting the adsorption energy between ethene and the outermost layer of Na_2O/Fe_5C_2 ($\varepsilon_{Fe\text{-}C}$ = 0.169 kcal/mol and $\sigma_{Fe\text{-}C}$ = 3.675 Å). Each MD simulation was equilibrated for 2×10^6 steps, then following 2×10^7 to 3×10^7 steps production for studying the diffusion behaviors of adsorbate molecules. The cutoff radius was 14 Å, and the trajectories were recorded every 1000 steps. 10 independent MD simulations were carried out for better statistics. All MD simulation were performed in the DL_POLY 2.0 code[13].

Mean square displacement and diffusion coefficient.

The mean square displacement (MSD) of ethene was defined as equation in the following:

$$MSD\left(\tau\right) = \frac{1}{N_m}\sum_{i}^{N_m}\frac{1}{N_\tau}\sum_{t_0}^{N_t}\left[r_i\left(t_0+\tau\right)-r_i\left(t_0\right)\right]^2 \qquad (1)$$

where N_m was the number of gas molecules, N_τ was the number of time origins used in calculating the average, and r_i was the coordinate of the i-th molecule. In addition, the slope of the *MSD* as a function of time determined the self-diffusion coefficient, D_s, which was defined according to the so-called Einstein relation.

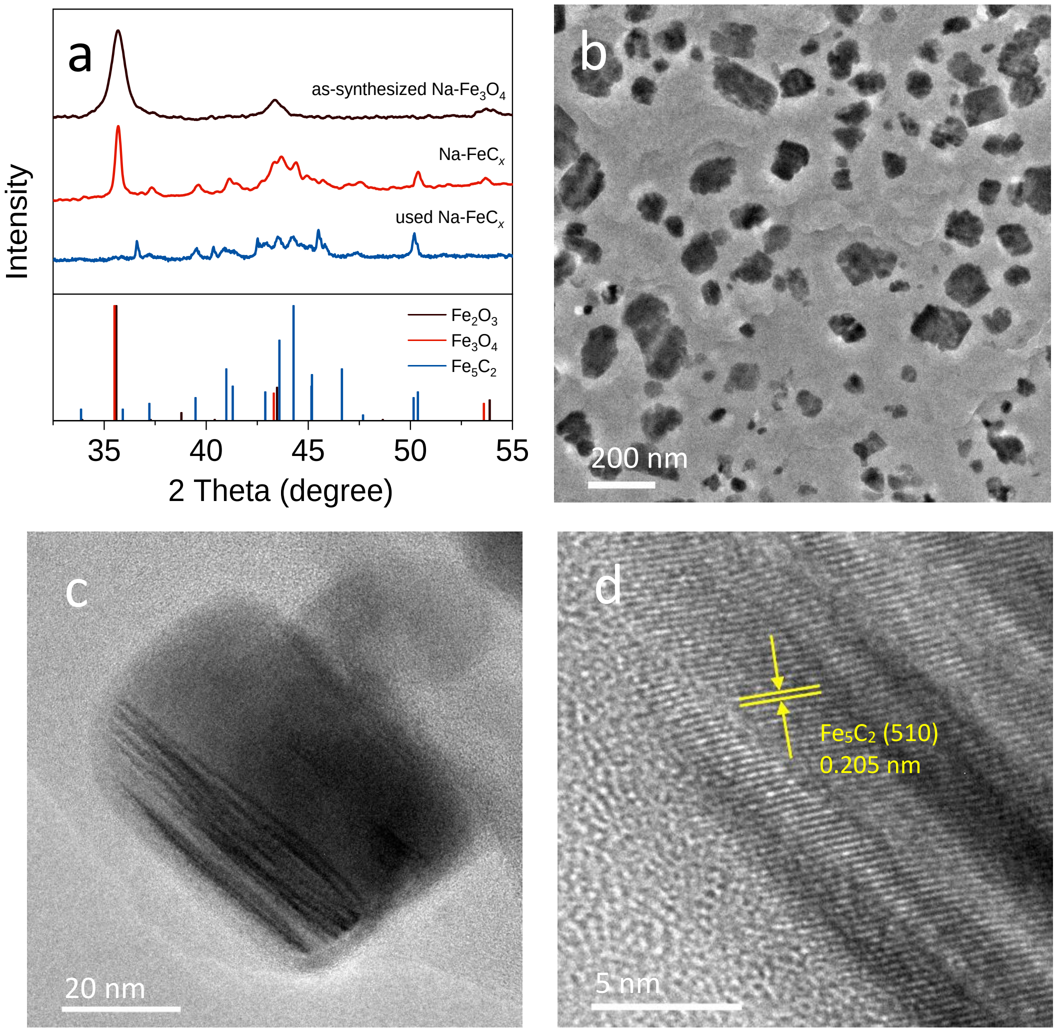
$$MSD\left(\tau\right) = 2nD_s\tau + b \qquad (2)$$

where n was the dimension of zeolites. In this work, $n = 1$ for the diffusion along with straight channel of ZSM-5. The reported *MSD* and D_s values were calculated as the average of 10 independent MD trajectories.

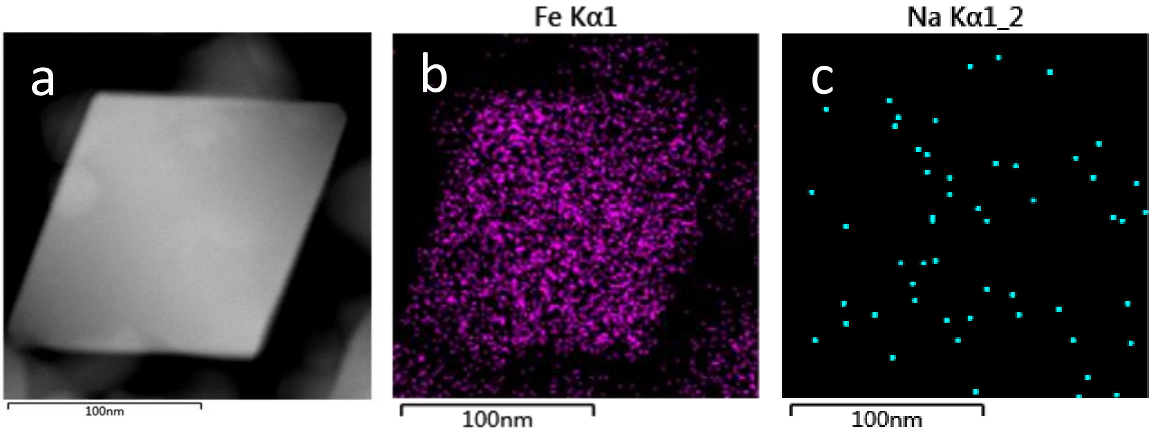
Interaction energy profiles and energy barriers

The interaction energy profiles[14,15] along with straight channel of *s*-ZSM-5 with different layers (*i.e,*1, 2, 4 and 8 layers) have been performed (Supplementary Fig. 60). In the calculations, an ethene molecule was firstly placed near the surface, and then systematically moved along with straight channel following the diffusion path in equidistant steps of 1 Å. The interaction energy between framework and molecule at each point was calculated and the energy barrier for crossing channel was determined by the difference between the local lowest and the highest energy along the diffusion pathway.

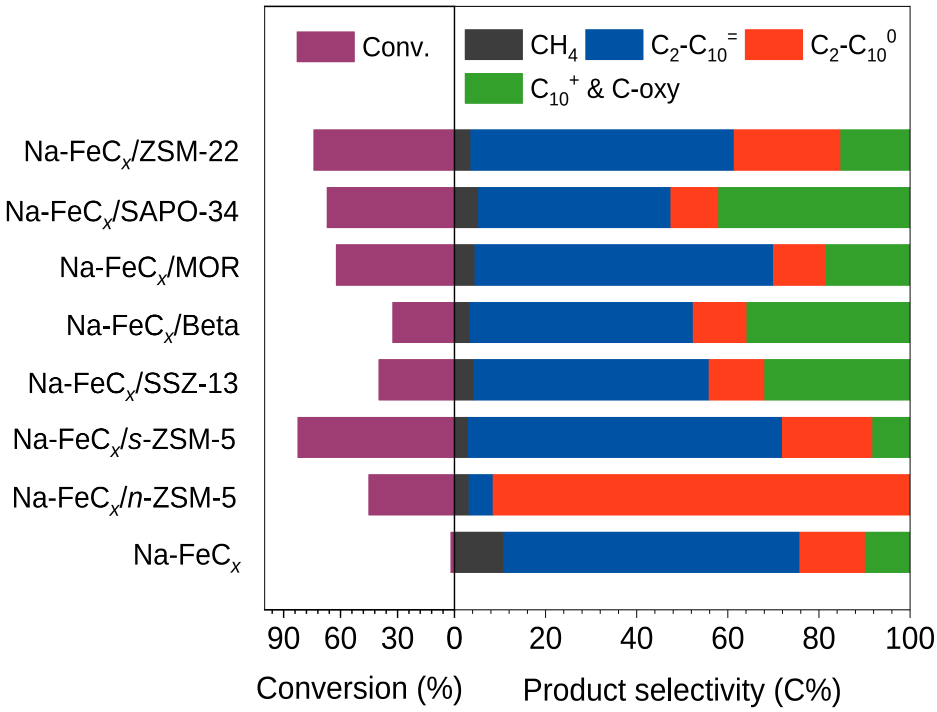
Supplementary Figures



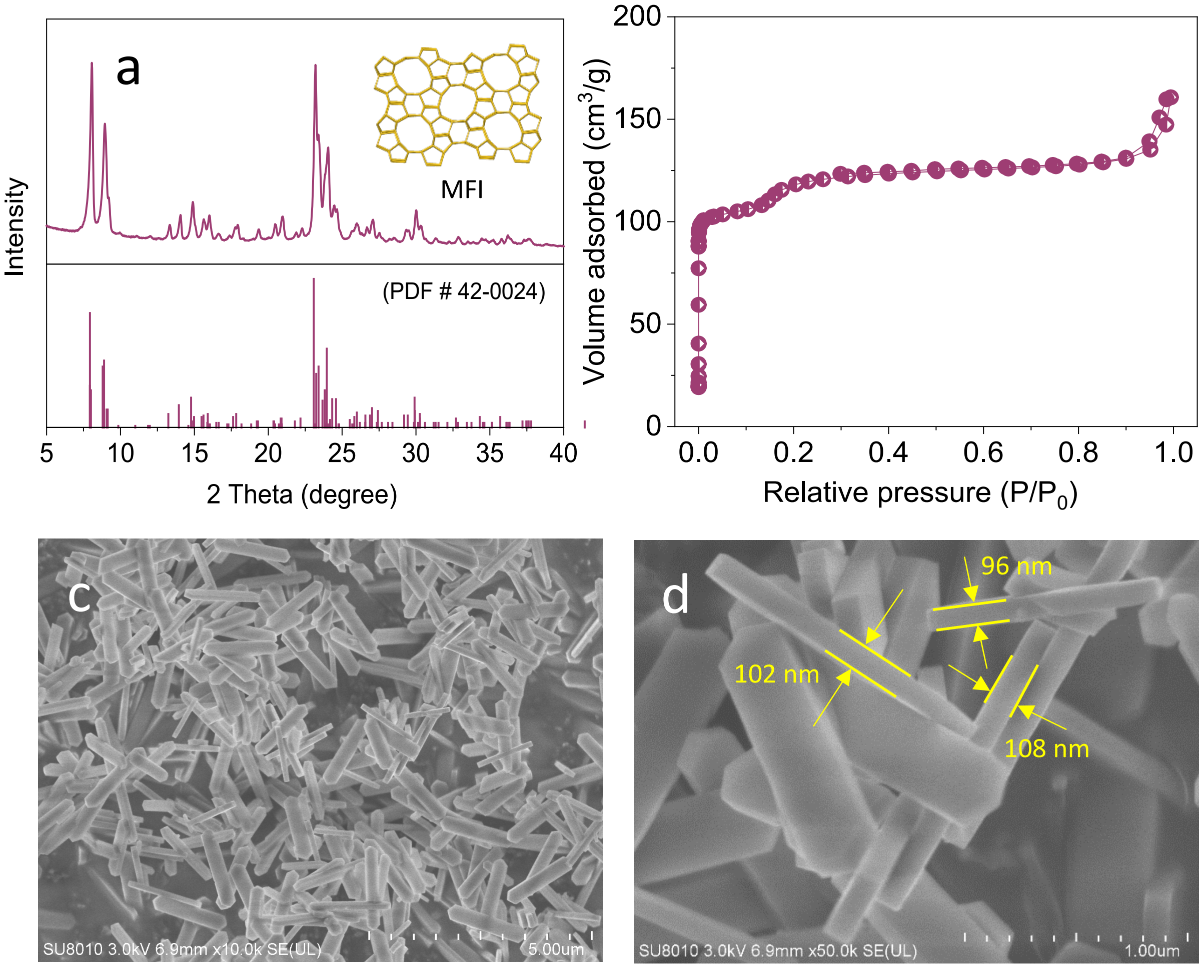
Supplementary Fig. 1. (a) XRD patterns of the as-synthesized Na-Fe_3O_4, Na-FeC_x, and used Na-FeC_x after LT-FTO reaction for 50 h, the standard XRD patterns of Fe_2O_3, Fe_3O_4, and Fe_5C_2 according to PDF # 04-0755, PDF # 03-0863, and PDF # 20-0509, respectively. The Na-FeC_x still has obvious signals of ferric oxides due to the incomplete carbonization. After reaction, the used Na-FeC_x shows sole signals of ferric carbide with almost undetectable oxide phase due to carbonization during the reaction. (b-d) TEM images of the Na-FeC_x.



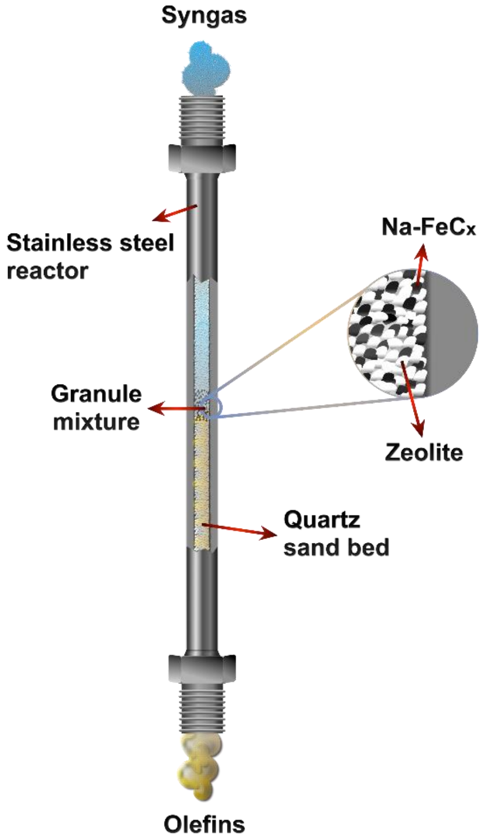
Supplementary Fig. 2. (a) HAADF-STEM image and EDS elemental maps of (b) Fe and (c) Na signals of the Na-FeC_x.



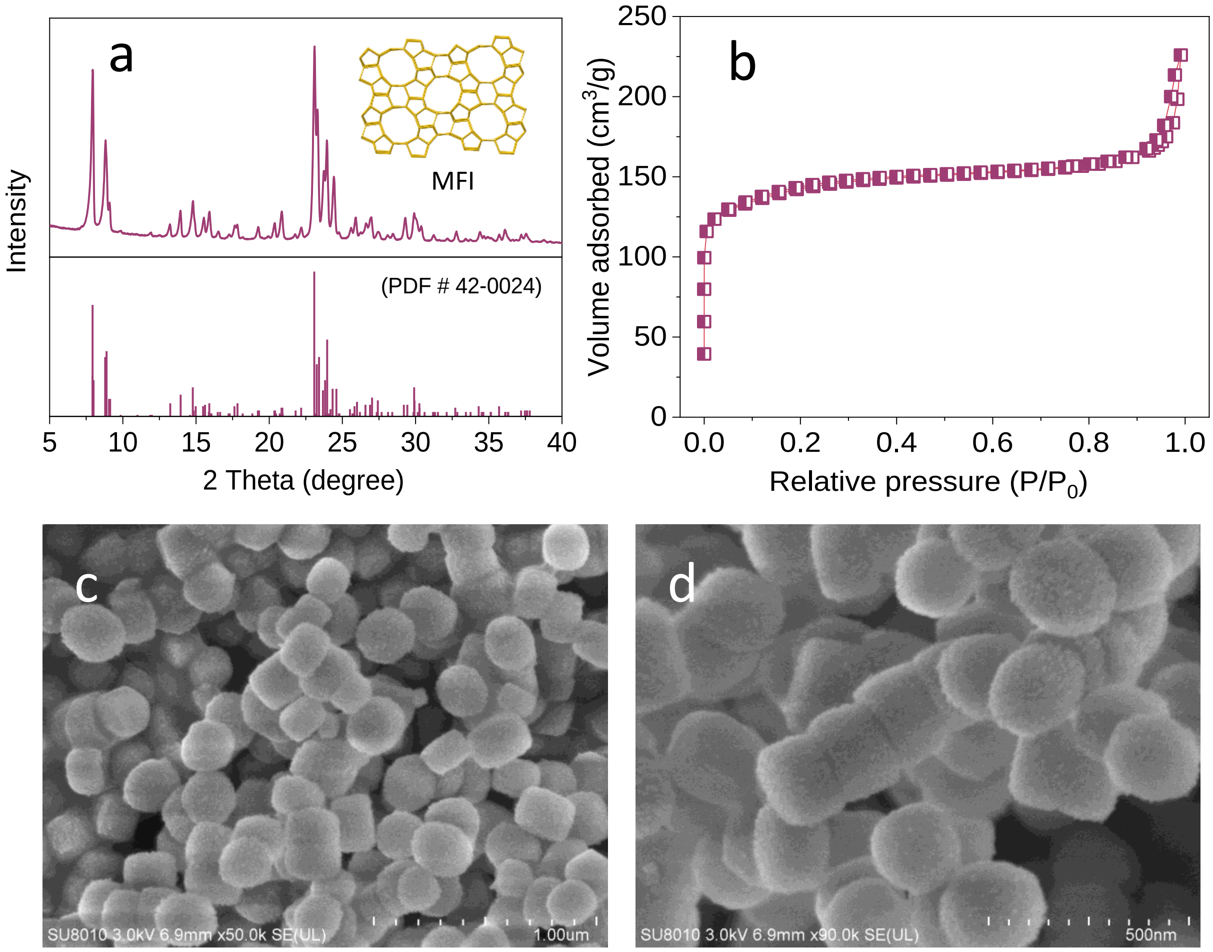
Supplementary Fig. 3. Catalytic data of various catalysts in the LT-FTO. Reaction conditions: weight ratio of zeolite to Na-FeC_x at 3, feed gas with ratio of CO/H_2/Ar at 45/45/10, 2400 mL·h^{-1}·g^{-1}, 2.0 MPa, 260 °C, and granule-mixing of the zeolite and Na-FeC_x individuals. C-oxy is the carbon-containing oxygenates. The aromatic products, if existed, were included in the C_2-C_{10}^0 and C_{10}^+ products.



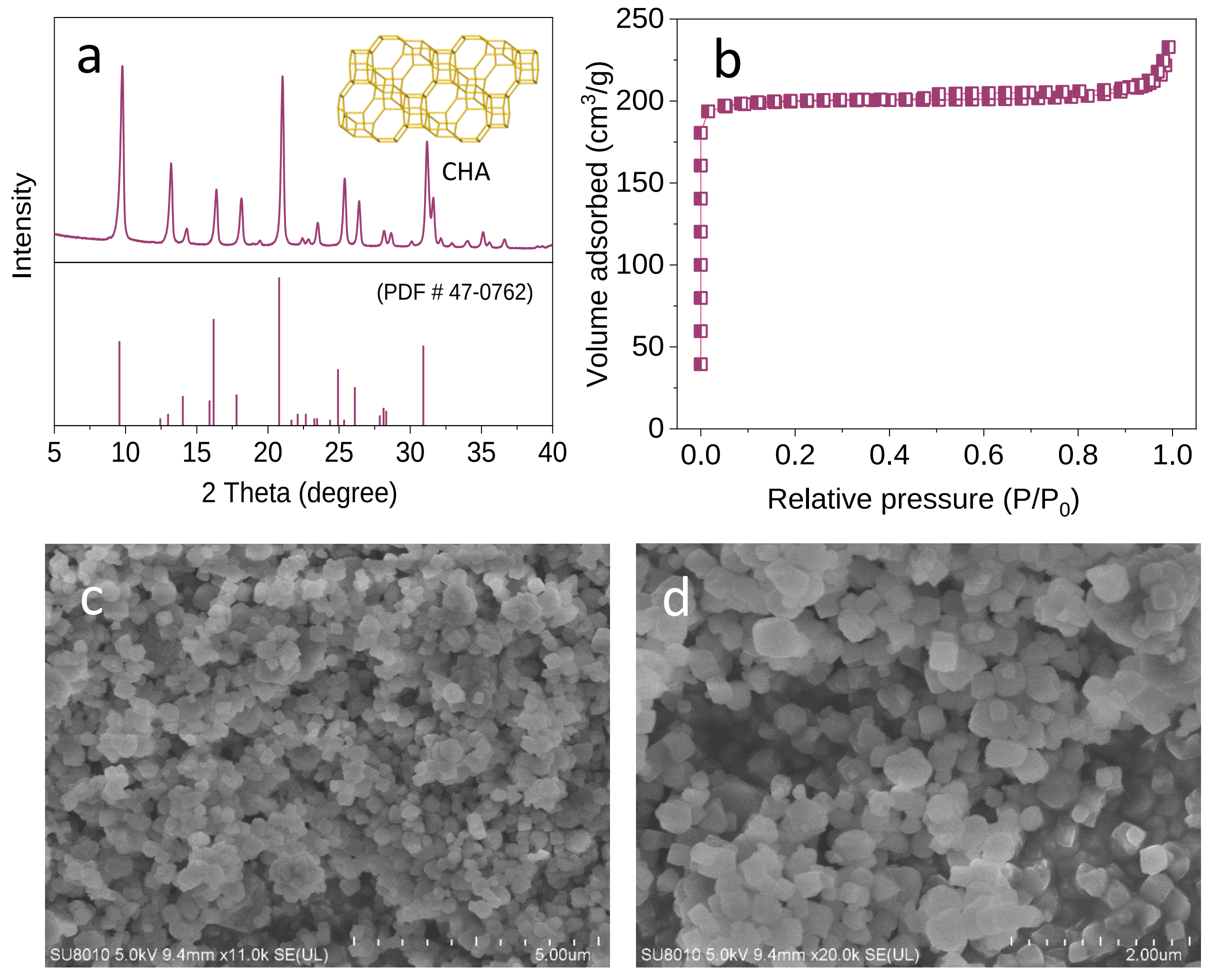
Supplementary Fig. 4. (a) XRD pattern, (b) N_2 sorption isotherms, and (c,d) SEM images of the *s*-ZSM-5 with Si/Al ratio of 26.2. Insets in (a) showed the framework and standard XRD pattern of MFI structure according to PDF # 42-0024.



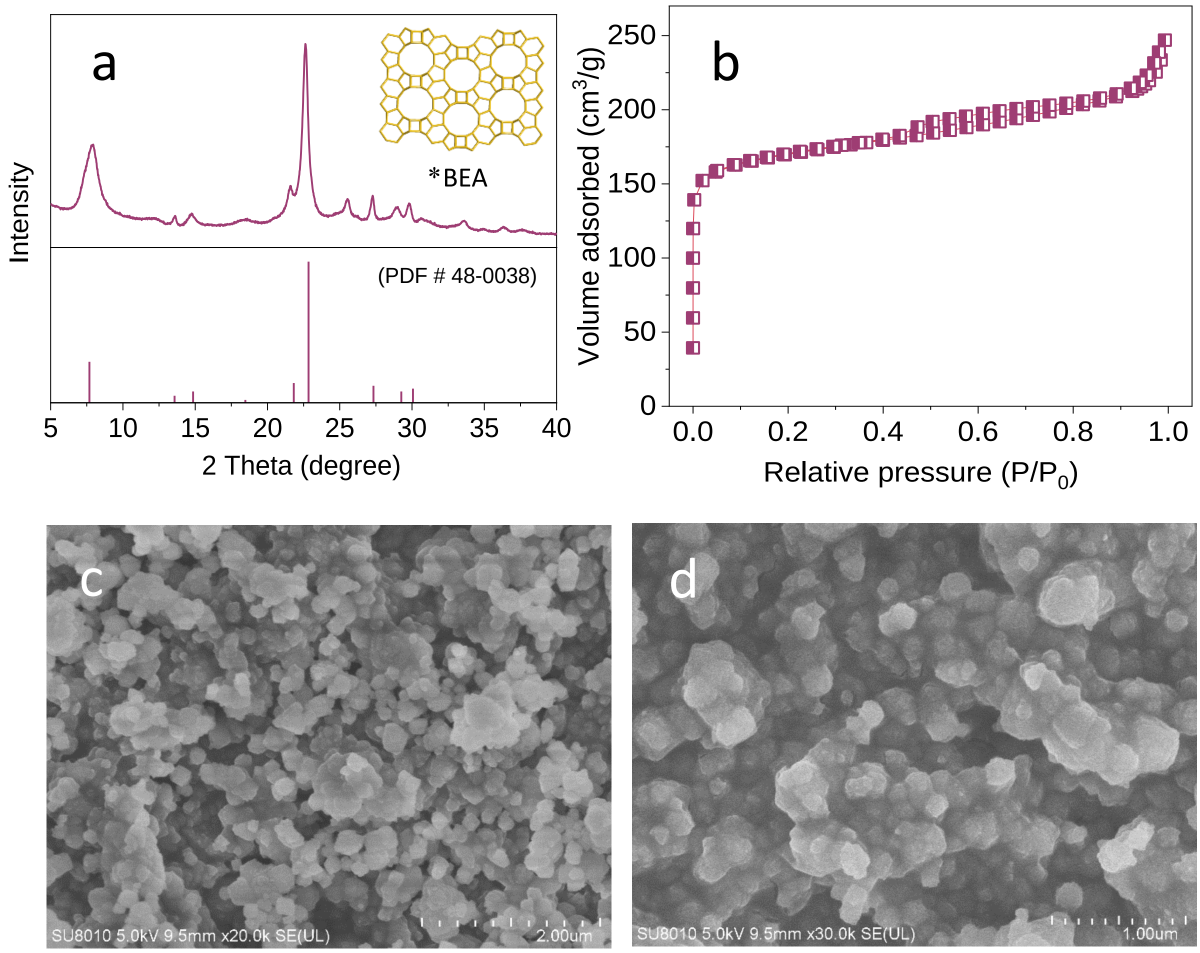
Supplementary Fig. 5. The Na-FeC_x and zeolite were made into granules with sizes at 20-40 mesh and mixed in the reactor (granule mixture).



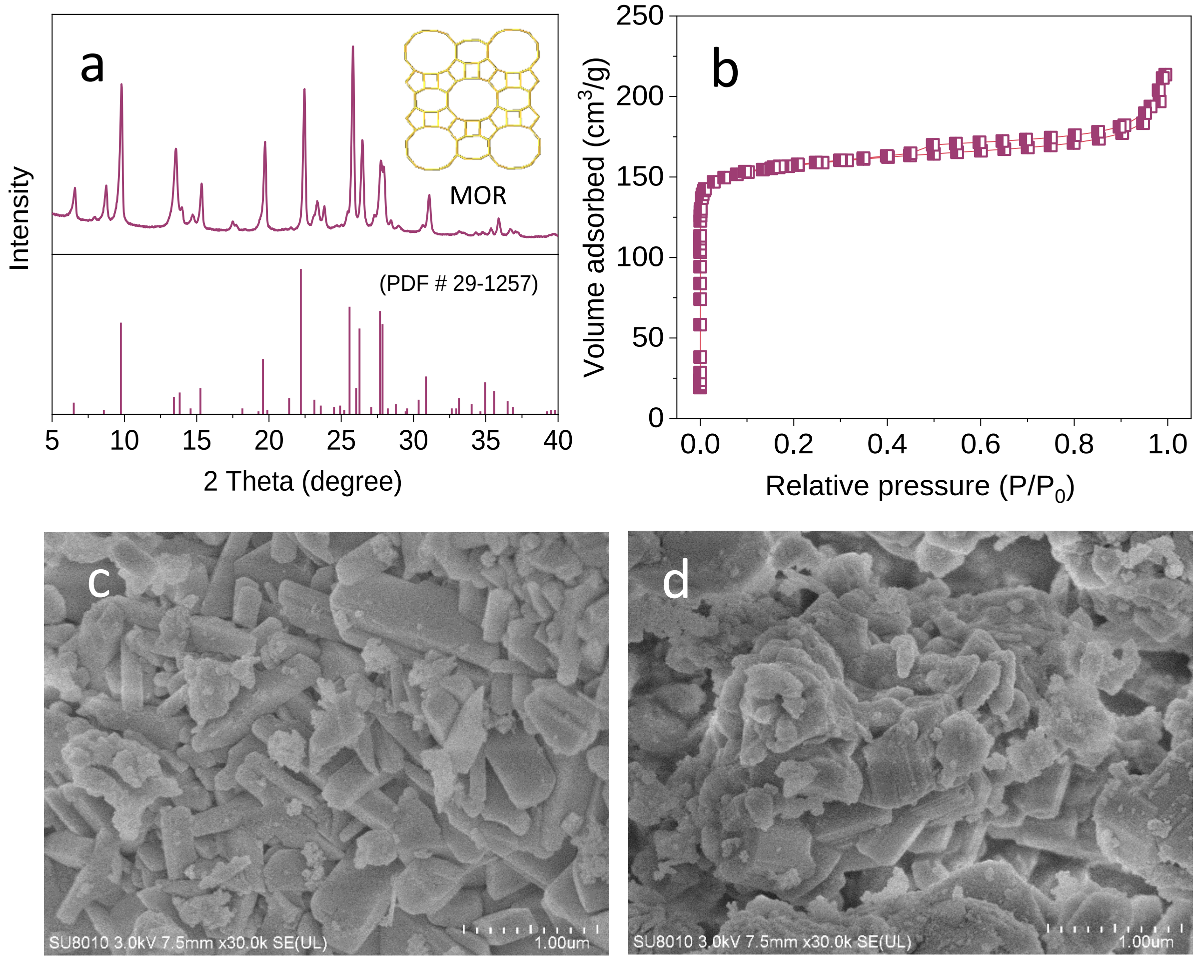
Supplementary Fig. 6. (a) XRD pattern, (b) N_2 sorption isotherms, and (c,d) SEM images of the *n*-ZSM-5 with Si/Al ratio of 30.5. Insets in (a) showed the framework and standard XRD pattern of MFI structure according to PDF # 42-0024.



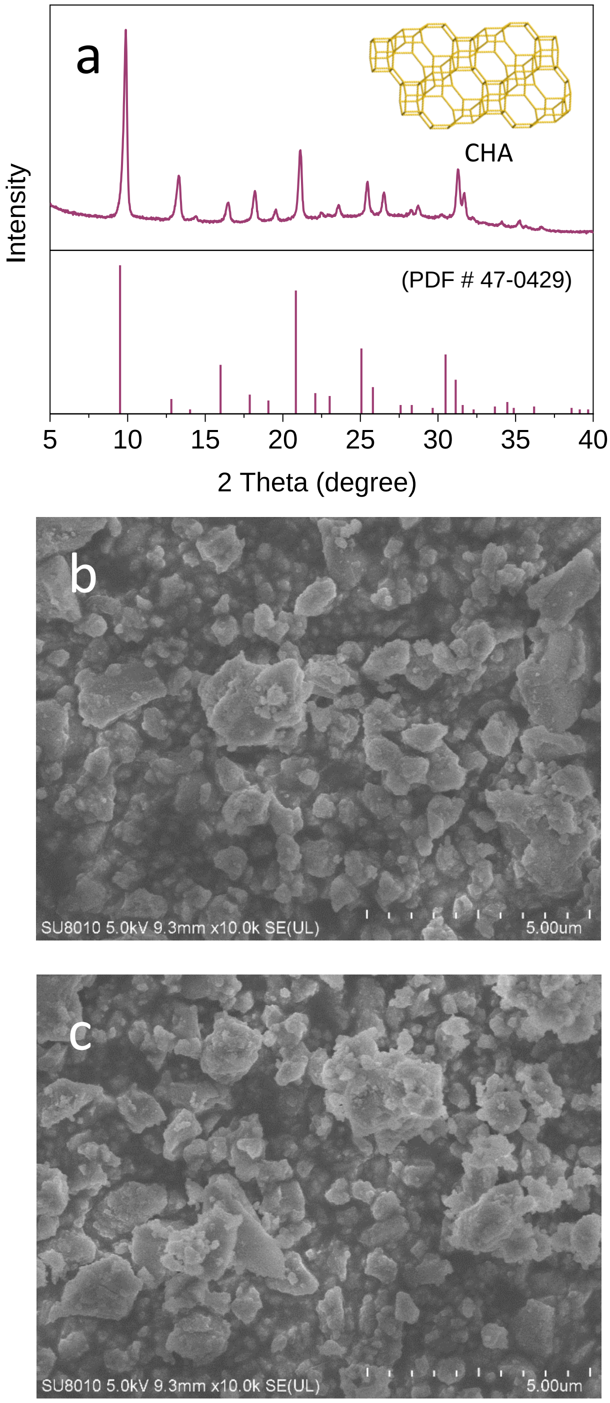
Supplementary Fig. 7. (a) XRD pattern, (b) N_2 sorption isotherms, and (c,d) SEM images of SSZ-13 with Si/Al ratio of 12.6. Insets in (a) showed the framework and standard XRD pattern of CHA structure according to PDF # 47-0762.



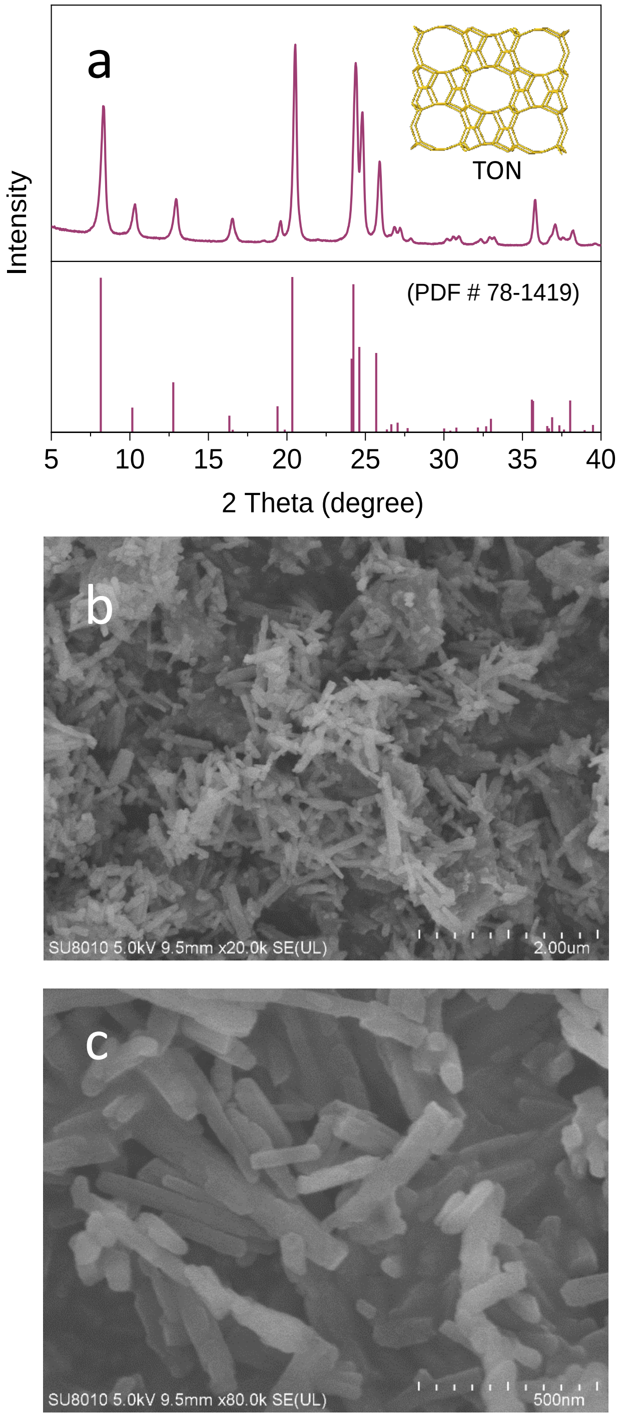
Supplementary Fig. 8. (a) XRD pattern, (b) N_2 sorption isotherms, and (c,d) SEM images of Beta zeolite with Si/Al ratio of 13.1. Insets in (a) showed the framework and standard XRD pattern of *BEA structure according to PDF # 48-0038.



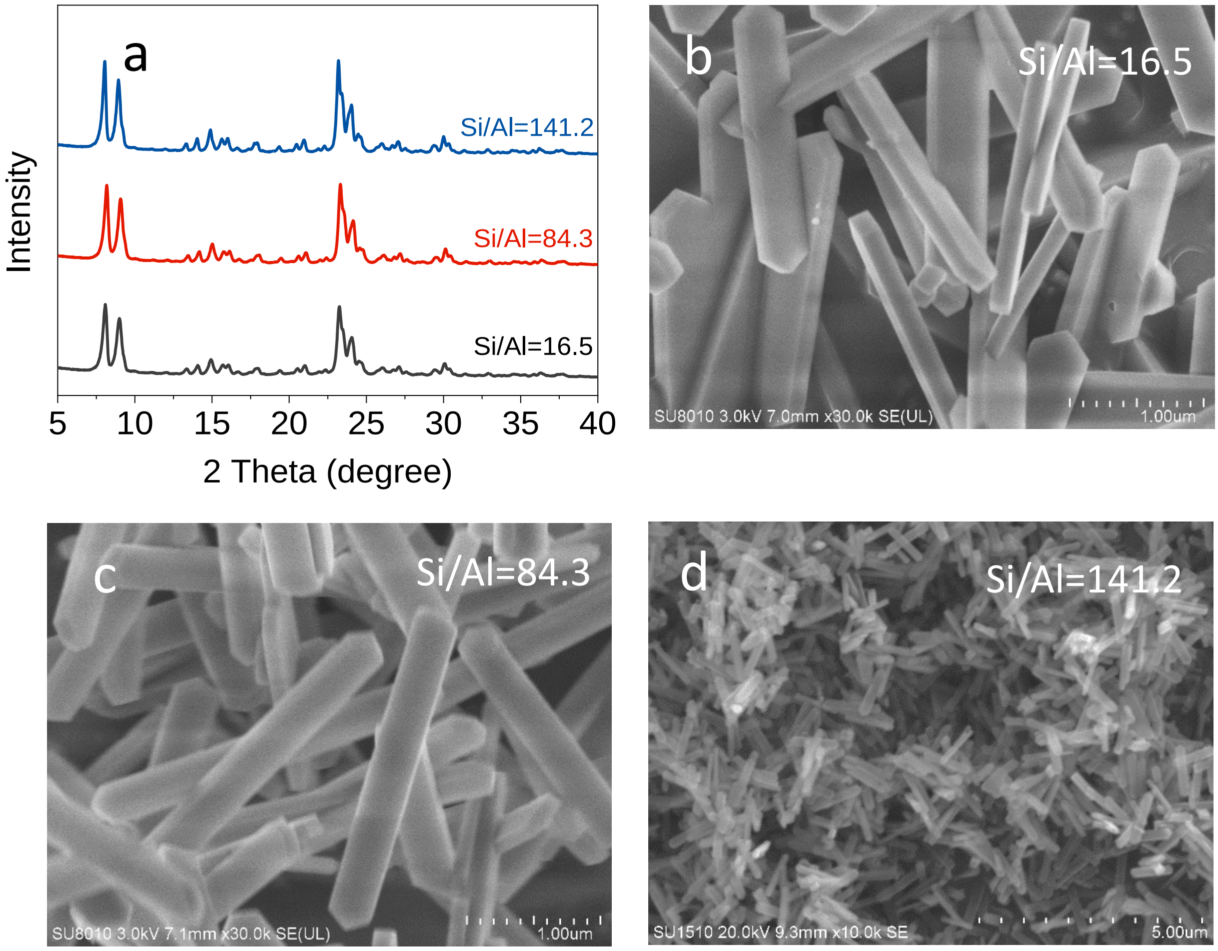
Supplementary Fig. 9. (a) XRD pattern, (b) N_2 sorption isotherms, and (c,d) SEM images of MOR zeolite with Si/Al ratio of 10.4. Insets in (a) showed the framework of MOR and standard XRD pattern of MOR structure according to PDF # 29-1257.



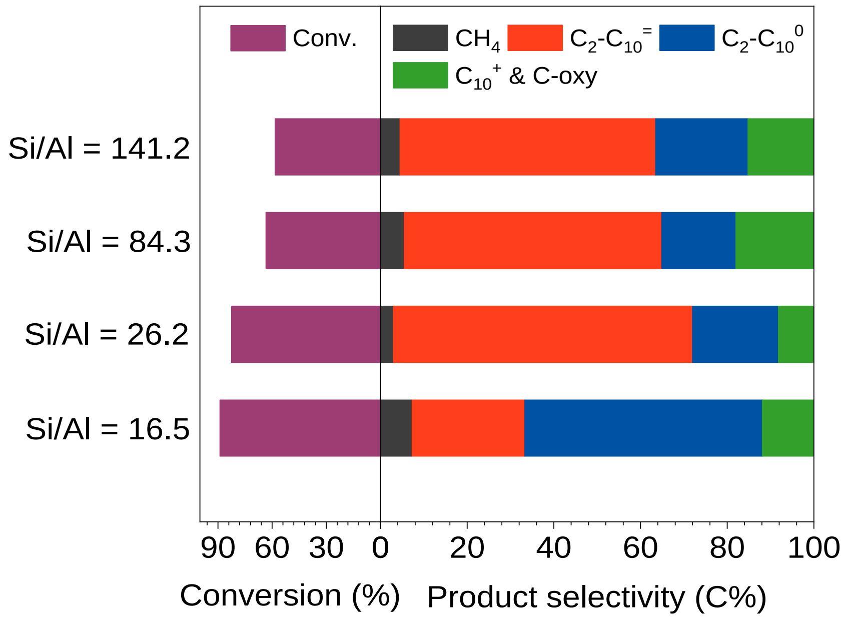
Supplementary Fig. 10. (a) XRD pattern and (b,c) SEM images of SAPO-34 with P/Al ratio of 0.76. Insets in (a) showed the framework of CHA and standard XRD pattern of CHA structure according to PDF # 47-0429.



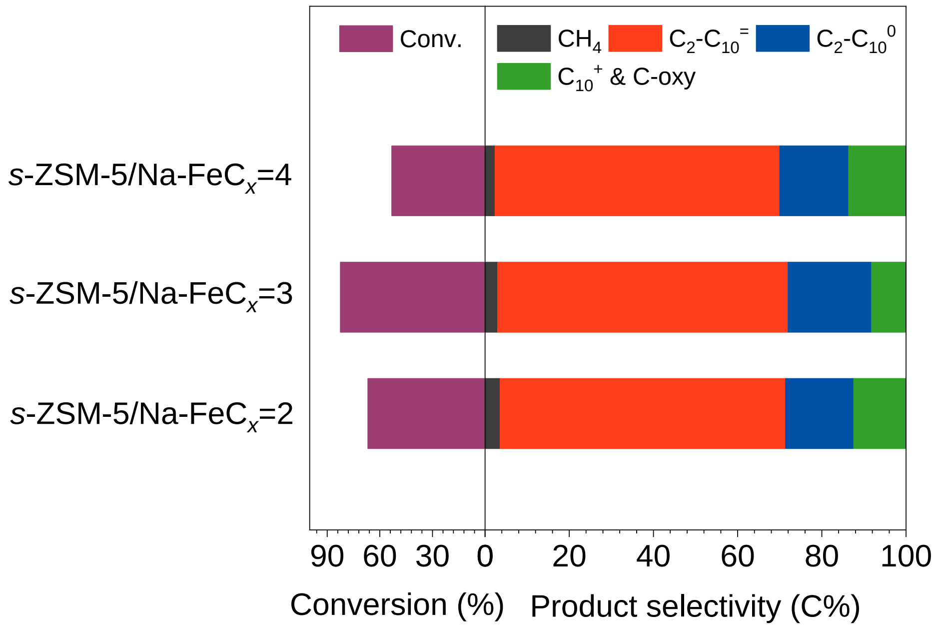
Supplementary Fig. 11. (a) XRD pattern and (b,c) SEM images of ZSM-22 zeolite with Si/Al ratio of 38.4. Insets in (a) showed the framework of TON and standard XRD pattern of TON structure according to PDF # 78-1419.



Supplementary Fig. 12. (a) XRD patterns and (b-d) SEM images of the *s*-ZSM-5 samples with different Si/Al ratios in the LT-FTO.

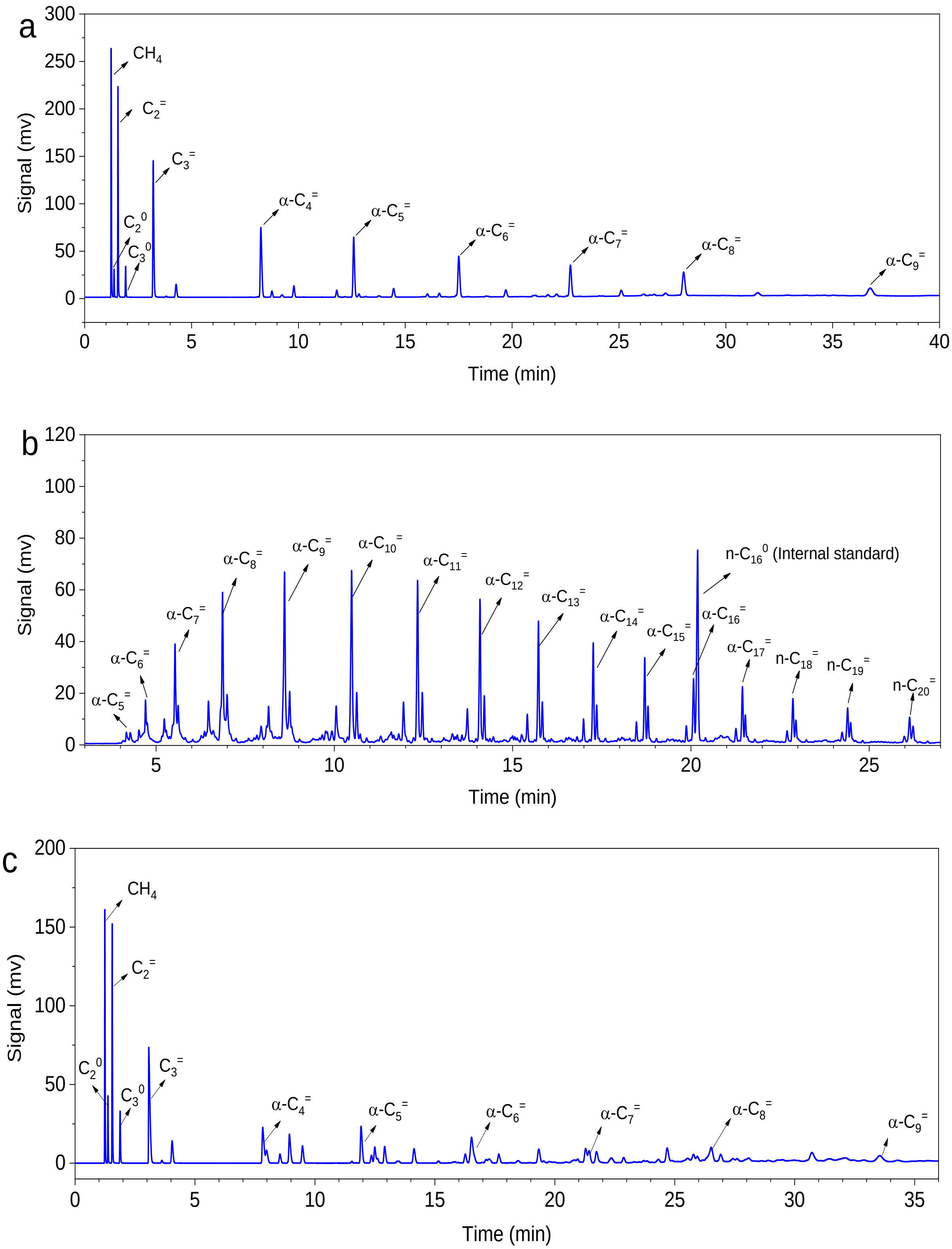


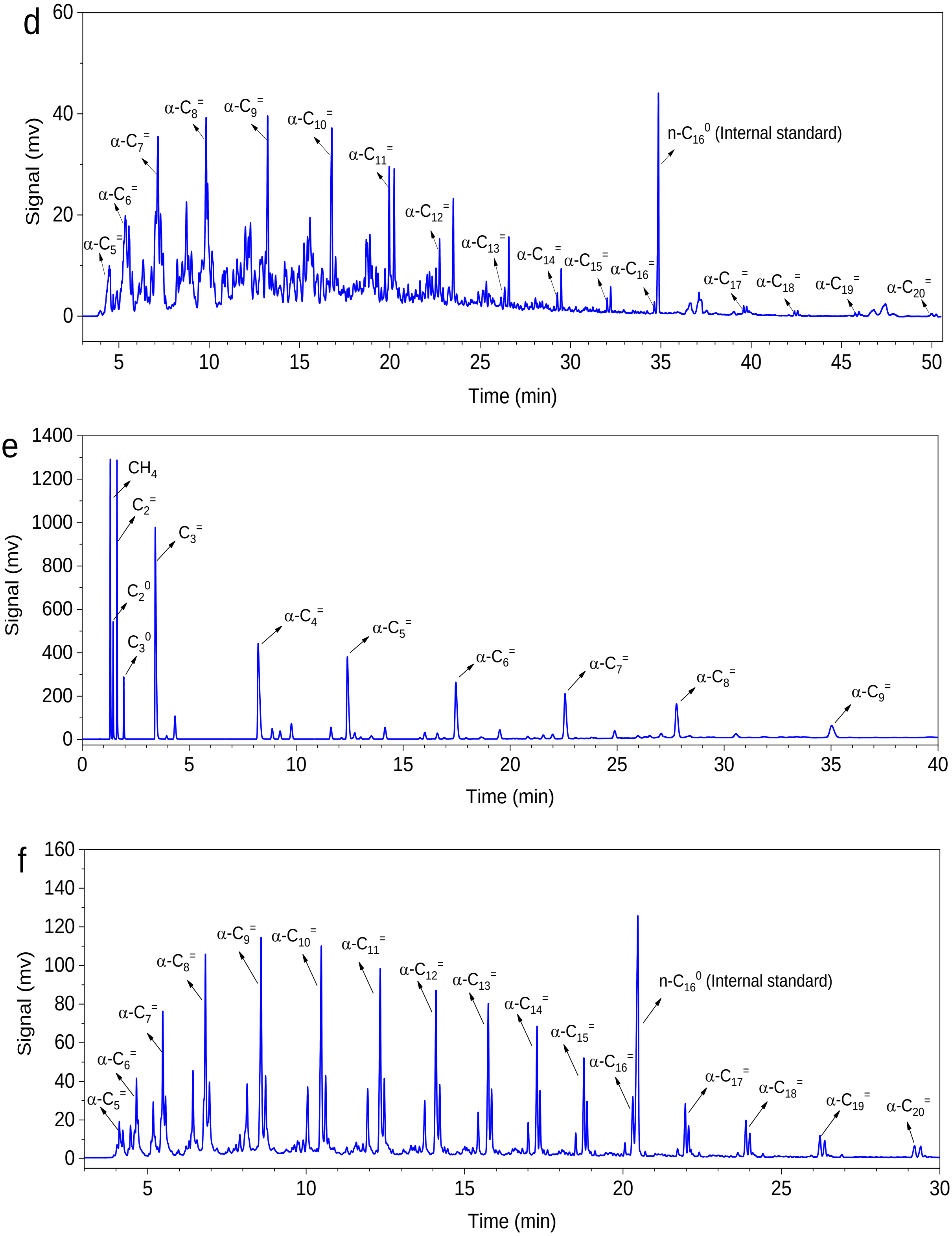
Supplementary Fig. 13. Catalytic performances of the Na-FeC$_x$/*s*-ZSM-5 catalyst with different Si/Al ratios in the LT-FTO.



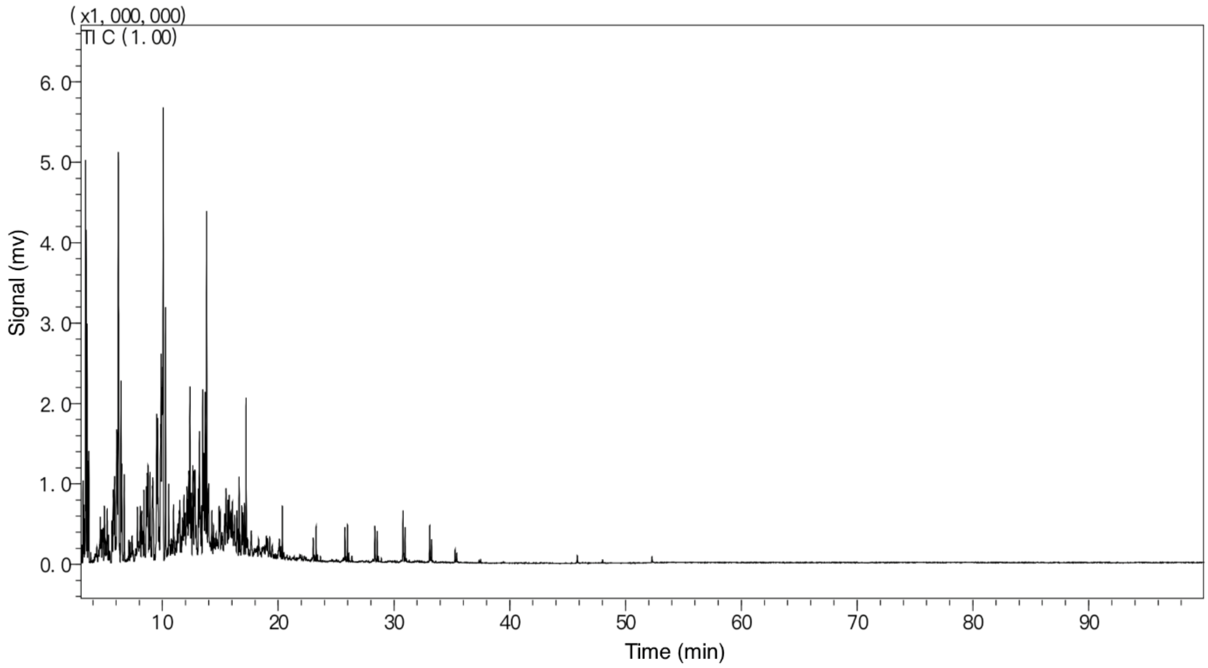
Supplementary Fig. 14. Catalytic data of the Na-FeC$_x$/*s*-ZSM-5 catalyst with zeolite/Na-FeC$_x$ mass ratios at 2, 3, and 4. Reaction conditions: CO/H_2/Ar at 45/45/10, 2400 mL·h^{-1}·g^{-1}, 2.0 MPa, 260 °C, and granule-mixing of the *s*-ZSM-5 and Na-FeC$_x$ individuals.

Note: We studied the influence of zeolite amount to the LT-FTO (Supplementary Fig.14, Supplementary Table 3). When a small amount of zeolite was added in the reaction system such as Na-FeC$_x$/*s*-ZSM-5 with mass ratio of zeolite/Na-FeC$_x$ at 0.5, the CO conversion at 27.3% was obtained. More zeolite increased the CO conversion and optimized the product selectivity. For example, Na-FeC$_x$/*s*-ZSM-5 with mass ratio of zeolite/Na-FeC$_x$ at 2 showed higher CO conversion of 66.8% and lower C_{10+} selectivity at 12.4%, where the C_2-C_{10} molecules with o/a at 4.2 were the dominant products. The best performance was obtained on the catalyst with mass ratio of zeolite/Na-FeC$_x$ at 3, where the C_{10+} product selectivity was as low as 8.2%. A further increase of zeolites led to a reduction of the CO conversion, giving 53.2% on the Na-FeC$_x$/*s*-ZSM-5 with *o/a* ratio at 4.1.


a
Signal (mv)
Time (min)
CH_4
$C_2^=$
$C_3^=$
C_2^0
C_3^0
α-$C_4^=$
α-$C_5^=$
α-$C_6^=$
α-$C_7^=$
α-$C_8^=$
α-$C_9^=$
b
Signal (mv)
Time (min)
α-$C_5^=$
α-$C_6^=$
α-$C_7^=$
α-$C_8^=$
α-$C_9^=$
α-$C_{10}^=$
α-$C_{11}^=$
α-$C_{12}^=$
α-$C_{13}^=$
α-$C_{14}^=$
α-$C_{15}^=$
α-$C_{16}^=$
α-$C_{17}^=$
n-C_{16}^0 (Internal standard)
n-$C_{18}^=$
n-$C_{19}^=$
n-$C_{20}^=$
c
Signal (mv)
Time (min)
CH_4
$C_2^=$
C_2^0
C_3^0
$C_3^=$
α-$C_4^=$
α-$C_5^=$
α-$C_6^=$
α-$C_7^=$
α-$C_8^=$
α-$C_9^=$

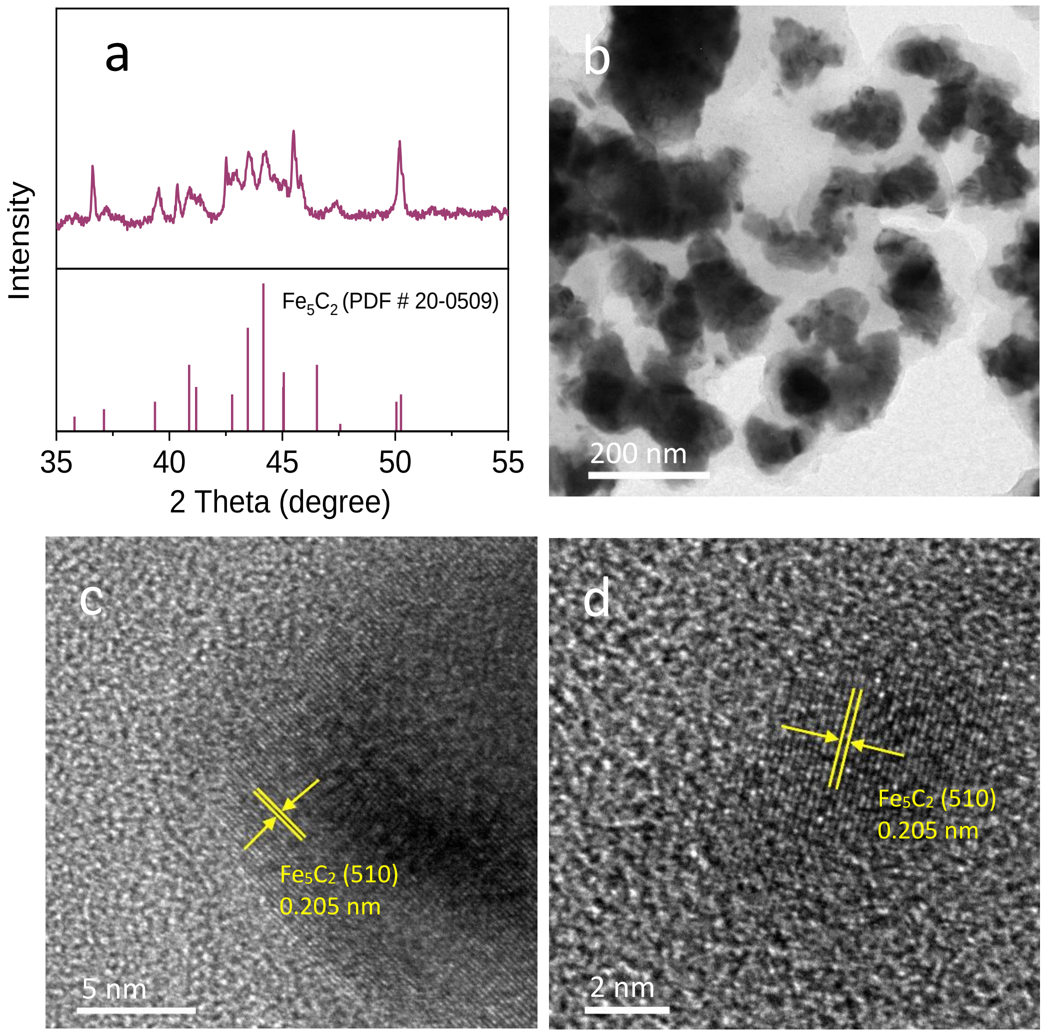


Supplementary Fig. 15. GC curves analyzing the gaseous (online analysis) and liquid products (offline analysis) of (a,b) Na-FeC$_x$, (c,d) Na-FeC$_x$/*s*-ZSM-5, and (e,f) Na-FeC$_x$/*s*-S1-OH. The reaction conditions were the same as those in Fig. 2a.

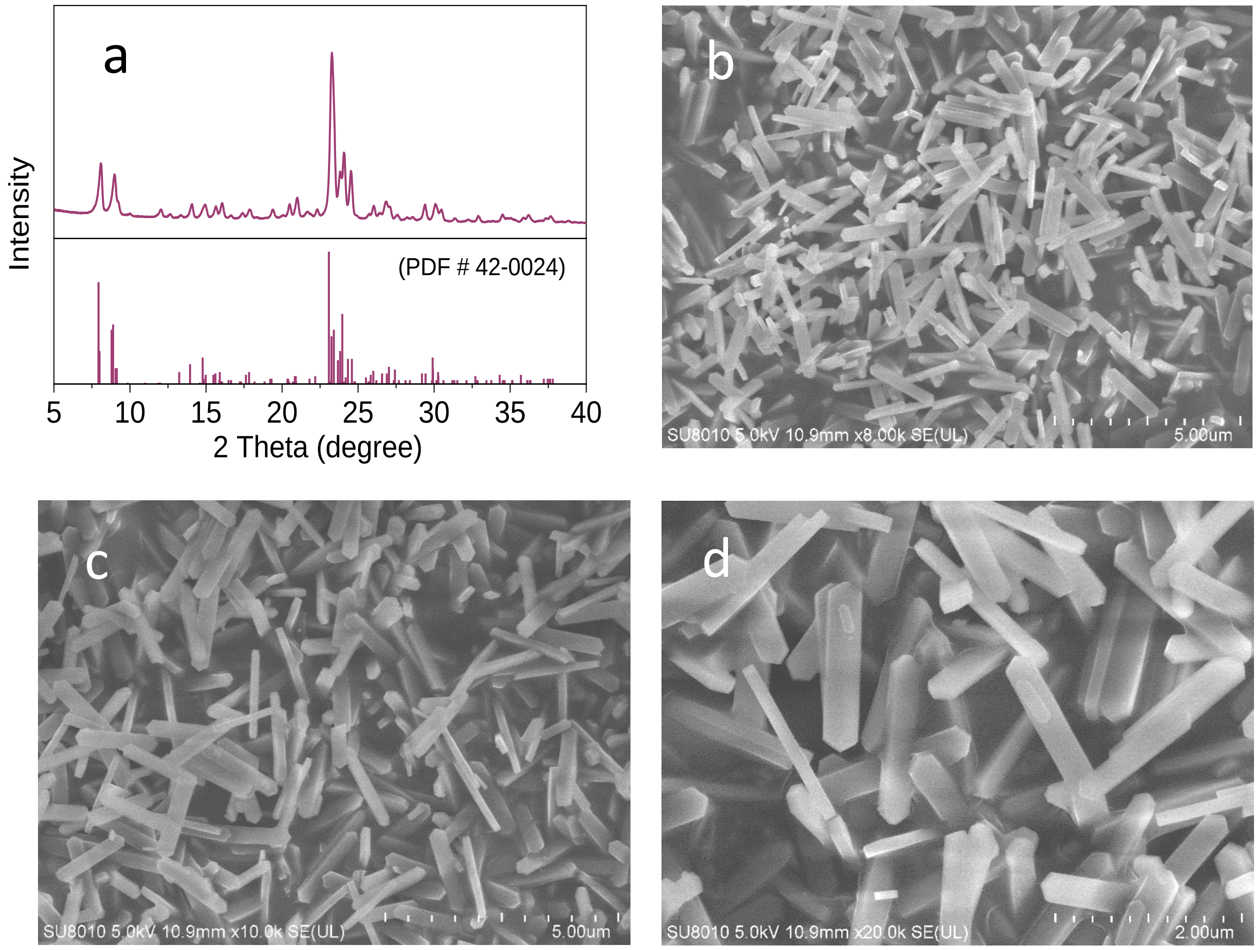


Supplementary Fig. 16. GC-MS analyzing the liquid products of the LT-FTO reaction over the Na-FeC$_x$/*s*-ZSM-5 catalyst. The reaction conditions were the same as those in Fig. 2a in the main text.

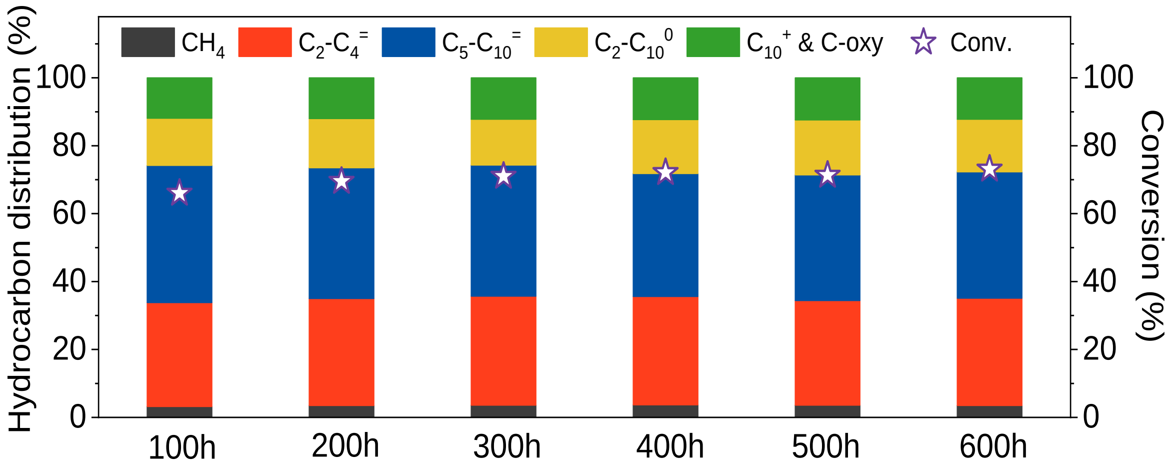
Note: The liquid products of the Na-FeC$_x$/*s*-ZSM-5 was analyzed by GC-MS. The aromatic compounds were almost undetectable with selectivity less than 0.5%.



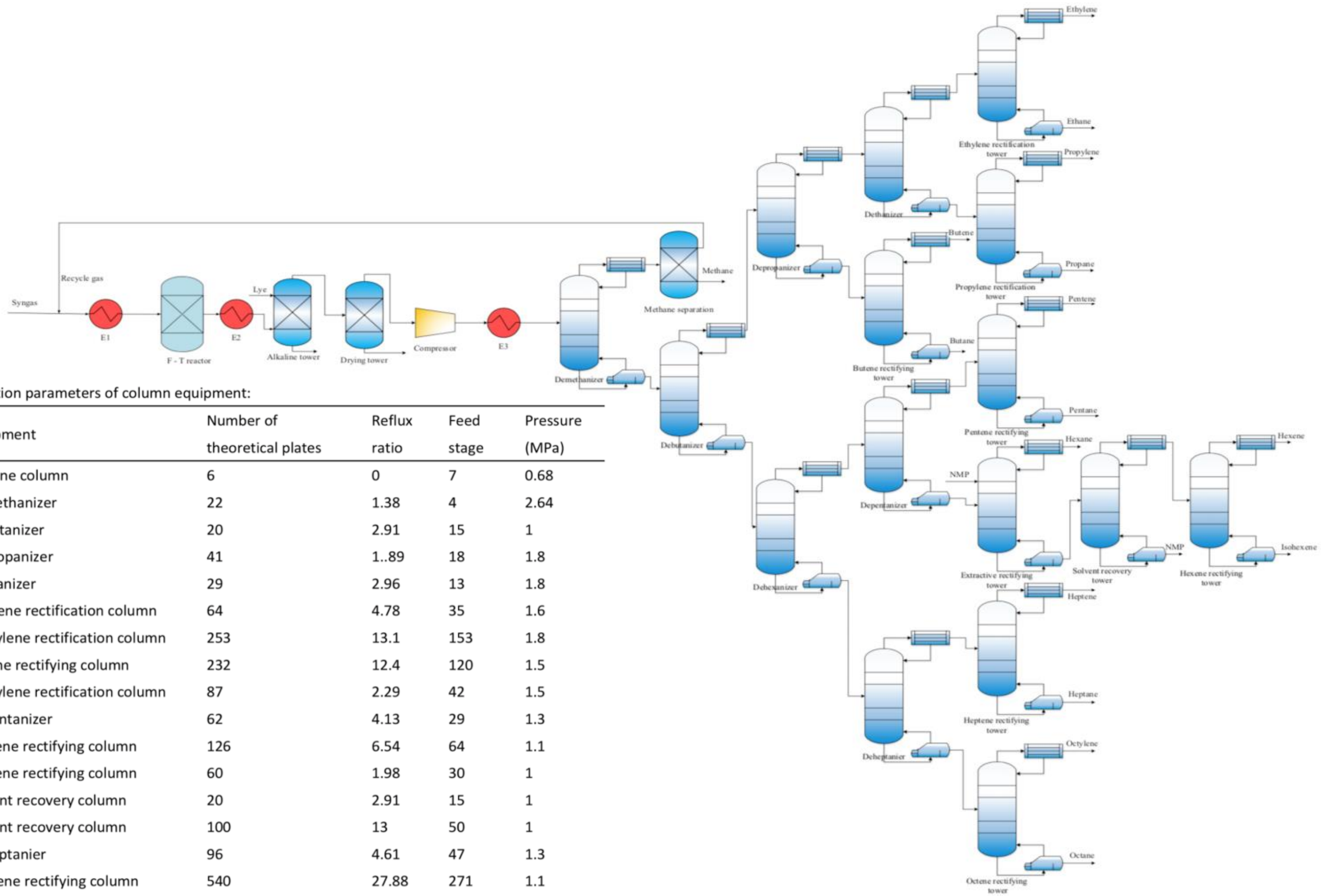
Supplementary Fig. 17. (a) XRD pattern and (b-d) TEM images of used Na-FeC$_x$ after the LT-FTO reaction for 600 h. The reaction conditions were the same to those in Fig. 2c. Inset in (a) showed standard XRD pattern of Fe_5C_2 structure according to PDF # 20-0509. After the reaction, the Na-FeC$_x$ and zeolite individuals were separated for further characterization.



Supplementary Fig. 18. (a) XRD pattern and (b-d) SEM image of used *s*-ZSM-5 after the LT-FTO reaction for 600 h. The reaction conditions were the same to those in Fig. 2c. Inset in (a) showed the standard XRD pattern of MFI structure according to PDF # 42-0024.



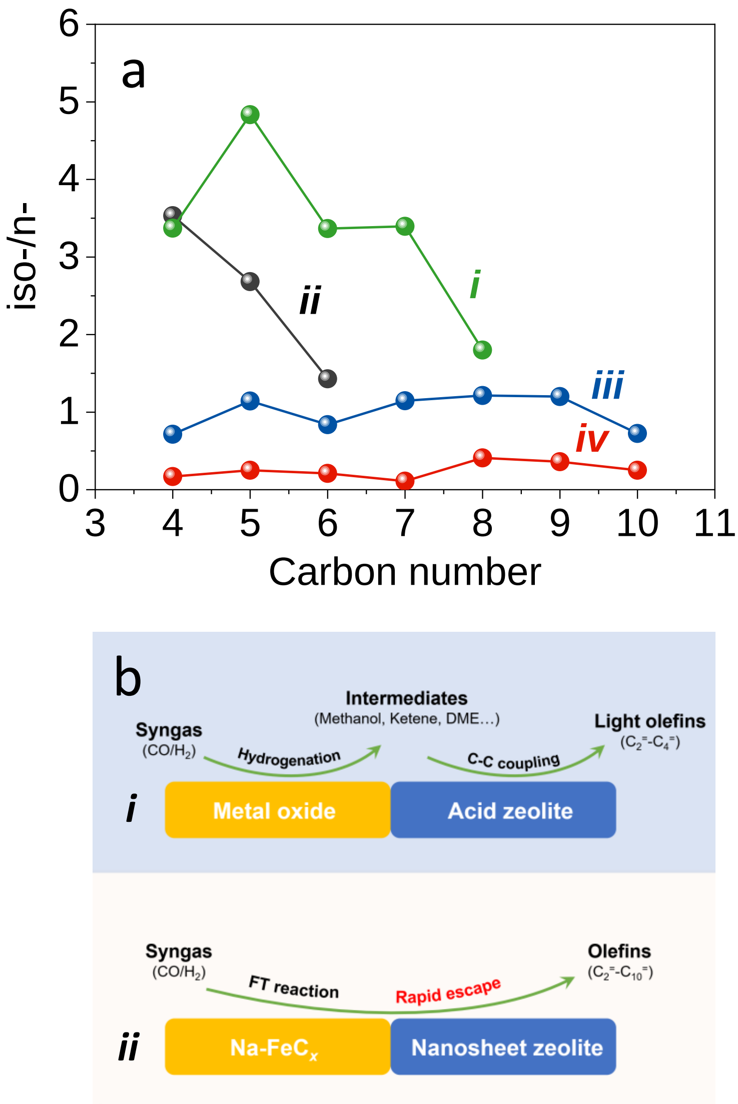
Supplementary Fig. 19. Catalytic data at different periods in the LT-FTO characterizing the durability of the Na-FeC$_x$/*s*-ZSM-5 catalyst. The reaction conditions were the same to those in Fig. 2c.



Operation parameters of column equipment:

Equipment	Number of theoretical plates	Reflux ratio	Feed stage	Pressure (MPa)
Alkaline column	6	0	7	0.68
Demethanizer	22	1.38	4	2.64
Debutanizer	20	2.91	15	1
Depropanizer	41	1..89	18	1.8
Dethanizer	29	2.96	13	1.8
Ethylene rectification column	64	4.78	35	1.6
Propylene rectification column	253	13.1	153	1.8
Butene rectifying column	232	12.4	120	1.5
Propylene rectification column	87	2.29	42	1.5
Depentanizer	62	4.13	29	1.3
Pentene rectifying column	126	6.54	64	1.1
Pentene rectifying column	60	1.98	30	1
Solvent recovery column	20	2.91	15	1
Solvent recovery column	100	13	50	1
Deheptanier	96	4.61	47	1.3
Heptene rectifying column	540	27.88	271	1.1
Heptene rectifying column	279	46.7	140	1.1

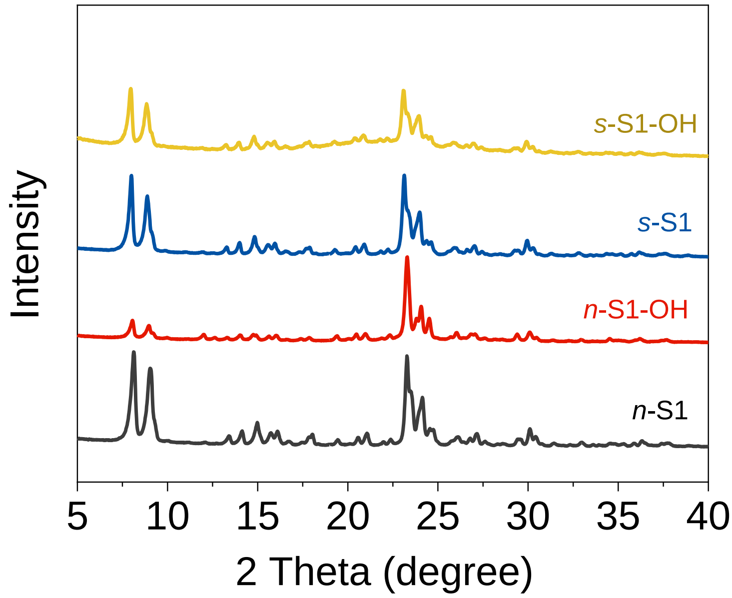
Supplementary Fig. 20. Process involving the LT-FTO reaction and product separation. Inset: Table showing the operation parameters of column equipment.



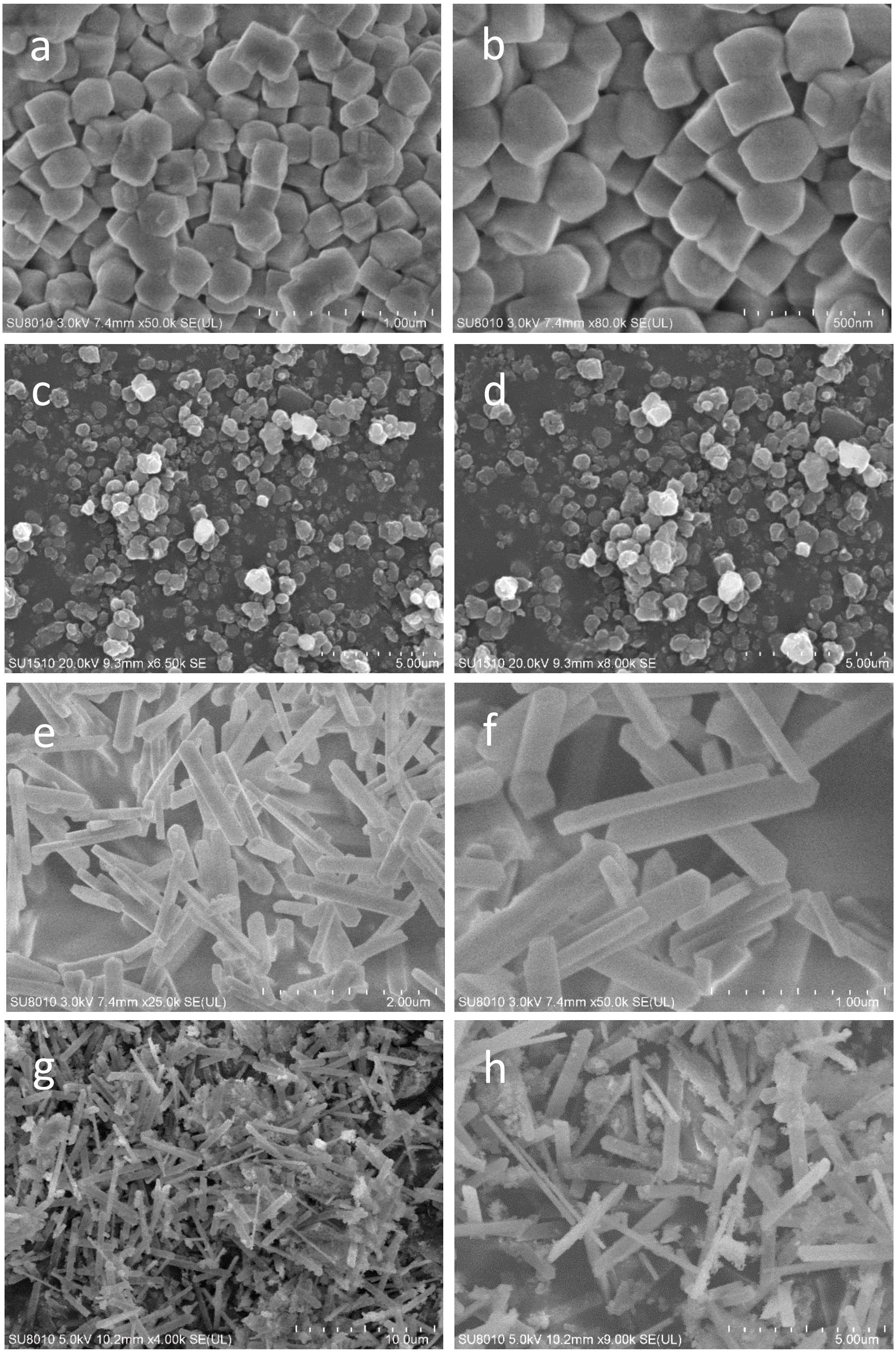
Supplementary Fig. 21. (a) The ratios of isomer to normal hydrocarbons in the products from different reactions. (i) 1-octene isomerization/cracking over *s*-ZSM-5, (ii) 1-hexene isomerization/cracking over *s*-ZSM-5, (iii) LT-FTO over Na-FeC$_x$/*s*-ZSM-5, and (iv) LT-FTO over Na-FeC$_x$/*s*-S1-OH. Reaction condition for 1-octene cracking: 0.2 g of catalyst, 260 °C, 0.1 MPa, 1-octene bubbling with a pure N_2 flow at 30 mL/min at 120 °C. Reaction condition for 1-hexene cracking: 0.2 g of catalyst, 260 °C, 0.1 MPa, and 1-hexene bubbling with a pure N_2 flow at 10 mL/min at 30 °C. (b) The strategies on (i) cascade reactions (OX-ZEO catalysts) and (ii) this work (Na-FeC$_x$/nanosheet zeolite) in FTO reaction.

Note: The acidic zeolites could catalyze the side reactions of isomerization and cracking of olefin products. We fed 1-butene, 1-hexene, and 1-octene into the *s*-ZSM-5 zeolite at 260 °C (temperature for the LT-FTO reaction, Supplementary Tables 4-6), where the isomerization occurred as a dominant reaction with extremely low selectivity to the cracked products. For example, the reaction using 1-butene feed showed 99.1% selectivity to the isomerized butene products with almost undetectable C_1-C_3 products (selectivity < 0.5%). Similar phenomena were also observed on the reaction using 1-hexene and 1-octene feeds, where the 91.5% and 93.4% selectivity to isomerized C_6 and C_8 products were obtained. This feature cannot meet the high α-olefin selectivity in the Na-FeC_x/*s*-ZSM-5 catalyzed LT-FTO (Supplementary Fig. 21a, Supplementary Tables 4-6). In addition, the cracking only occurred slightly because of the low reaction temperature, which would not obviously change the carbon number distribution of the products in LT-FTO.

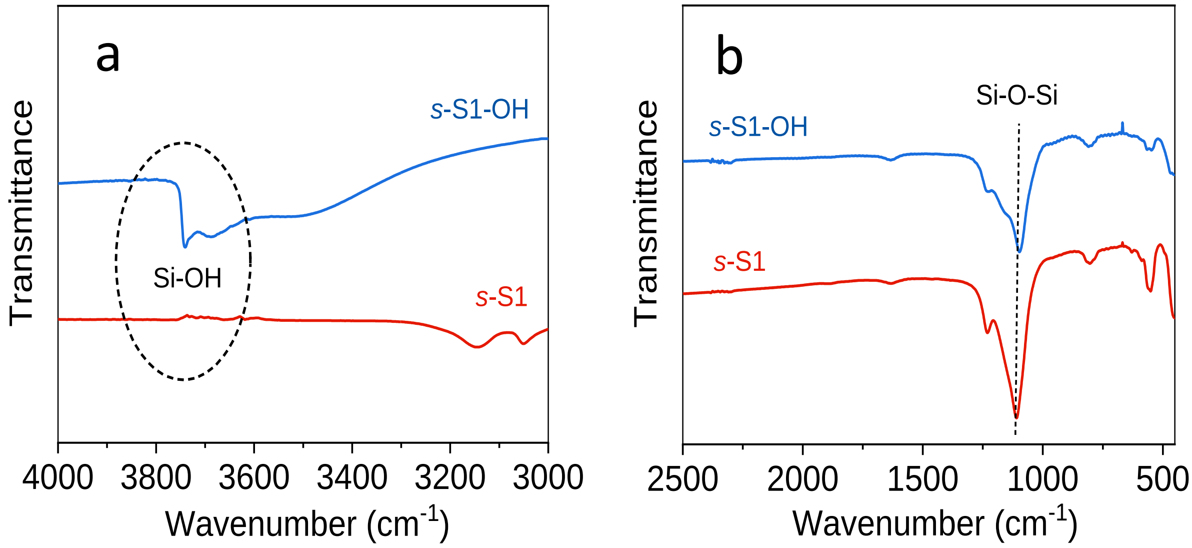
In order to avoid the isomerization/cracking side reaction and improve the selectivity to desired alfa-olefins, we reasonably developed a siliceous *s*-S1-OH zeolite without acidic sites for the isomerization and cracking (e.g. 1-hexene conversion <1% when fed to *s*-S1-OH). The FeC_x/*s*-S1-OH still exhibited significantly enhanced CO conversions, similar to that using *s*-ZSM-5 catalyst, but giving molar ratio of iso-/normal-olefins (ethylene and propylene were not included in the calculation) at ~0.2 because of the reduced side reactions of isomerization (Supplementary Fig. 21).



Supplementary Fig. 22. XRD patterns of the *n*-S1, *n*-S1-OH, *s*-S1, and *s*-S1-OH.

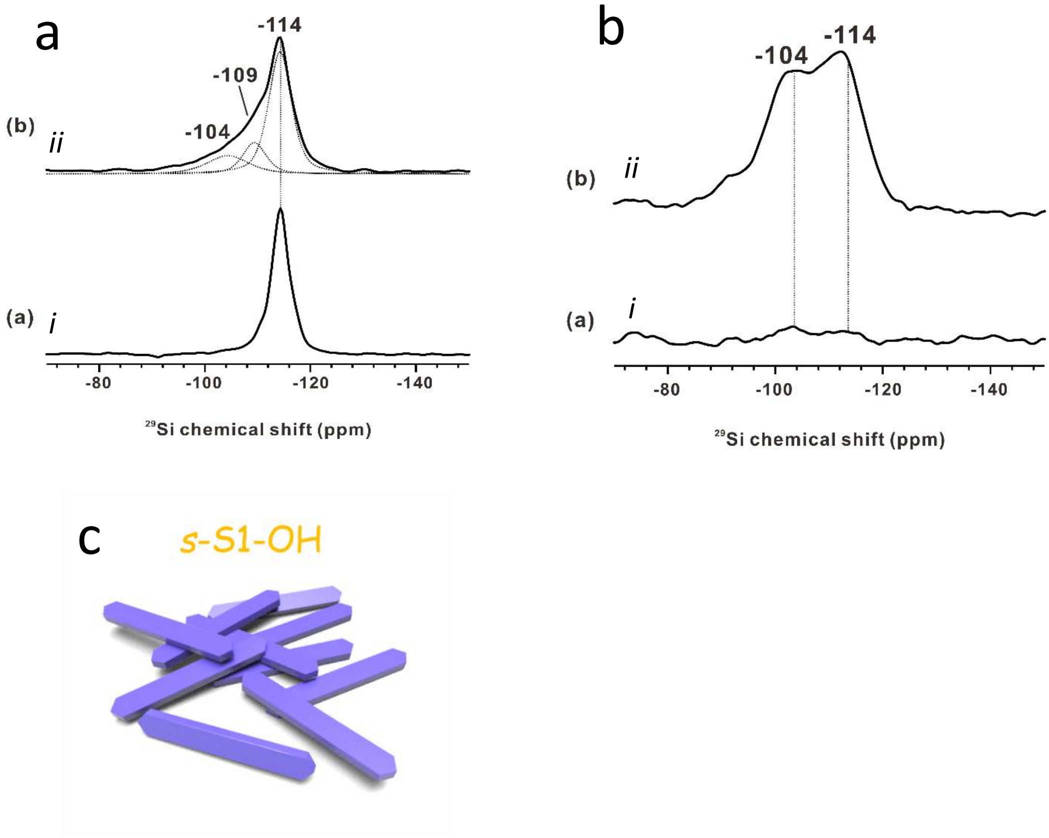


Supplementary Fig. 23. SEM images of the (a,b) *n*-S1, (c,d) *n*-S1-OH, (e,f) *s*-S1, and (g,h) *s*-S1-OH.



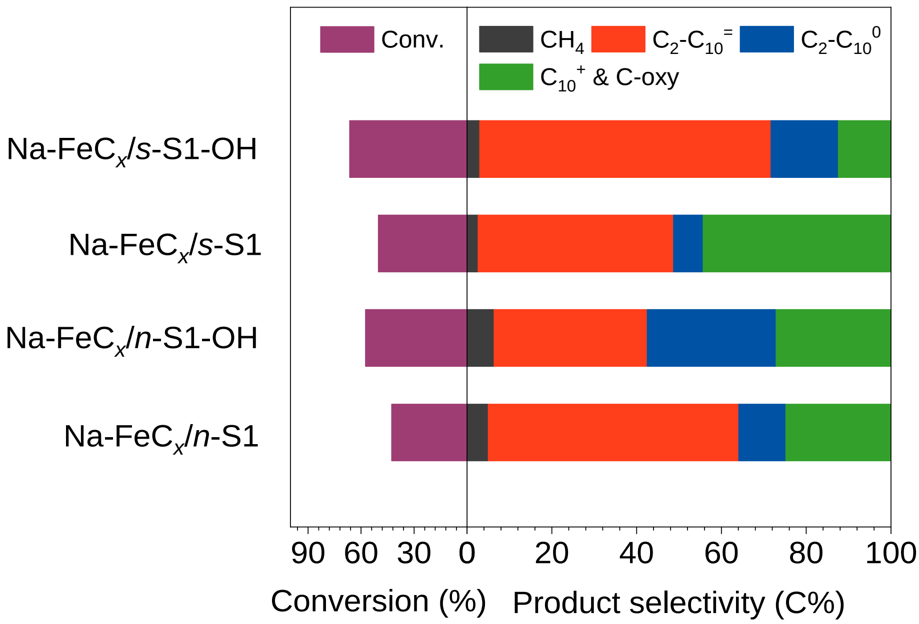
Supplementary Fig. 24. IR spectra of the *s*-S1 and *s*-S1-OH samples in the regions at (a) 3000-4000 cm^{-1} and (b) 450-2500 cm^{-1}.

Note: The *s*-S1 and *s*-S1-OH samples gave a peak at ~1100 cm^{-1}, which is assigned to the Si-O-Si band of zeolite framework. Compared with *s*-S1, the *s*-S1-OH gave much stronger peaks at 3680 and 3730 cm^{-1} in Supplementary Fig. 24a, confirming the abundant silanol species[16,17].



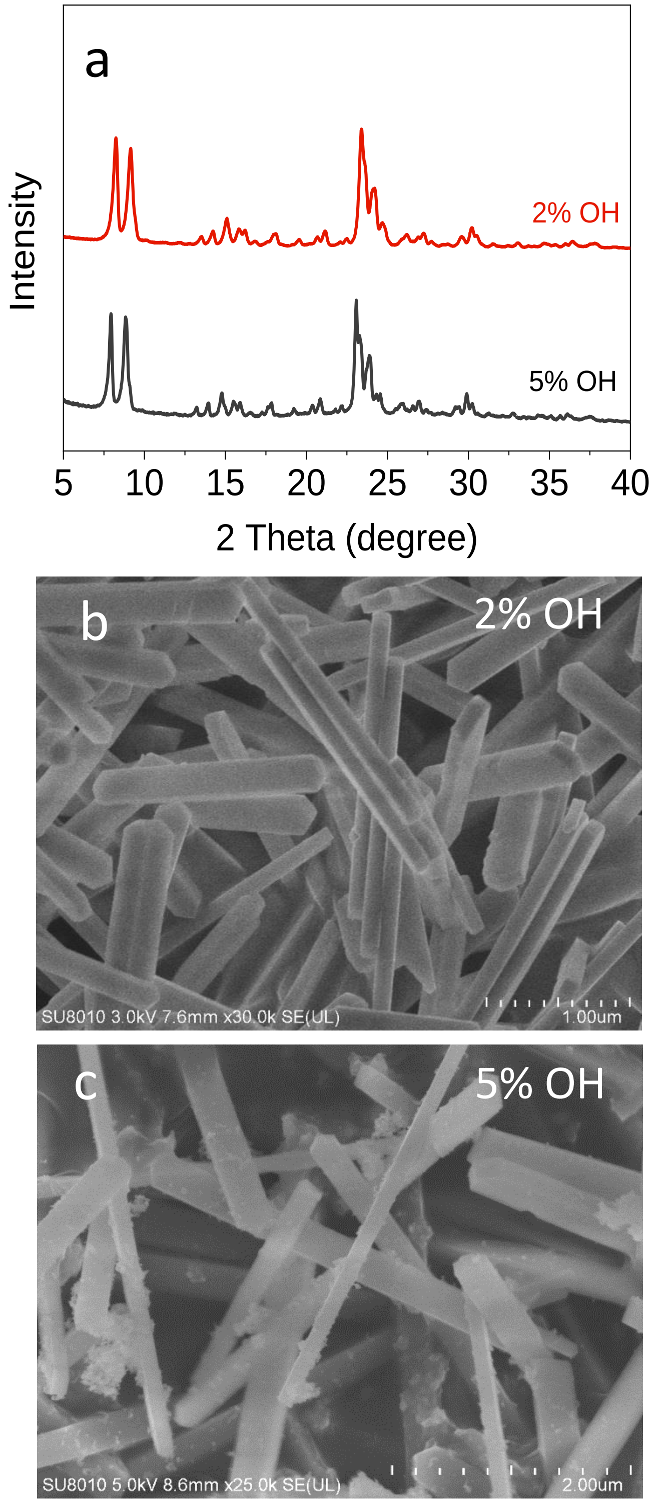
Supplementary Fig. 25. (a) ^{29}Si MAS NMR spectra and (b) ^{29}Si-^{1}H CP-MAS spectra of (i) *s*-S1 and (ii) *s*-S1-OH. (c) Model showing the *s*-S1-OH with short *b* axis.

Note: In Supplementary Fig. 25a, both *s*-S1 and *s*-S1-OH displayed Q_4 species [$Si(OSi)_4$] at -114 ppm. Compared with *s*-S1, the *s*-S1-OH gave additional signals at −104 and -109 ppm, indicating the presence of Q_3 sites [$Si(OSi)_3(OH)$] on siliceous zeolite crystals[17-19]. The ^{29}Si-^{1}H CP-MAS spectra of *s*-S1-OH showed the signals at -104 and -114 ppm, while no signals were observed on the ^{29}Si-^{1}H CP-MAS spectra of *s*-S1 (Supplementary Fig. 25b), indicating the presence of hydrogen atoms around silicon atoms on the *s*-S1-OH zeolite crystals (Q_3 sites for siliceous zeolite), and the absence of hydrogen atoms around silicon atoms on the *s*-S1 zeolite crystals[18,19]. These results confirm the successful functionalization of hydroxyl groups on the *s*-S1-OH zeolite framework, in good agreement with the IR results.



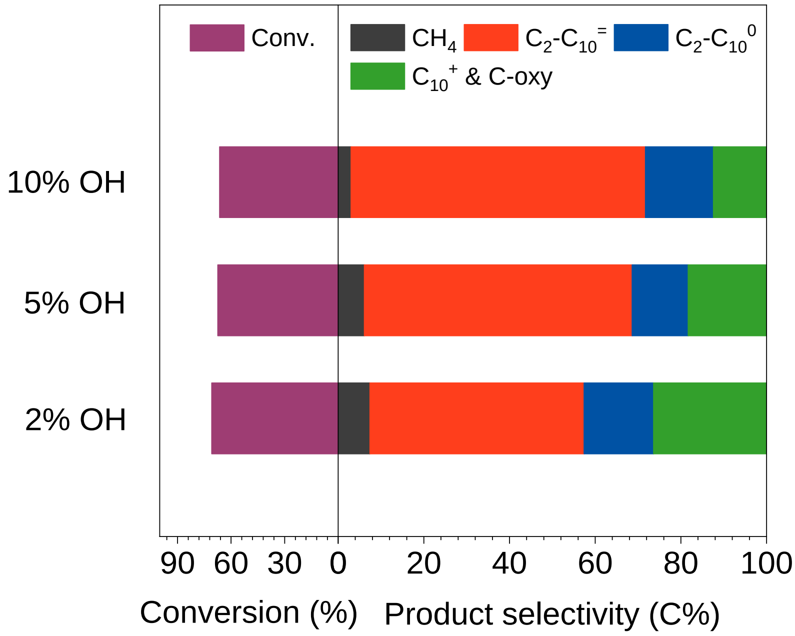
Supplementary Fig. 26. Catalytic data of the Na-FeC$_x$/*n*-S1, Na-FeC$_x$/*s*-S1, and Na-FeC$_x$/*s*-S1-OH in the LT-FTO. Reaction conditions: weight ratio zeolite to Na-FeC$_x$ at 3, feed gas with ratio of CO/H_2/Ar at 45/45/10, 2400 mL·h^{-1}·g^{-1}, 2.0 MPa, 260 °C, and granule-mixing of the zeolite and Na-FeC$_x$ individuals.

Note: The Na-FeC$_x$/*n*-S1 and Na-FeC$_x$/*s*-S1 showed the CO conversion at 42.7% and 50.2% in the LT-FTO, which were higher than that of the bare Na-FeC$_x$ catalyst under the equivalent reaction conditions. In addition, these catalysts still gave high C_{10+} product selectivity at 24.7% and 44.3%, respectively (Supplementary Fig. 26). In contrast, the Na-FeC$_x$/*s*-S1-OH showed enhanced activity and reduced selectivity for C_{10+} products, giving CO conversion and C_{10+} product selectivity at 66.5% and 12.4%, respectively. In this case, the o/a ratio appeared at 4.3 for the C_2-C_{10} molecules.

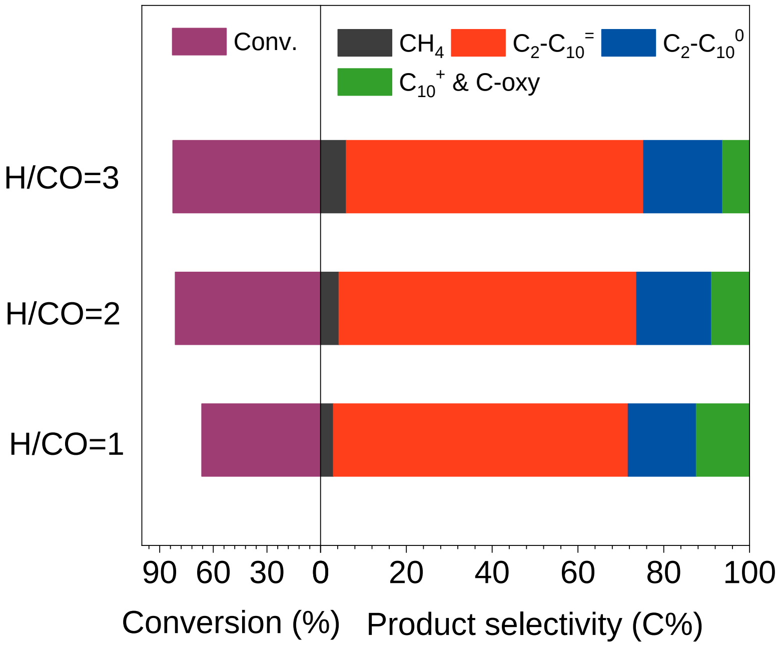


Supplementary Fig. 27. (a) XRD patterns and (b and c) SEM images of *s*-S1-OH zeolites with different silanol concentrations in the LT-FTO.

Note: The silanol concentrations are described by the molar ratio of DEMS to the total amount of silica species in the starting gel for zeolite synthesis.

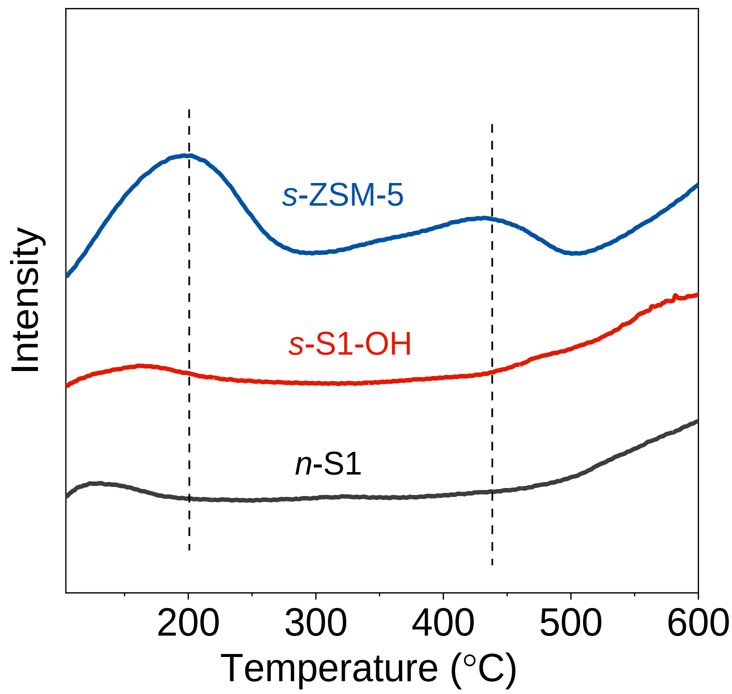


Supplementary Fig. 28. Catalytic performances of Na-FeC$_x$/*s*-S1-OH catalyst with different silanol concentrations in the LT-FTO. If it was specially noted, the *s*-S1-OH means the zeolite with silanol concentration at 10%. Reaction conditions: weight ratio zeolite to Na-FeC$_x$ at 3, feed gas with ratio of CO/H_2/Ar at 45/45/10, 2400 mL·h^{-1}·g^{-1}, 2.0 MPa, 260 °C, and granule-mixing of the zeolite and Na-FeC$_x$ individuals.



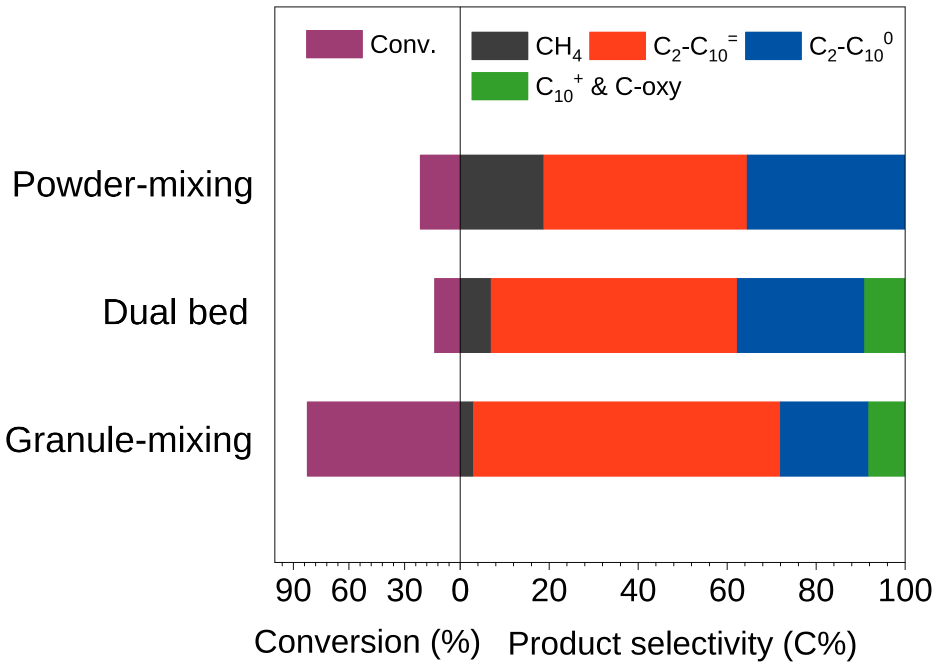
Supplementary Fig. 29. Catalytic data of the Na-FeC$_x$/*s*-S1-OH in the LT-FTO with different H_2/CO ratios at 1, 2, and 3. Reaction conditions: weight ratio of *s*-S1-OH to Na-FeC$_x$ at 3, 2400 mL·h^{-1}·g^{-1}, 2.0 MPa, 260 °C, and granule-mixing of the *s*-S1-OH and Na-FeC$_x$.

Note: We optimized the ratio of H_2/CO in the feed gas of LT-FTO over the Na-FeC$_x$/*s*-S1-OH. As shown in Supplementary Fig. 29, by adjusting the H_2/CO ratio to 2, the CO conversion could be 81.3%. When this ratio was 3, the CO conversion could reach to 82.6%. More importantly, the Na-FeC$_x$/*s*-S1-OH still showed a narrow distribution with dominant C_2-C_{10} molecules with o/a at 3.8, where the methane and C_{10+} product selectivities are 6.0% and 6.3% (Supplementary Table 8), respectively.

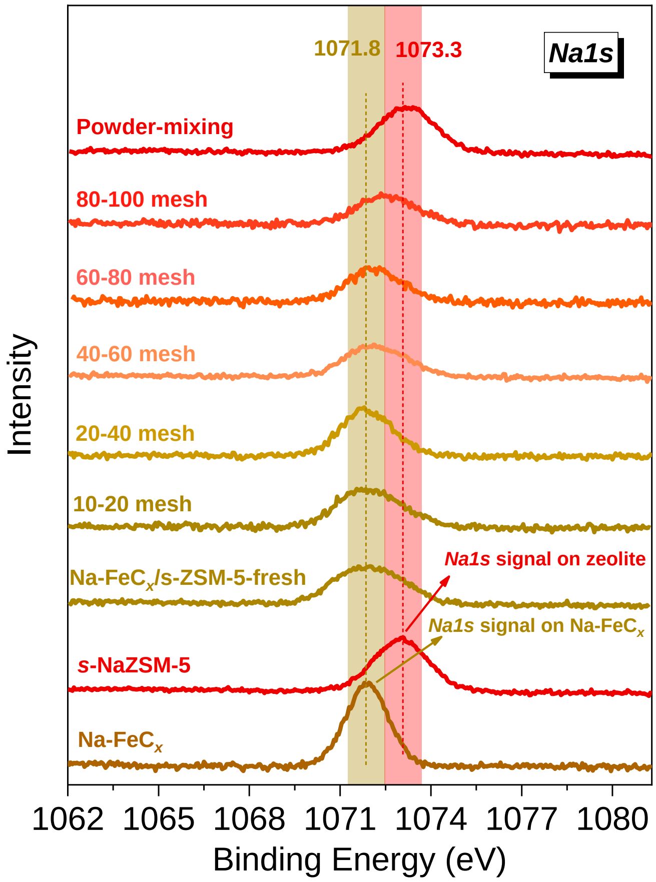


Supplementary Fig. 30. NH_3-TPD profiles of the *n*-S1, *s*-S1-OH, and *s*-ZSM-5.

Note: NH_3-TPD curve of *s*-ZSM-5 showed the peaks centered at 200 and 440 °C, which were assigned to the desorption of ammonia that was adsorbed on the weak and strong acid sites[20,21], respectively. In contrast, the signals of desorbed ammonia were almost undetectable on the *n*-S1 and *s*-S1-OH, confirming the absence of acidic sites on the siliceous zeolites.



Supplementary Fig. 31. Catalytic data of the Na-FeC_x/*s*-ZSM-5 with different mixing manners of Na-FeC_x and *s*-ZSM-5 individuals. Reaction conditions: weight ratio of *s*-ZSM-5 to Na-FeC_x at 3, ratios of CO/H_2/Ar at 45/45/10, 2400 mL·h^{-1}·g^{-1}, 2.0 MPa, and 260 °C.



Supplementary Fig. 32. Na1s XPS spectra of Na-FeC$_x$, *s*-NaZSM-5 (Na^+-exchanged zeolite as standard), and the granule- and power-mixing Na-FeC$_x$/*s*-ZSM-5 catalysts. These catalysts were used in LT-FTO for 50 h before the XPS characterization. The Na-FeC$_x$/*s*-ZSM-5-fresh is the fresh sample as standard without catalyzing LT-FTO.

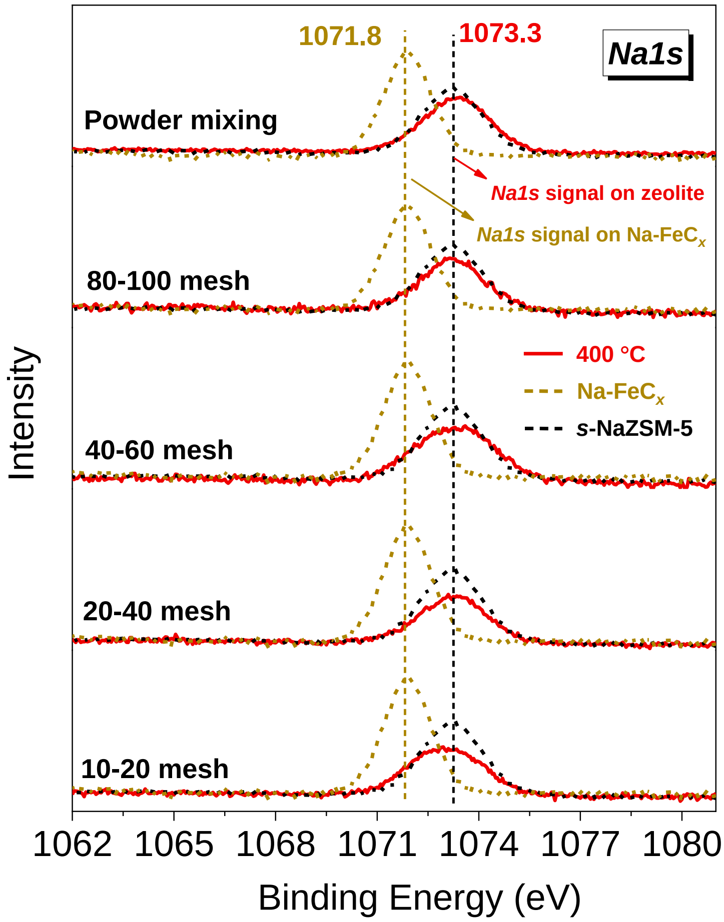
Note: The sodium migration from sodium-modified catalyst to the zeolite might occur in many reactions using physically mixed catalysts reported previously[22-27], but this migration is strongly related to the proximity of the catalyst granules. For example, de Jong and co-workers[22] reported sodium migration from ferric carbide to the zeolite on the granule-mixing catalyst with granule sizes at ~100 μm for the syngas-to-aromatic conversion at 400 °C, as confirmed by the continuously decreased aromatic selectivity because of poisoning the zeolite acid in the syngas-to-aromatic conversions. For the catalyst with granules sizes at ~500 μm, the catalytic performances were very similar to that over spatially separated dual-bed catalyst, which is regarded as a standard manner with prevented sodium migration in lab research. This result indicate the relatively hindered sodium migration on the catalyst with granules sizes at ~500 μm, which is quite different from the catalyst with granule sizes at ~100 μm. In contrast, the Na-

FeC_x/*s*-ZSM-5 catalyst in this work with average granule sizes at ~600 μm (~0.6 mm, 20-40 mesh) was performed at 260 °C, where the relatively long distance between the granules and lower reaction temperature should be unfavorable for the sodium migration.

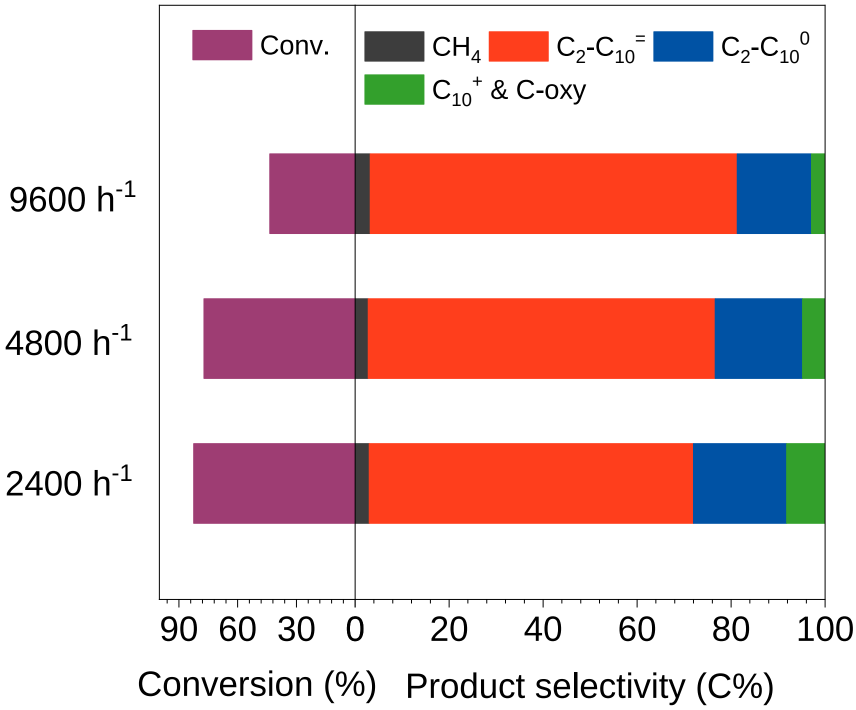
We performed Na1s XPS study to evaluate the sodium migration from Na-FeC_x to zeolite with different proximities, the granule- and powder-mixed Na-FeC_x/*s*-ZSM-5 catalysts were tested in LT-FTO for 50 h at 260 °C (average diameters: ~1.4 mm for 10-20 mesh, ~0.6 mm for 20-40 mesh, ~0.3 mm for 40-60 mesh, and ~0.16 mm for 80-100 mesh), and then characterized by XPS. Because the Na species on FeC_x and zeolite exhibited distinguishable Na1s peaks at 1071.8 and 1073.3 eV, the sodium migration could be identified by the Na1s peak shift. The catalyst with granules at 10-20 and 20-40 meshes exhibited the Na1s peak centralized at 1071.8 eV, which is very similar to that of the fresh mixture of Na-FeC_x and zeolite, suggesting the undetectable Na migration by XPS technique (Supplementary Fig. 32). For the catalyst with small granules, such as 80-100 mesh, the Na1s peak of the used catalysts appeared at 1072.4 eV with an obvious shift from the peak of Na species on Na-FeC_x (1071.8 eV) because the Na migration occurred. For the Na-FeC_x/*s*-ZSM-5 catalyst with a very close proximity in powder-mixing manner, the Na1s peak appeared at 1073.3 eV that is similar to that of the Na species on zeolite (Supplementary Fig. 32), suggesting massive sodium migration. Such migration would result into Na-FeC_x catalyst with insufficient sodium species and abundant alkane products in the FTO (Supplementary Table 9).

These results support that the Na migration is strongly influenced by the proximity between the Na-FeC_x and zeolite, where smaller granules would benefit the Na migration, in good agreement with the previous results[22]. Further studies were performed by treating the Na-FeC_x/*s*-ZSM-5 catalysts with different granule sizes and powder mixture at 400 °C, giving the Na1s peak at ~1073.3 eV for all the samples (Supplementary Fig. 33). These results confirm the obvious Na migration at higher temperature, even for the catalysts with large granules (e.g. 10-20 mesh).

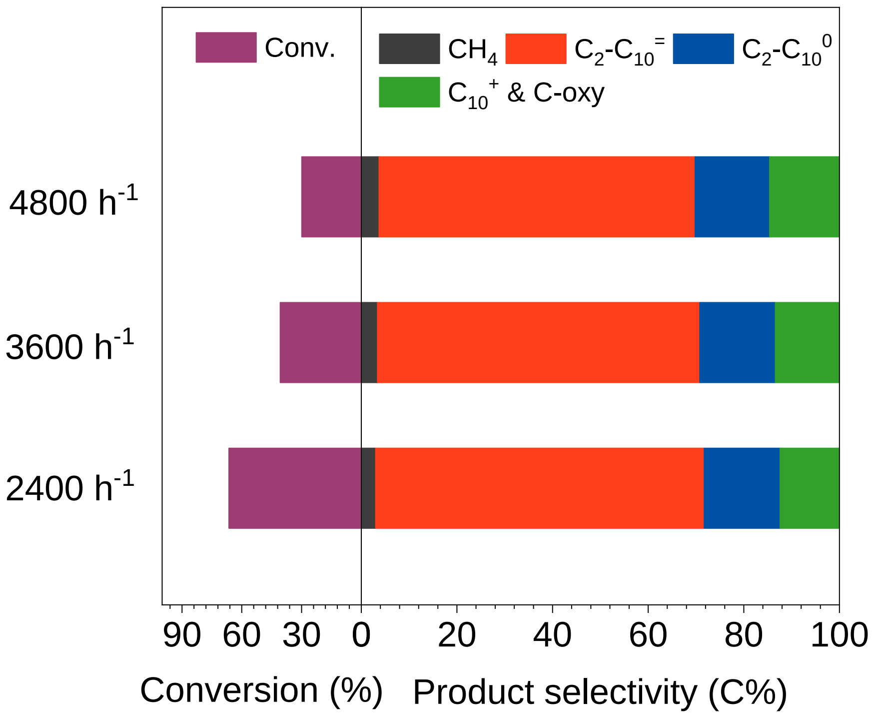
In sum, these data confirm that the sodium migration is strongly related to the proximity of catalyst granules and reaction temperature, where closer proximity and higher reaction temperature would benefit more sodium migration.



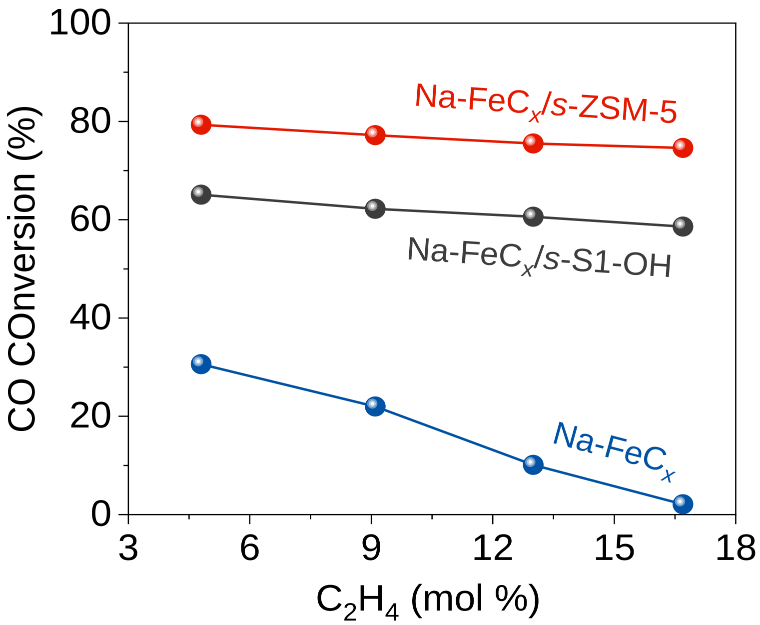
Supplementary Fig. 33. Na1s XPS spectra of Na-FeC$_x$, *s*-NaZSM-5 (Na$^+$-exchanged zeolite as standard), and the granule- and powder-mixing Na-FeC$_x$/*s*-ZSM-5 catalysts (These samples were treated at 400 °C in flowing nitrogen for 10 h before the XPS analysis). The dotted lines show the standard spectra of Na-FeC$_x$ and *s*-NaZSM-5 for comparison.



Supplementary Fig. 34. Catalytic data of the Na-FeC$_x$/*s*-ZSM-5 catalyst in the LT-FTO with different GHSV at 2400, 4800, and 9600 mL·h^{-1}·g^{-1}. Reaction conditions: weight ratio of *s*-ZSM-5 to Na-FeC$_x$ at 3, ratios of CO/H_2/Ar at 45/45/10, 2.0 MPa, 260 °C, and granule mixture of *s*-ZSM-5 and Na-FeC$_x$.



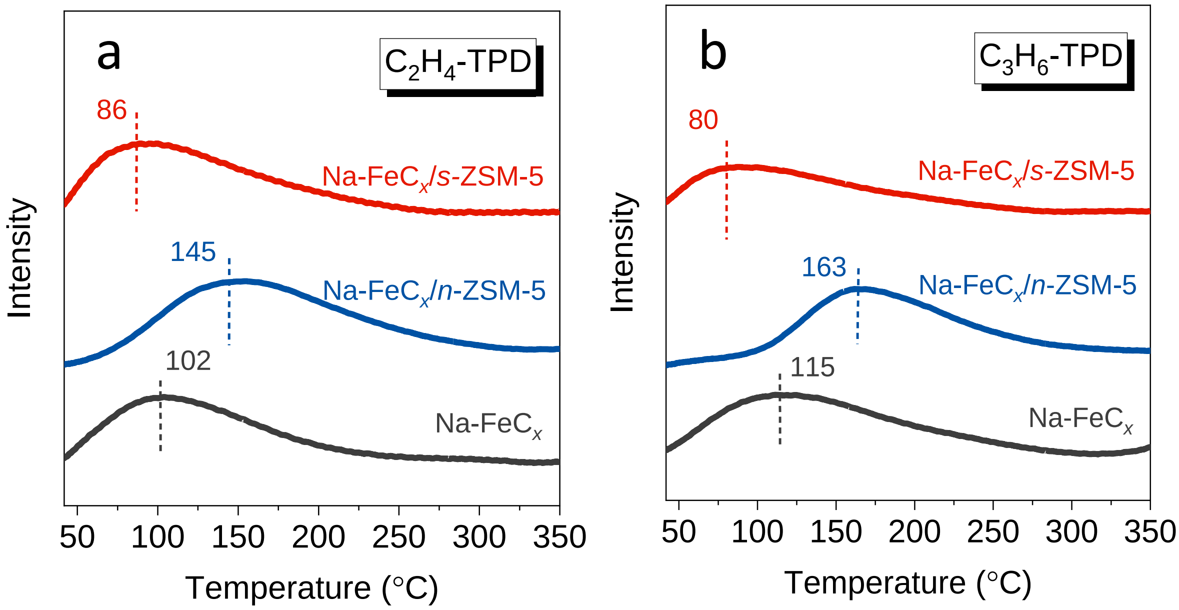
Supplementary Fig. 35. Catalytic data of the Na-FeC_x/*s*-S1-OH with different GHSV at 2400, 3600, and 4800 mL·h^{-1}·g^{-1}. Reaction conditions: weight ratio of *s*-ZSM-5 to Na-FeC_x at 3, ratio of CO/H_2/Ar at 45/45/10, 2.0 MPa, 260 °C, and granule mixture of *s*-S1-OH and Na-FeC_x.



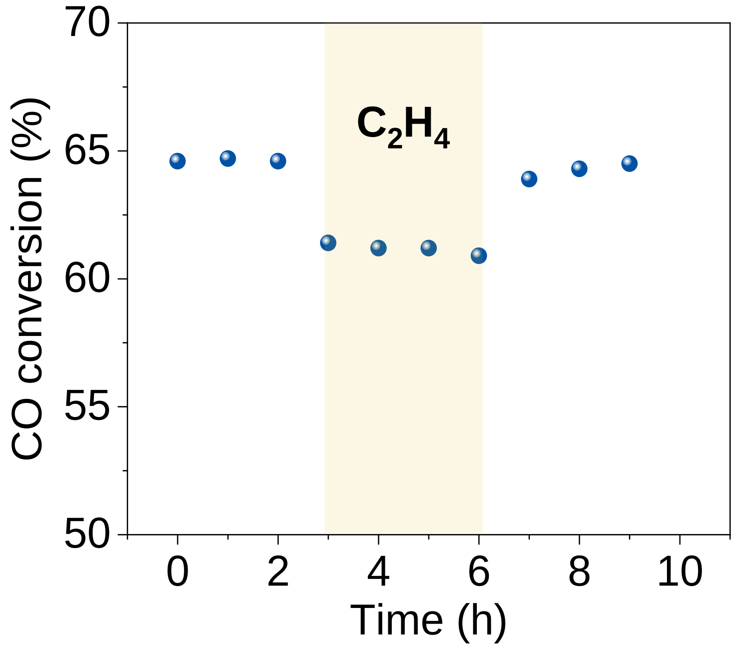
Supplementary Fig. 36. Dependences of CO conversion on ethene concentrations in the feed gas of the LT-FTO reaction over the Na-FeC$_x$, Na-FeC$_x$/*s*-S1-OH and Na-FeC$_x$/*s*-ZSM-5 catalysts. Reaction conditions: ratio of H_2/CO at 1.0, 2400 mL·h^{-1}·g^{-1}, 2.0 MPa, reaction temperature at 260 °C for Na-FeC$_x$/*s*-S1-OH and Na-FeC$_x$/*s*-ZSM-5 catalysts, 300 °C for Na-FeC$_x$ catalyst, and granule mixture of zeolite and Na-FeC$_x$. Because of poor activity of the Na-FeC$_x$ in the LT-FTO at 260 °C, study over the Na-FeC$_x$ without zeolite was performed at 300 °C for evaluating the influence of ethene to the CO conversion.

Note: *Discussion to exclude the effect of zeolite influenced water diffusion to improve the catalysis*. Water is also slightly produced in the syngas conversion reactions, but the zeolite in this work should modulate the olefin sorption rather than water because of the significantly distinguishable phenomenon compared with the previous hydrophobic Fe-based catalysts[22]. The previous studies have revealed that once the water products were rapidly escaped from the Fe-based catalyst surface, the water-gas shift side reaction would be efficiently hindered to reduce the CO_2 selectivity in FTO. In our case, the CO_2 selectivity was still at 45-50% that is in constant with the general FTO but quite different from the FTO over the hydrophobic catalysts. As well, the rapid water desorption could not obviously increase the CO conversion and adjust the carbon number distribution of the olefin products[22], while the zeolite promoter in this work simultaneously boosted the CO conversion and optimize the carbon number distribution. These features are completely different from the hydrophobic catalysts.

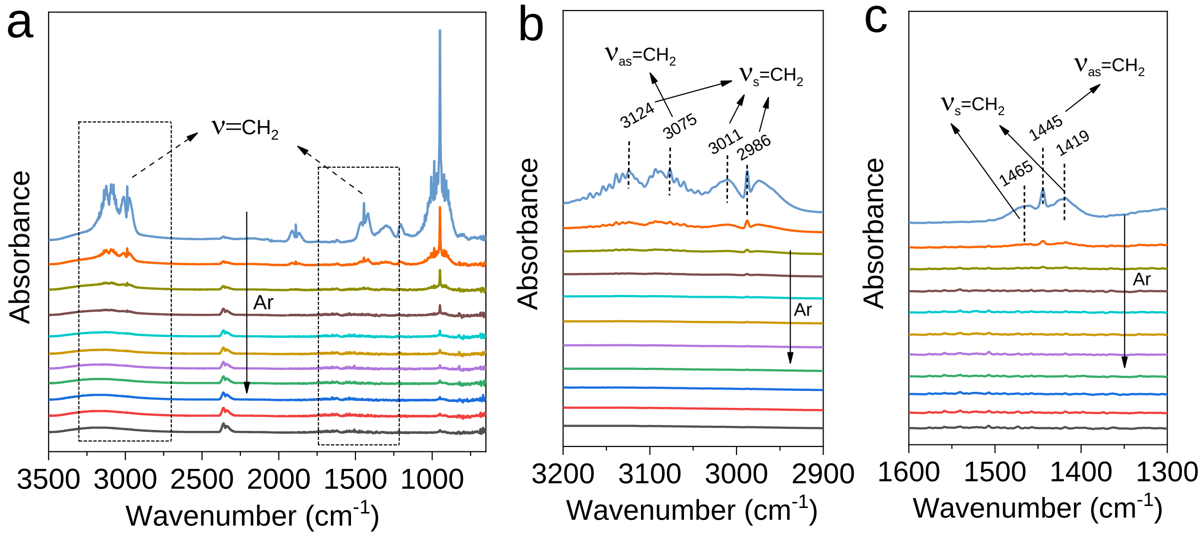
Another result to confirm the zeolite influenced olefin sorption more than water is the different catalytic behaviors of Na-FeC$_x$ mixed with *s*-ZSM-5 and *n*-ZSM-5. These two zeolites have the same microporous environment but different *b*-axis distance (different diffusion distance for molecules). In the LT-FTO, the two catalysts exhibited quite different product selectivities. The Na-FeC$_x$/*s*-ZSM-5 gave olefins as dominant products, while the Na-FeC$_x$/*n*-ZSM-5 favored to produce alkanes (olefin/paraffin molar ratio at only 0.1, Supplementary Table 1). This phenomenon was because of the long retention time of olefins within the micropores of *n*-ZSM-5 that leads to deep hydrogenation to form alkanes, which also confirms the zeolite adjusted olefin diffusion.



Supplementary Fig. 37. Temperature programed desorption (TPD) curves of (a) ethene and (b) propene over the Na-FeC$_x$, Na-FeC$_x$/*n*-ZSM-5, and Na-FeC$_x$/*s*-ZSM-5.

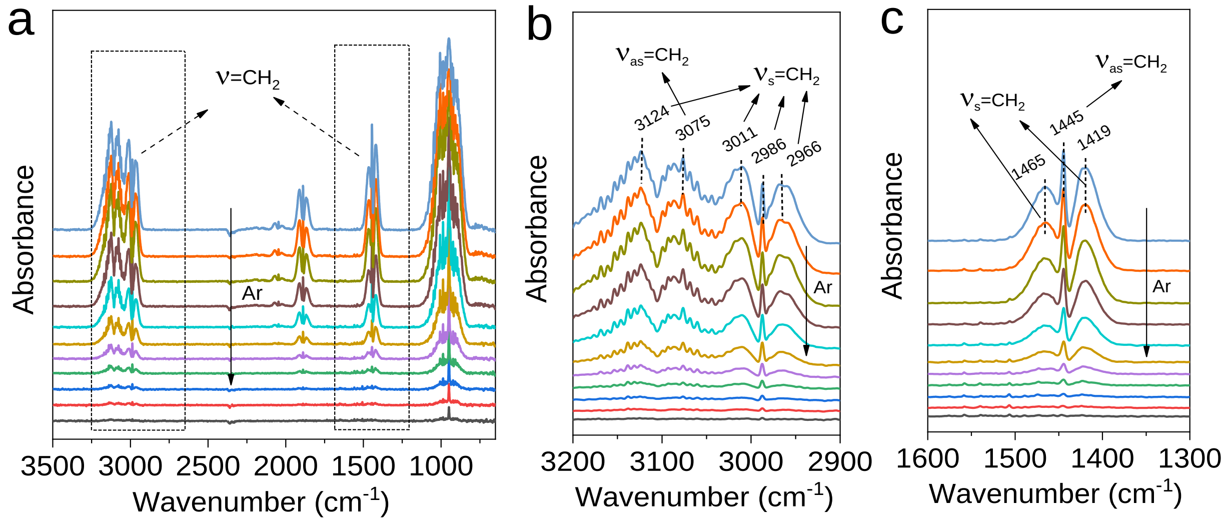


Supplementary Fig. 38. Data showing the influence of ethene to the CO conversion in the LT-FTO over the Na-FeC_x/*s*-S1-OH catalyst. Reaction conditions: weight ratio of *s*-S1-OH to Na-FeC_x at 3, GHSV at 2880 mL·h^{-1}·g^{-1}, ratio of ethene/H_2/CO at 0.4/1/1, 2.0 MPa, 260 °C, and granule mixture of Na-FeC_x and zeolite.



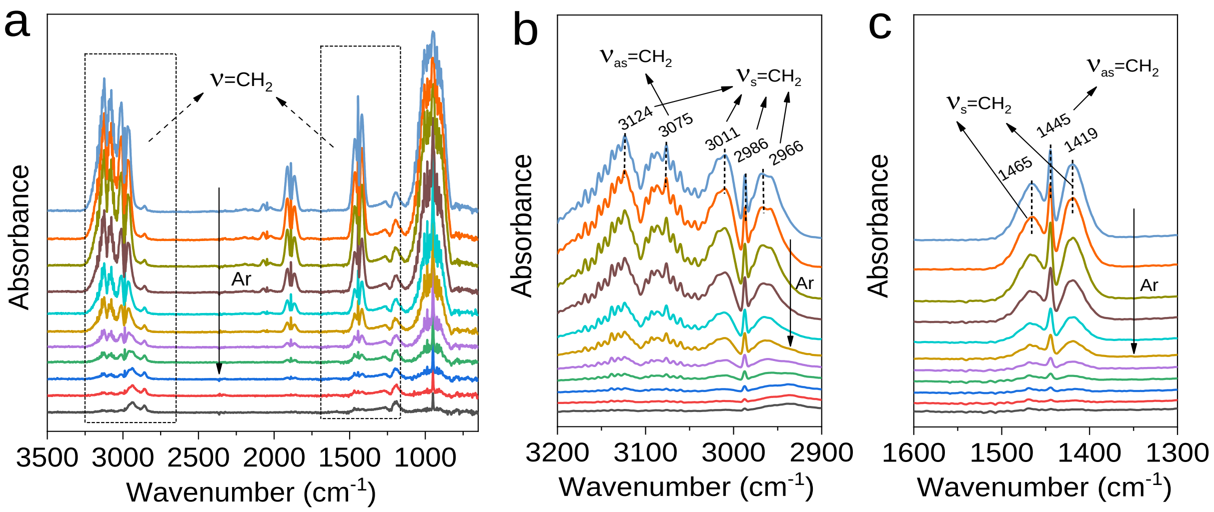
Supplementary Fig. 39. (a) C_2H_4-desorption DRIFT spectra of the *s*-ZSM-5. (b) Enlarge view of the region at 2900-3200 cm^{-1}. (c) Enlarge view of the region at 1300-1600 cm^{-1}. The IR spectra were recorded at intervals of two minutes.

Note: We preformed the ethene- and propene-desorption DRIFT spectra on *s*-ZSM-5 zeolite without Na-FeC_x. As shown in Supplementary Figs. 39 and 43, most of the ethene and propene molecules would rapidly desorb within the initial several minutes. For example, over 91% of the ethene and 96% of the propene were removed in 4 min (Supplementary Tables 13 and 14). In contrast, their adsorption on Na-FeC_x were much stronger as confirmed by the obvious DRIFT signals after desorption for longer time (Supplementary Figs. 40 and 44). In the olefin desorption test over the Na-FeC_x/zeolite samples, a pre-desorption operation was performed for 4 min before recording the signals in Figure 3c in the main text, which would remove the olefins pre-adsorbed in zeolite to avoid their influences to the study.

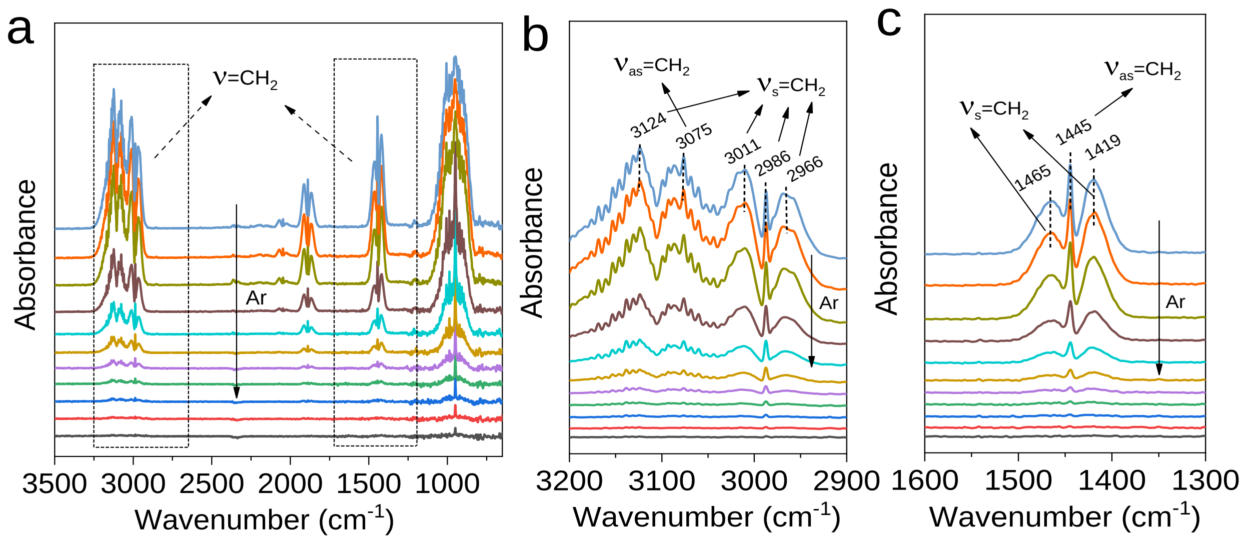


Supplementary Fig. 40. (a) C_2H_4-desorption DRIFT spectra of the Na-FeC_x. (b) Enlarge view of the region at 2900-3200 cm^{-1}. (c) Enlarge view of the region at 1300-1600 cm^{-1}. The IR spectra were recorded at intervals of two minutes.

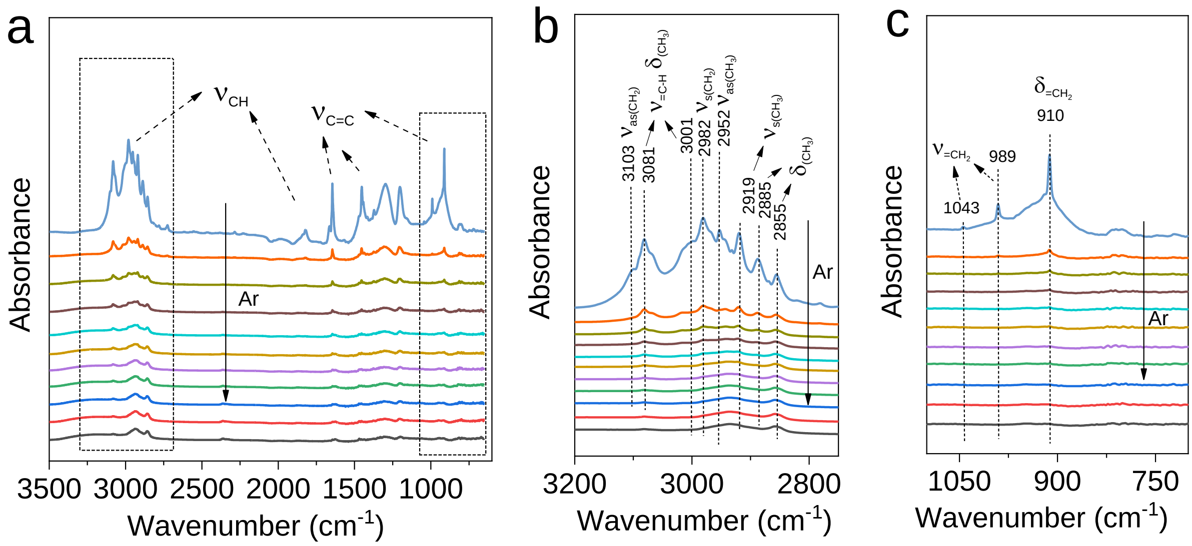
Note: The data in Fig. 3c were selected from the spectra in Supplementary Figs. 40-42.



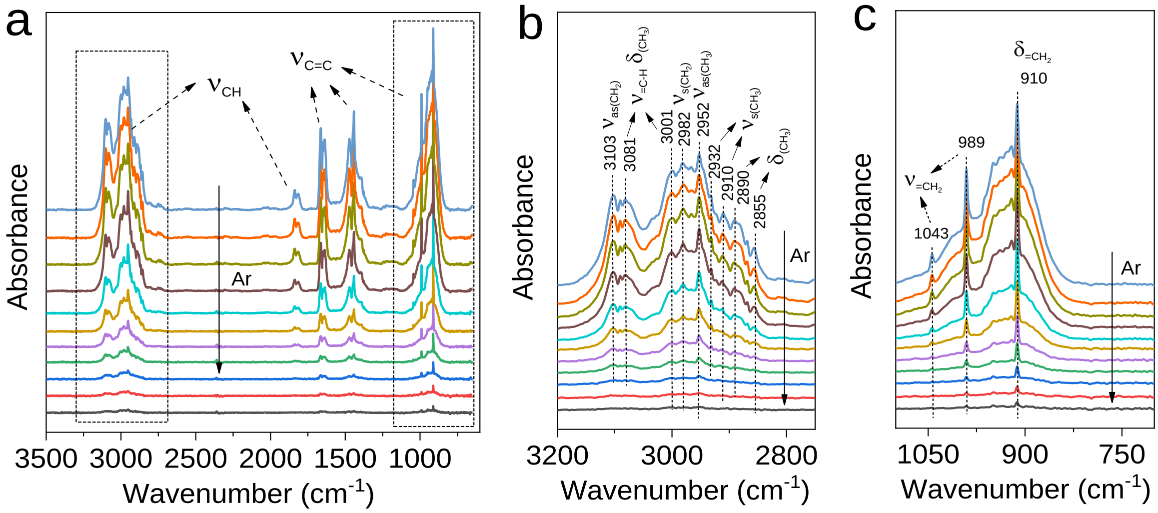
Supplementary Fig. 41. (a) C_2H_4-desorption DRIFT spectra of the Na-FeC$_x$/*n*-ZSM-5. (b) Enlarge view of the region at 2900-3200 cm^{-1}. (c) Enlarge view of the region at 1300-1600 cm^{-1}. The IR spectra were recorded at intervals of two minutes.



Supplementary Fig. 42. (a) C_2H_4-desorption DRIFT spectra of the Na-FeC_x/*s*-ZSM-5. (b) Enlarge view of the region at 2900-3200 cm^{-1}. (c) Enlarge view of the region at 1300-1600 cm^{-1}. The IR spectra were recorded at intervals of two minutes.

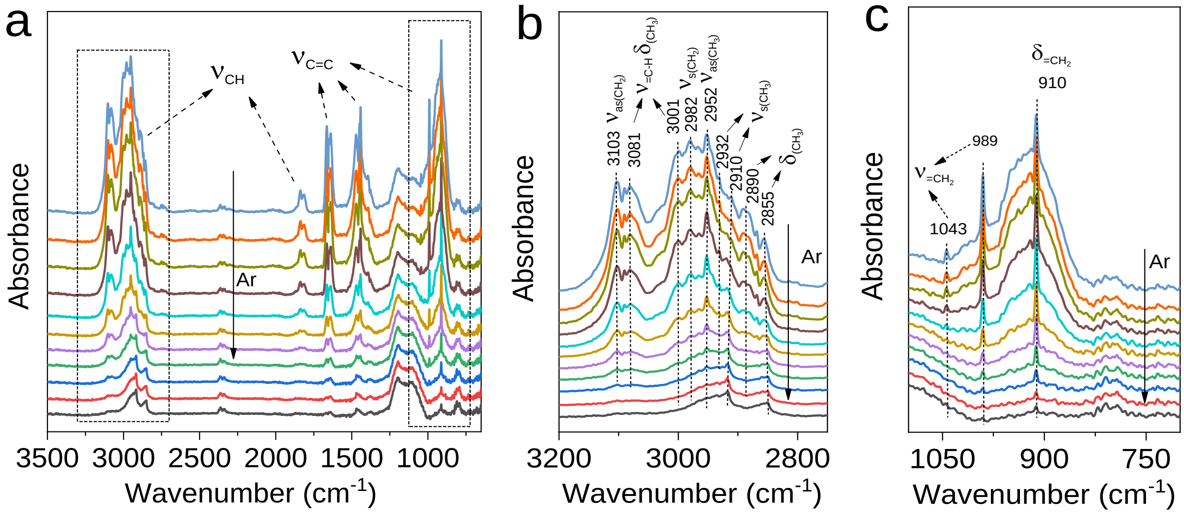


Supplementary Fig. 43. (a) C_3H_6-desorption DRIFT spectra of the *s*-ZSM-5. (b) Enlarge view of the region at 2900-3200 cm^{-1}. (c) Enlarge view of the region at 1300-1600 cm^{-1}. The IR spectra were recorded at intervals of two minutes.

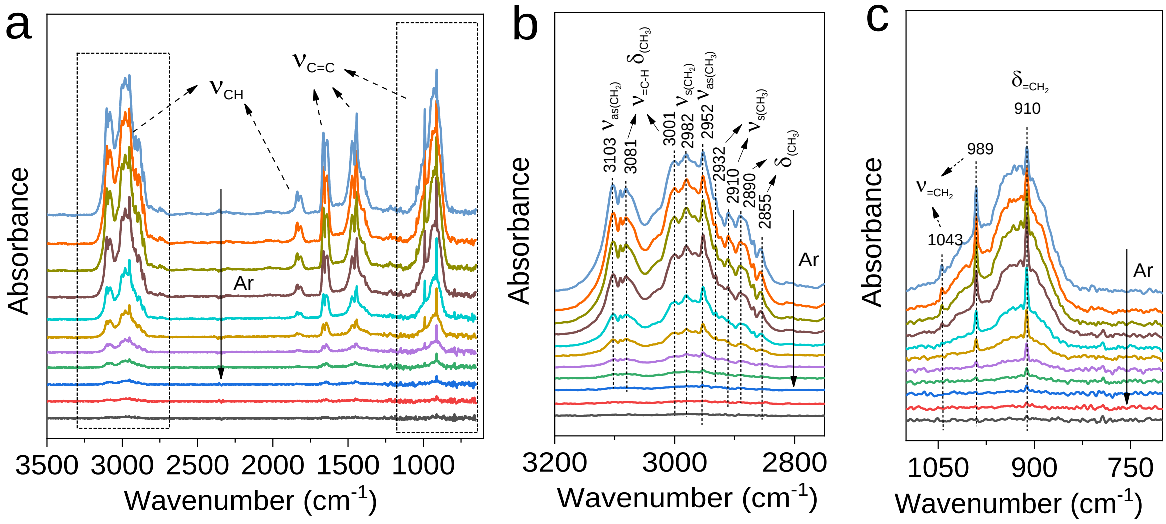


Supplementary Fig. 44. (a) C_3H_6-desorption DRIFT spectra of the Na-FeC_x. (b) Enlarge view of the region at 2750-3200 cm^{-1}. (c) Enlarge view of the region at 700-1100 cm^{-1}. The IR spectra were recorded at intervals of two minutes.

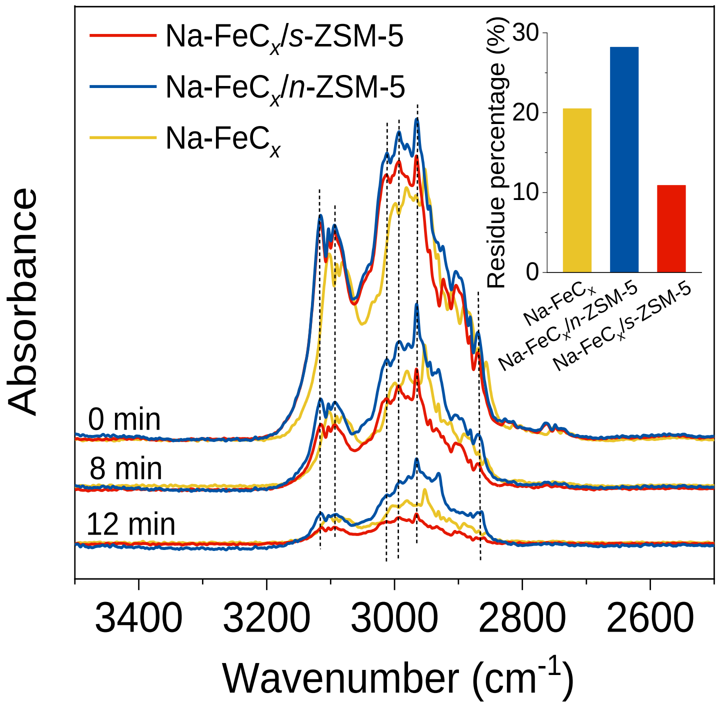
Note: As shown in Supplementary Figs. 43-46, after pre-adsorption treatment, all samples appeared the bands assigning to symmetrical and asymmetrical stretching vibration of $=CH_2$ (signals at 989, 1043, 2982, and 3103 cm^{-1}), symmetrical and asymmetrical stretching vibration of $-CH_3$ (signals at 2910, 2932, and 2952 cm^{-1}), deformation vibration of CH_2 (signals at 910 cm^{-1}), and deformation vibration of CH_3 (signals at 2855, 2890, 3001, and 3081 cm^{-1}). After flowing Ar to the samples, the IR spectra were recorded every 2 min. According to the peak intensity changes as a function of time, it was evaluated that the propene desorption rates over the Na-FeC_x/*s*-ZSM-5 and Na-FeC_x/*s*-S1-OH were higher than Na-FeC_x (Supplementary Figs. 47 and 49).



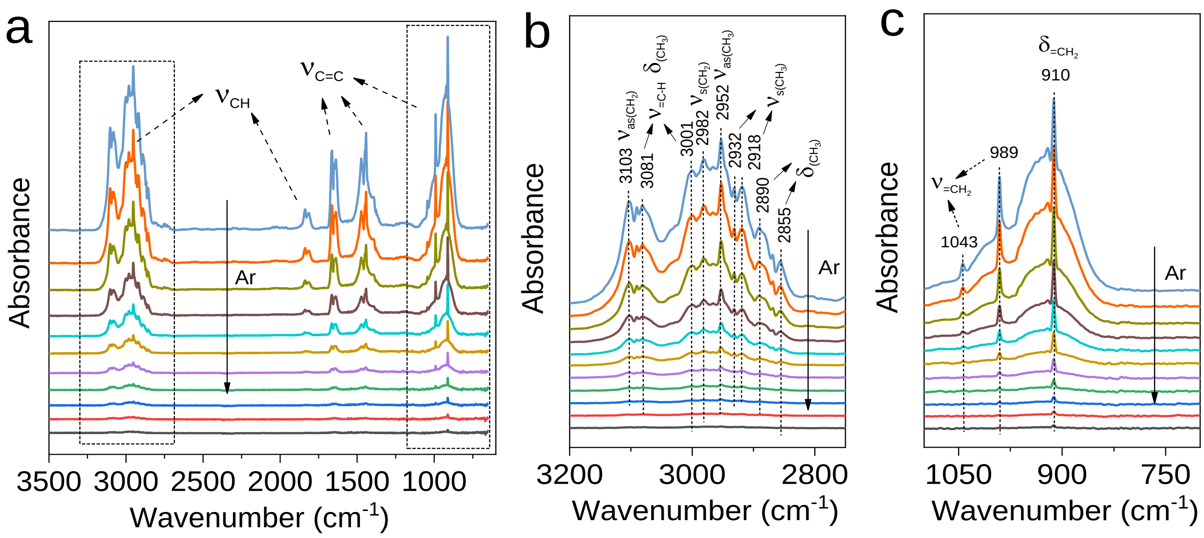
Supplementary Fig. 45. (a) C_3H_6-desorption DRIFT spectra of the Na-FeC$_x$/*n*-ZSM-5. (b) Enlarged view of the region at 2750-3200 cm^{-1}. (c) Enlarged view of the region at 700-1100 cm^{-1}. The IR spectra were recorded at intervals of two minutes.



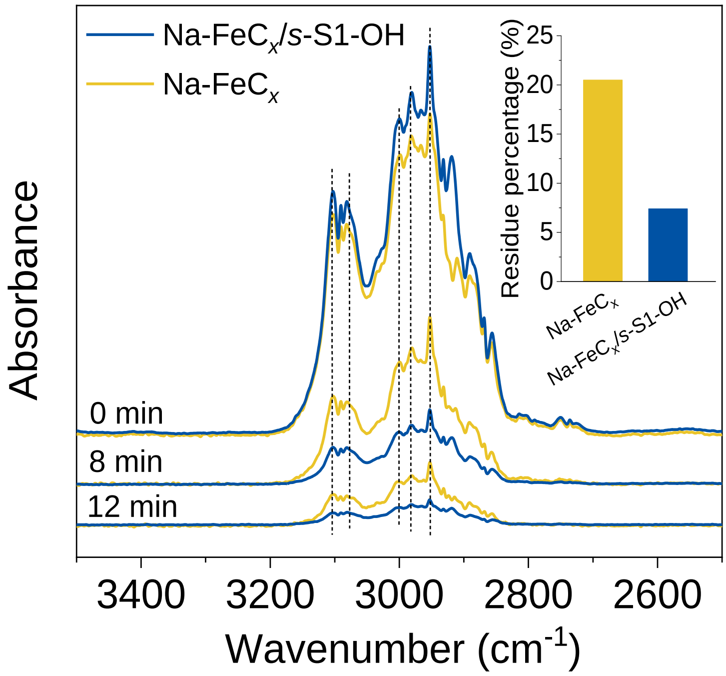
Supplementary Fig. 46. (a) C_3H_6-desorption DRIFT spectra of the Na-FeC$_x$/*s*-ZSM-5. (b) Enlarge view of the region at 2750-3200 cm^{-1}. (c) Enlarge view of the region at 700-1100 cm^{-1}. The IR spectra were recorded at intervals of two minutes.



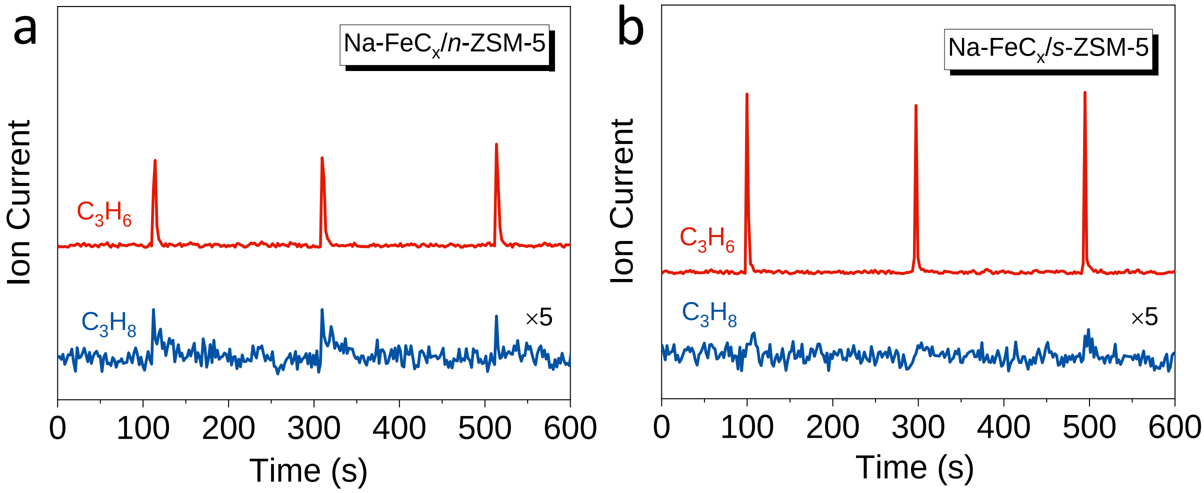
Supplementary Fig. 47. Typical spectra at different time during the C_3H_6-desorption IR tests over the Na-FeC$_x$/*s*-ZSM-5, Na-FeC$_x$/*n*-ZSM-5, and Na-FeC$_x$ catalysts. These typical spectra were selected from the data in Supplementary Figs. 44-46. The inset figure gave the residue percentage of propene according to the peak intensity changes (signal at 2952 cm^{-1}) over the various catalysts during 0-12 min (data calculated from Supplementary Table 14).



Supplementary Fig. 48. (a) C_3H_6-desorption DRIFT spectra of the Na-FeC$_x$/*s*-S1-OH. (b) Enlarge view of the region at 2750-3200 cm^{-1}. (c) Enlarge view of the region at 700-1100 cm^{-1}. The IR spectra were recorded at intervals of two minutes.

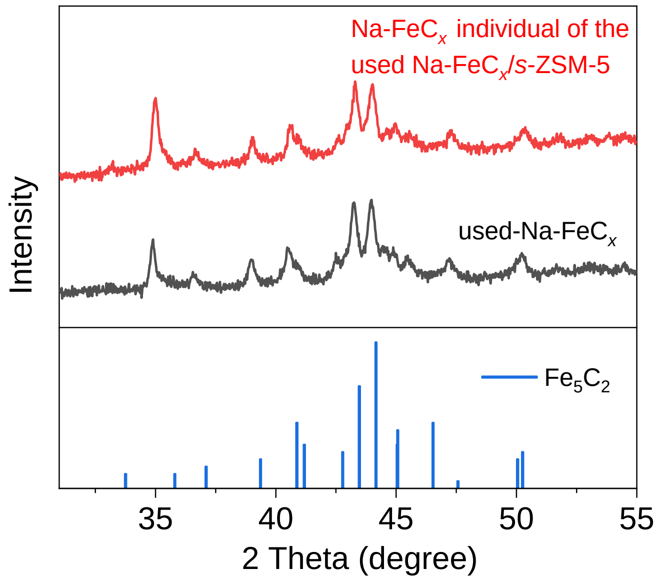


Supplementary Fig. 49. Typical spectra at different time during the C_3H_6-desorption DRIFT tests over the Na-FeC$_x$/*s*-S1-OH and Na-FeC$_x$ catalysts. These typical spectra were selected from the data in Supplementary Figs. 44 and 48. The inset figure gave the residue percentage of propene according to the peak intensity changes (signal at 2952 cm^{-1}) over the various catalysts during 0-12 min (data calculated from Supplementary Table 14).

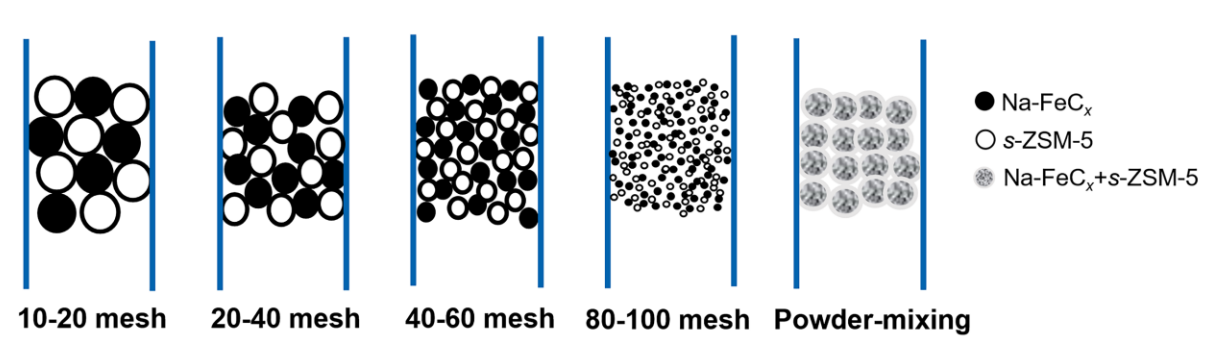


Supplementary Fig. 50. Mass spectra obtained by pulsing 1 mL of 10% C_3H_6/He into a pure H_2 flow (200 mL/min) over the (a) Na-FeC$_x$/*n*-ZSM-5 and (b) Na-FeC$_x$/*s*-ZSM-5 catalysts. Reaction conditions: weight ratio of zeolite to Na-FeC$_x$ at 3, 0.1 MPa, 260 °C, and the granule-mixing of the zeolite and Na-FeC$_x$ individuals.

Note: We used pulse experiments to explore the reactivity of olefins on the Na-FeC$_x$/*n*-ZSM-5 and Na-FeC$_x$/*s*-ZSM-5. More propane was formed over the Na-FeC$_x$/*n*-ZSM-5 than that over the Na-FeC$_x$/*s*-ZSM-5. This phenomenon might be explained by stronger hydrogen spillover and longer retention time of propene molecules in the *n*-ZSM-5 zeolite than that in the *s*-ZSM-5 zeolite, which benefit the deep hydrogenation of propene to propane.

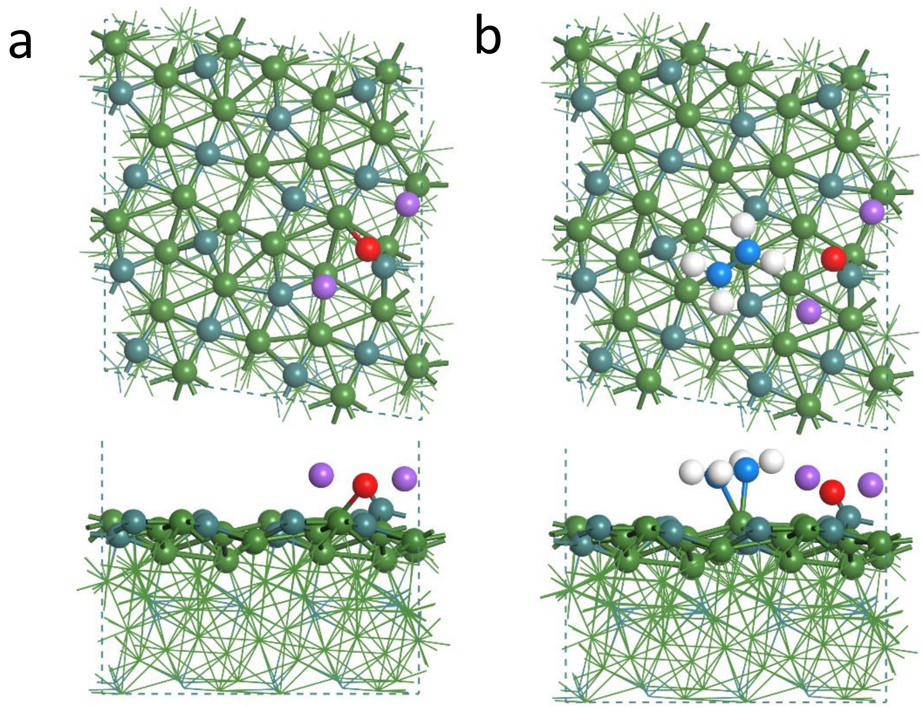


Supplementary Fig. 51. XRD patterns of the used Na-FeC$_x$ in the reactions with and without zeolites. The samples exhibited almost the same XRD patterns with Fe_5C_2 as a dominant phase.

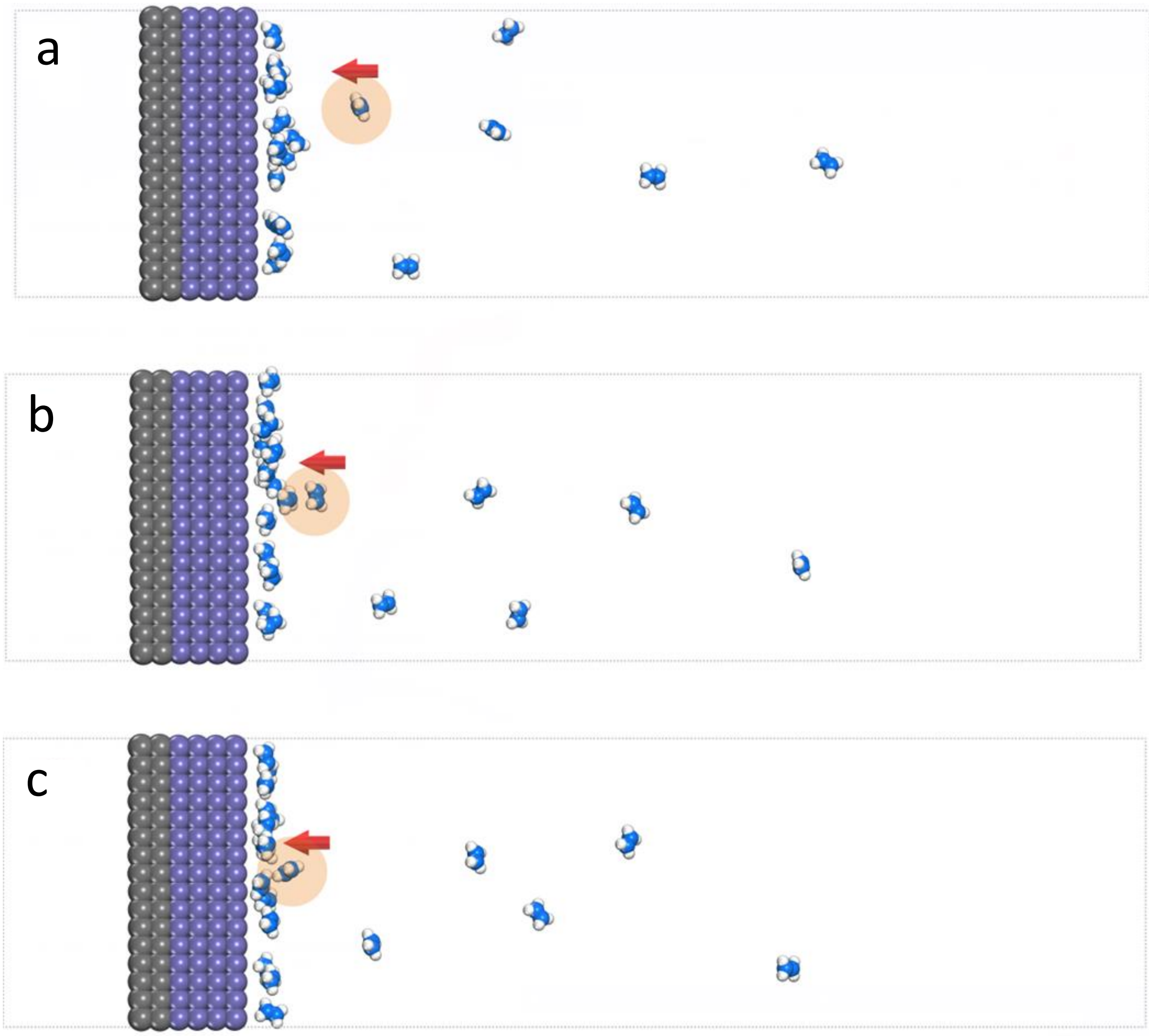


Supplementary Fig. 52. Models of Na-FeC$_x$/*s*-ZSM-5 catalysts with different proximities between Na-FeC$_x$ and *s*-ZSM-5.

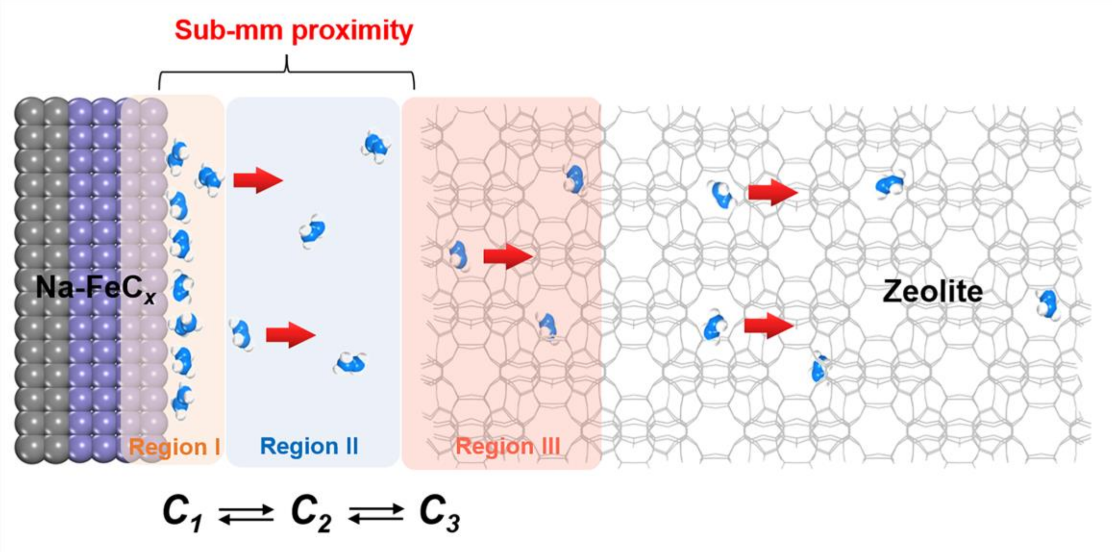
Note： We evaluated the Na-FeC$_x$/*s*-ZSM-5 catalysts with different granule sizes (average diameters: ~1.4 mm for 10-20 mesh, ~0.6 mm for 20-40 mesh, ~0.3 mm for 40-60 mesh, and ~0.16 mm for 80-100 mesh) and the powder-mixing Na-FeC$_x$/*s*-ZSM-5 catalyst (Supplementary Fig. 52). As shown in Supplementary Table 10, the Na-FeC$_x$/*s*-ZSM-5 catalysts with different granule sizes at ~0.3-1.4 mm exhibited the CO conversions at 77.1-88.3%, which are obviously higher than that over the Na-FeC$_x$ catalyst without zeolite promoter. In these reactions over granule-mixing Na-FeC$_x$/*s*-ZSM-5 catalysts, the *o/a* ratios were 3.5-4.0. The test using the catalyst with closer proximity with granules at ~0.16 mm (80-100 mesh) still exhibited significant promotion effect of zeolite, giving the CO conversion at 69.8%. This value is lower than that with larger granules, which might be due to stronger sodium migration with close proximity as confirmed by the Na1s XPS study (Supplementary Fig. 32). These data show that the granule mixture with a scale proximity at ~0.16 mm could boost the syngas conversion and maintain the high olefin selectivity, but the proximity at ~0.3 mm is suggested for obtaining better performance.



Supplementary Fig. 53. The top and side view structures of (a) Na-FeC_x, and (b) one ethene adsorbed on the Na-FeC_x (C_2H_4: C atoms in blue and H atoms in white; Na-FeC_x: O atoms in red, Na atoms in purple, Fe atoms in green and C atoms in dark cyan). The adsorption energy values are -0.58 and -0.28 eV for the first and second ethene molecules, respectively.

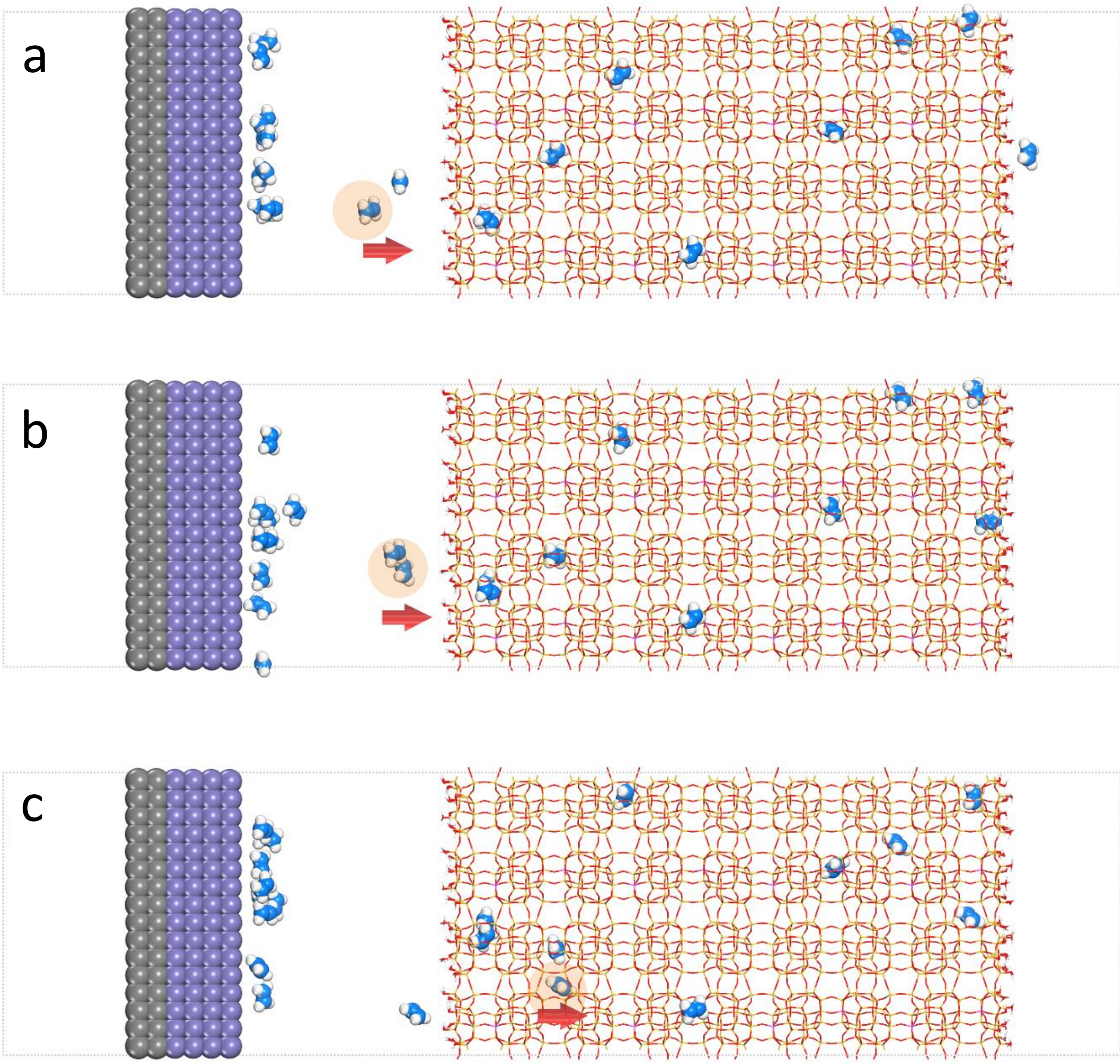


Supplementary Fig. 54. Scheme showing 20 ethene molecules (blue-white balls) desorbing from Na-FeC_x (black-purple surfaces) following the time of a → b → c. It shows free ethene preferring to re-adsorb on the Na-FeC_x surface that hinders the desorption process.

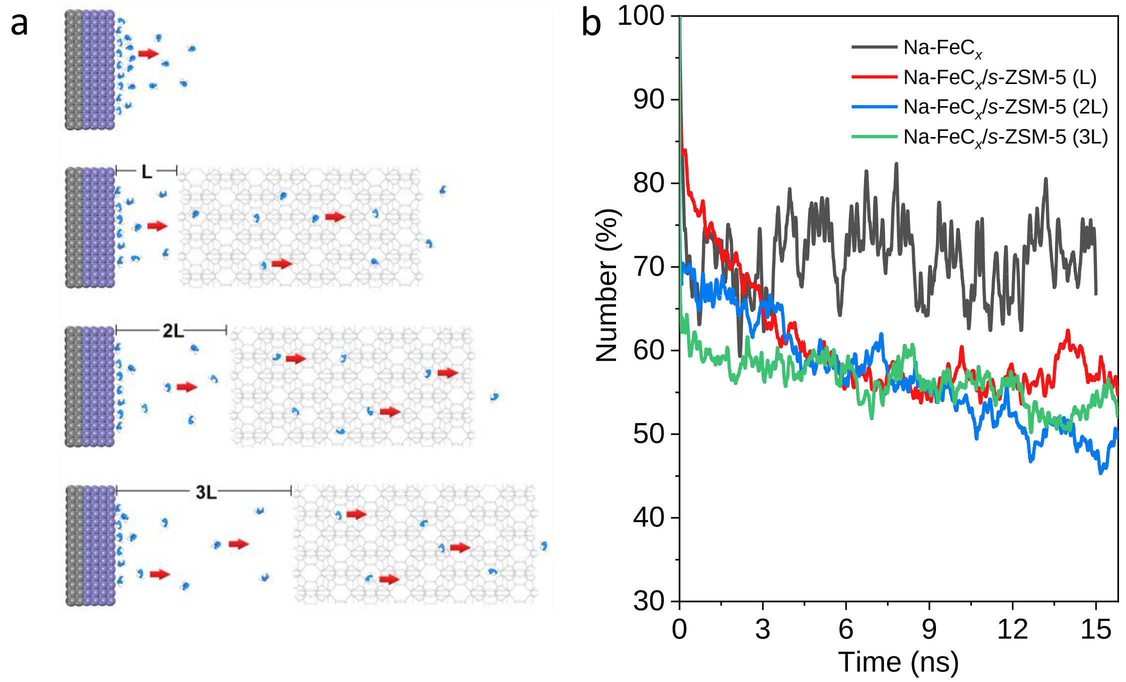


Supplementary Fig. 55. Scheme showing the diffusion of olefins from the Na-FeC$_x$ surface to the zeolite *via* different regions.

Note: With the sub-mm proximity, the diffusion of olefins from the Na-FeC$_x$ surface to the zeolite could be described in the model, the olefins formed on the Na-FeC$_x$ surface (region I) could desorb into the free region between Na-FeC$_x$ and zeolite (region II), where the olefin concentration on Na-FeC$_x$ surface (region I) and in free region (region II) could be described as C_1 and C_2, respectively. On the Na-FeC$_x$ catalyst, the LT-FTO reaction was greatly hindered because of the olefin desorption and re-adsorption on the Na-FeC$_x$ surface described by the equilibrium between C_1 and C_2 ($^*C_nH_{2n} \rightleftarrows {}^* + C_nH_{2n}$, * means the Na-FeC$_x$ surface sites). On the zeolite side, the free olefins in region II could be rapidly adsorbed by zeolite (region III) to obviously reduce C_2, which would efficiently shift the equilibrium of $^*C_nH_{2n} \rightleftarrows {}^* + C_nH_{2n}$ on Na-FeC$_x$ surface.

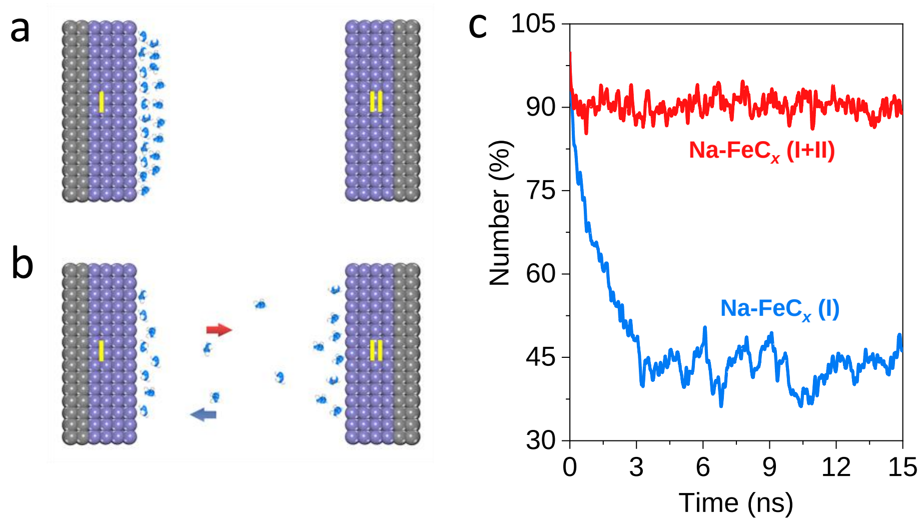


Supplementary Fig. 56. Scheme showing 20 ethene molecules (blue-white balls) desorbing from Na-FeC_x (black-purple surfaces)/*s*-ZSM-5 (red-yellow) following the time of a → b → c. It shows free ethene preferring to adsorb on the *s*-ZSM-5 that facilitates ethene desorption from Na-FeC_x surface.



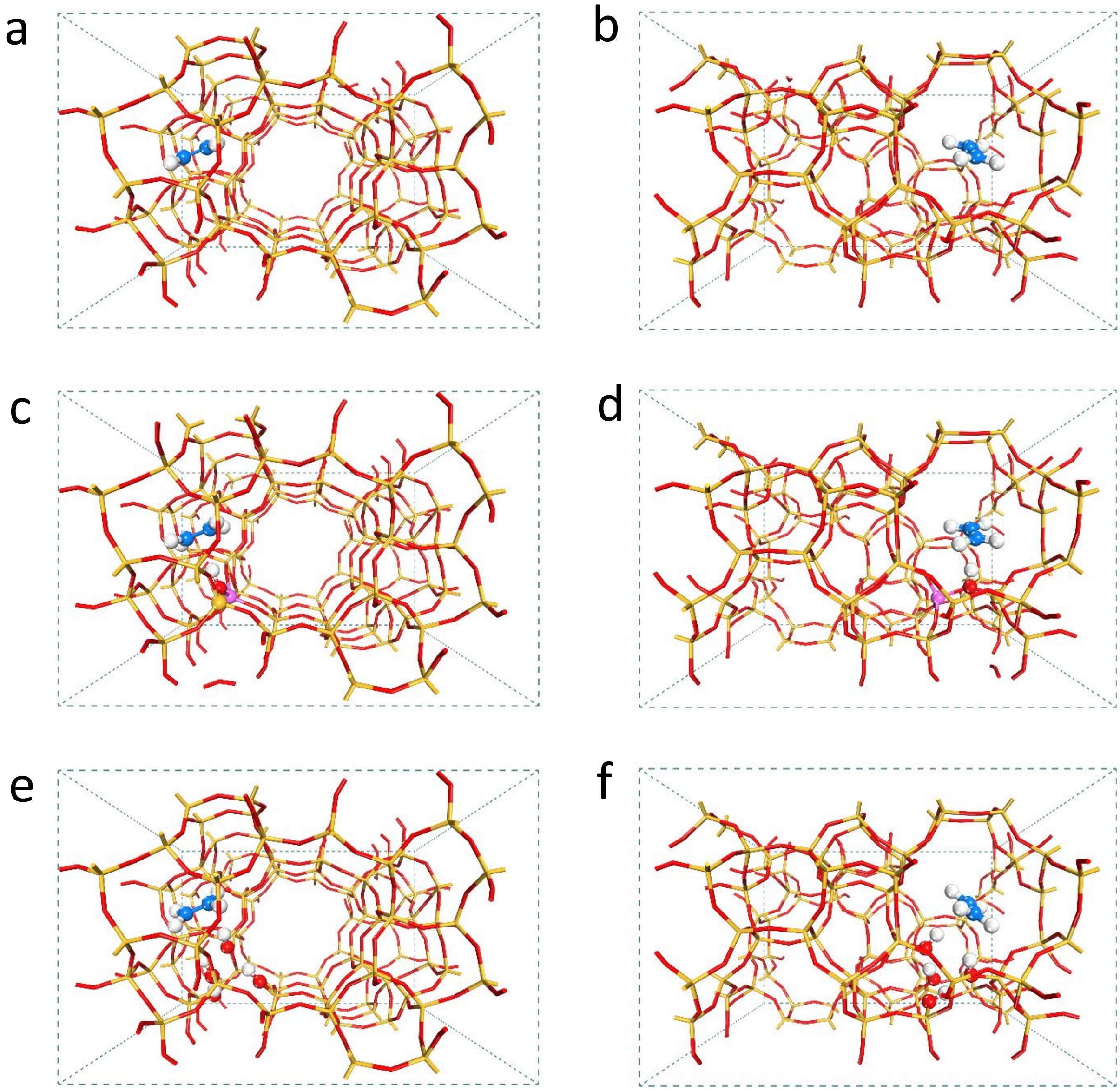
Supplementary Fig. 57. (a) The models and (b) the percentages of ethene molecules adsorption on Na-FeC$_x$ surface during the desorption process with different distance between Na-FeC$_x$ and Na-FeC$_x$/*s*-ZSM-5. Ethene molecules: Blue-white balls. Na-FeC$_x$: Black-purple surfaces. *s*-ZSM-5: Gray grids.

Note: We have performed the theoretical investigation on ethene desorption/adsorption on the model of Na-FeC$_x$ and zeolite component with varied distance. 20 ethene molecules were localized on the Na-FeC$_x$ surface, and their desorption processes were qualitatively predicted as a function of desorption time by the MD simulation at 260 °C. As shown in Supplementary Fig. 57, the Na-FeC$_x$/*s*-ZSM-5 with different proximities exhibited the comparable ethene desorption efficiency, as suggested by the residue ethene molecules on the Na-FeC$_x$ surface with a quantitative equilibrium at ~50%. This result is well consistent with the experimental results, where the Na-FeC$_x$/*s*-ZSM-5 catalysts with different proximities (e.g. 10-20, 20-40, 40-60, and 80-100 mesh) could boost the CO conversion, because the zeolite optimized the olefin escape from Na-FeC$_x$ surface (region I in Supplementary Fig. 55) through capturing the olefins from the free region (region II) rather than directly interact with the Na-FeC$_x$. The proximities between Na-FeC$_x$ and zeolite (diameter of region II) within sub-mm level are enough for olefin escaping, but the Na migration would deactivate the catalysts with a very close proximity (e.g. powder mixing). Therefore, an appropriate proximity is still suggested in the experimental tests.

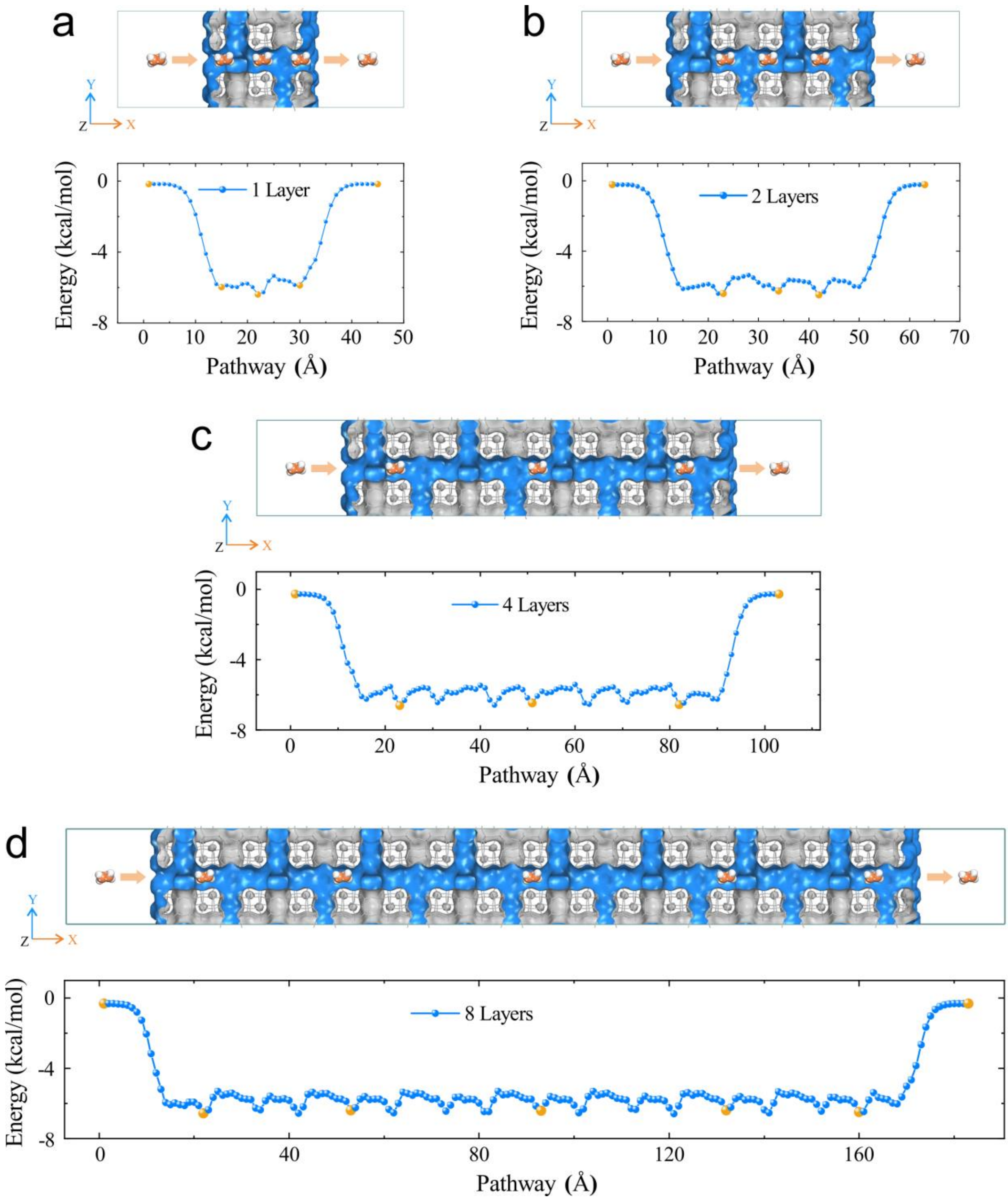


Supplementary Fig. 58. (a) The initial state of 20 ethene molecules adsorbed on Na-FeC$_x$ (I) surface with Na-FeC$_x$ (II). (b) Scheme showing ethene molecules desorbing from Na-FeC$_x$ surface (I) to the other (II). Ethene molecules: Blue-white balls. Na-FeC$_x$: Black-purple surfaces. (c) Percentages of ethene molecules adsorption on Na-FeC$_x$ surface (I) and (I+II) during the desorption process.

Note: We performed the theoretical investigation on ethene desorption/adsorption on models of Na-FeC$_x$ and Na-FeC$_x$ (20 molecules were localized on the Na-FeC$_x$ surface, and their desorption processes were qualitatively predicted as a function of desorption time by the MD simulation at 260 °C). The results showed that the number of ethene molecules on the Na-FeC$_x$ surface (I in Supplementary Fig. 58) would desorb rapidly and re-adsorb on another Na-FeC$_x$ surface (II in Supplementary Fig. 58). In this case, the quantitative equilibrium gave about 90% of the ethene molecules adsorbing on the Na-FeC$_x$ surface (I and II in Supplementary Fig. 58), which is much higher than 48% in the case using zeolite (Supplementary Fig. 62, Fig. 4d). This result demonstrates the importance of zeolite for the product escape from the Na-FeC$_x$ surface.

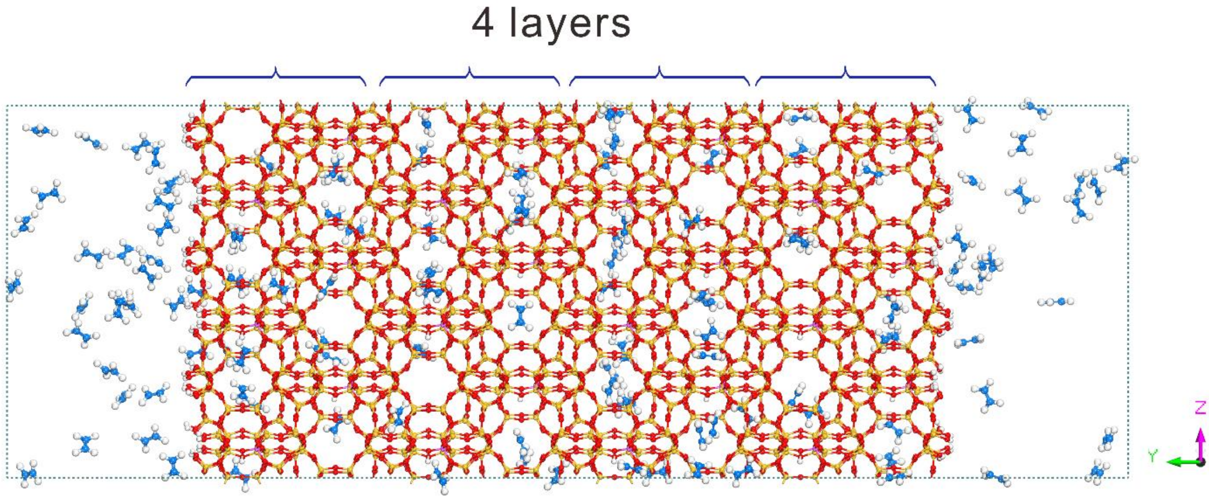


Supplementary Fig. 59. Structures showing ethene adsorption within the micropores of MFI structures. Siliceous MFI (S-1) along (a) Y and (b) X directions, proton-form aluminosilicate MFI (ZSM-5) along (c) Y and (d) X directions, and silanol modified siliceous MFI (S1-OH) along (e) Y and (f) X directions. Zero damping DFT-D3 method of Grime was used for calculating the adsorption energy.

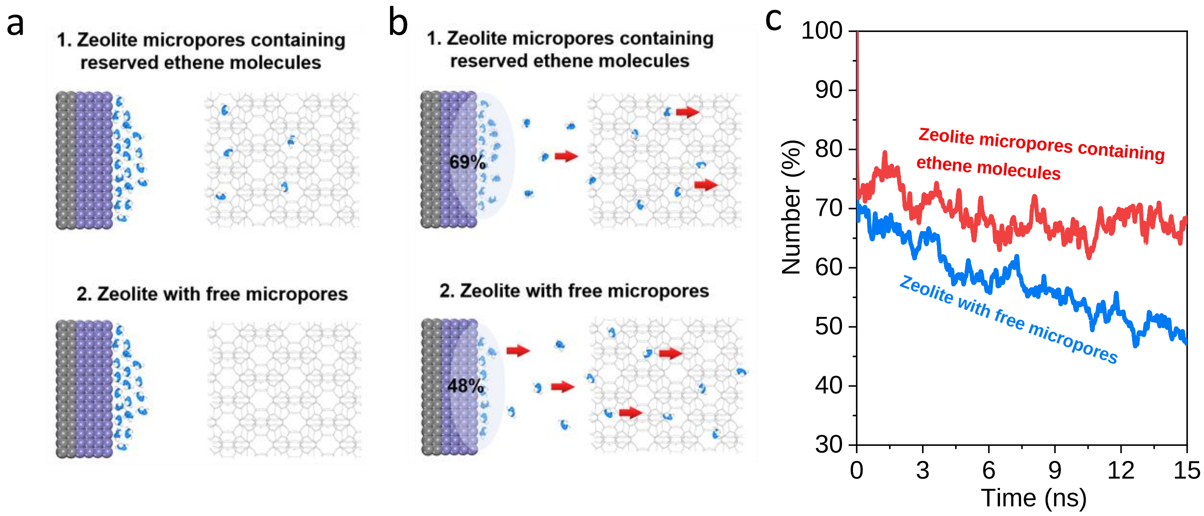


Supplementary Fig. 60. Definition of ethene diffusion pathways and according interaction energy profiles along with straight channels of *s*-ZSM-5 with (a) 1, (b) 2, (c) 4, and (d) 8 layers of units. These data were obtained using a molecular dynamics simulation.

Note: In this work, the efficacy of olefin diffusion within zeolite micropores is important for the rapidly olefin product escaping. Theoretical results on MFI zeolite structures with the same components have revealed that the olefin diffusion within zeolite micropores proceeds a site-by-site process, and the migration from one site to another proceeds the same energy barriers (Supplementary Fig. 60). Therefore, for the MFI zeolites with similar Si/Al ratios (*e.g. s-ZSM-5 vs n-ZSM-5*), the diffusion efficiency of olefin within one zeolite crystal is not determined by the energy barriers, but strongly related to the number of adsorption sites within the zeolite micropores (e.g. proton sites within aluminosilicate zeolites and silanol sites within siliceous zeolites). Considering the MFI zeolites with the same components (e.g. Si/Al ratios), the number of adsorption sites is linearly related to the length of the microporous channels in one zeolite crystal. According to the theoretical simulations, the efficacy of olefin diffusion in one MFI zeolite crystal is strongly related to the distance of diffusion channels, which could be adjusted by the zeolite crystal morphology (e.g. *s*-ZSM-5 *vs* *n*-ZSM-5).



Supplementary Fig. 61. Model showing 128 ethene molecules adsorbed in *s*-ZSM-5 structure with 4-unit layers. The number of ethene molecules at 32, 64 and 256 were used for calculating in *s*-ZSM-5 models with 1-, 2- and 8-unit layers, respectively.



Supplementary Fig. 62. (a) The initial state of 20 ethene molecules adsorbed on Na-FeC$_x$ surface with zeolite micropores containing reserved ethene molecules and zeolite with free micropores. (b) Scheme showing ethene molecules desorbing from Na-FeC$_x$ surface to the zeolite micropores containing reserved ethene molecules and zeolite with free micropores. Ethene molecules: Blue-white balls. Na-FeC$_x$: Black-purple surfaces. *s*-ZSM-5: Gray grids. (c) Percentages of ethene molecules adsorption on Na-FeC$_x$ surface during the desorption process over zeolite micropores containing ethene molecules and zeolite with free micropores.

Note: 20 ethene molecules were localized on the Na-FeC$_x$ surface, and their desorption processes were qualitatively predicted as a function of desorption time by the MD simulation at 260 °C. In this case, the number of ethene molecules on the Na-FeC$_x$ surface maintained a quantitative equilibrium of about 69%, much higher than 48% in the case using zeolite with free micropores (Supplementary Fig. 62). These results would explain the requirement of zeolite with short b-axis thickness for the efficient LT-FTO over Na-FeC$_x$/zeolite catalysts. In fact, the enhancement of molecule diffusion by employing nano-sized zeolite crystals (and zeolite containing mesopores and/or macropores) in the catalyst granules has been extensively reported[28-33] and industrially applied[28,32-35]. These results might help to understand zeolite with short *b*-axis thickness benefiting the rapid olefin escape compared with the general zeolites.

Supplementary Tables

Supplementary Table 1. Catalytic data of various catalysts in the LT-FTO.[a]

Catalyst	CO Conv. (%)	CO_2 Sel. (%)	Selectivity (%)						o/a
			CH_4	C_{2-4}^0	$C_{2-4}^=$	C_{5-10}^0	$C_{5-10}^=$	C_{10}^+ & C-oxy	
Na-FeC$_x$	1.9	30.3	10.8	6.7	33.0	7.7	32.0	9.8	4.4
Na-FeC$_x$[b]	25.6	45.3	9.2	4.5	23.3	8.0	32.7	22.3	4.5
Na-FeC$_x$/*n*-ZSM-5	45.1	38.5	3.2	47.7	1.0	43.6	4.3	0.2	0.1
Na-FeC$_x$/*s*-ZSM-5	82.5	46.6	3.0	8.9	29.1	10.9	39.9	8.2	3.5
Na-FeC$_x$/SSZ-13	39.8	49.0	4.2	7.3	27.1	4.9	24.6	31.9	4.2
Na-FeC$_x$/Beta	32.5	49.5	3.5	7.2	27.8	4.6	21.1	35.8	4.1
Na-FeC$_x$/MOR	62.2	46.5	4.5	6.7	23.5	4.8	42.0	18.5	5.7
Na-FeC$_x$/SAPO-34	67.1	46.6	5.3	5.1	18.6	5.4	23.6	42.0	4.0
Na-FeC$_x$/ZSM-22	74.1	49.2	3.6	10.6	28.7	12.8	29.1	15.2	2.5

[a] Reaction conditions: weight ratio of zeolite/Na-FeC$_x$ at 3, CO/H_2/Ar at 45/45/10, 2400 mL·h^{-1}·g^{-1}, 2.0 MPa, 260 °C, granule mixture of zeolite and Na-FeC$_x$.

[b] Reaction temperature at 300 °C.

Supplementary Table 2. Structural parameters of various samples.

Catalyst	S_{BET} (m^2 g^{-1})	V (cm^3 g^{-1})	Si/Al ratio by ICP[a]
n-ZSM-5	388.7	0.13	30.5
s-ZSM-5	337.3	0.21	26.2
SSZ-13	542.9	0.27	12.6
Beta	471.1	0.18	13.1
MOR	502.7	0.21	10.4

[a] The Si/Al ratio was obtained by ICP analysis with error abounds at ±3%, which is obtained by repeating the measurement for multiple times.

Supplementary Table 3. Catalytic data of the Na-FeC$_x$/*s*-ZSM-5 catalyst with different zeolite/Na-FeC$_x$ weight ratios of 2, 3, and 4.[a]

Zeolite/Na-FeC$_x$	CO Conv. (%)	CO_2 Sel. (%)	Selectivity (%)						o/a
			CH_4	C_{2-4}^{0}	$C_{2-4}^{=}$	C_{5-10}^{0}	$C_{5-10}^{=}$	C_{10}^{+} & C-oxy	
2	66.8	47.4	3.6	7.8	30.8	8.4	37.0	12.4	4.2
3	82.5	46.6	3.0	8.9	29.1	10.9	39.9	8.2	3.5
4	53.2	45.4	2.4	7.3	25.4	9.1	42.2	13.6	4.1

[a] Reaction conditions: CO/H_2/Ar at 45/45/10, 2400 mL·h^{-1}·g^{-1}, 2.0 MPa, 260 °C, granule mixture of zeolite and Na-FeC$_x$.

Supplementary Table 4. Data showing the performances of *s*-ZSM-5 zeolite for 1-butene isomerization/cracking.[a]

Item	Tempt. (℃)	Conv. (%)	Product selectivity (%)						
			CH_4	C_2	$C_3^{=}$	C_3^{0}	i-$C_4^{=}$	n-C_4^{0}	i-C_4^{0}
1-butene	260	97.8	0	0	0.5	<0.1	99.1	0.1	0.3

[a] Reaction conditions: 0.2 g of *s*-ZSM-5, flow rate at 20 mL/min, feed gas was 1% of C_4H_8, 0.1 MPa, 260 °C.

Supplementary Table 5 Data showing the performances of *s*-ZSM-5 zeolite for 1-hexene isomerization/cracking.[a]

Item	Temp. (℃)	Conv. (%)	Product selectivity (%)								
			CH_4	C_2	$C_3^=$	C_3^0	n-$C_4^=$	i-$C_4^=$	n-$C_5^=$	i-$C_5^=$	i-$C_6^=$
1-hexene	260	61.9	0	0	1.4	0	0.7	2.4	1.1	2.9	91.5

[a] Reaction conditions: 0.2 g of *s*-ZSM-5, 1-hexene bubbling with a pure N_2 flow at 20 mL/min at 120 °C, 0.1 MPa, 260 °C.

Supplementary Table 6. Data showing the performances of *s*-ZSM-5 zeolite for 1-octene isomerization/cracking.[a]

Item	Tempt. (℃)	Conv. (%)	Product selectivity (%)												
			CH_4	C_2	$C_3^=$	C_3^0	n-$C_4^=$	i-$C_4^=$	n-$C_5^=$	i-$C_5^=$	n-$C_6^=$	i-$C_6^=$	n-$C_7^=$	i-$C_7^=$	i-$C_8^=$
1-octene	260	51.8	0	0	0.3	0	0.2	0.6	0.4	0.7	0.6	1.2	1.1	1.5	93.4

[a] Reaction conditions: 0.2 g of *s*-ZSM-5, 1-octene bubbling with a pure N_2 flow at 20 mL/min at 120 °C, 0.1 MPa, 260 °C.

Supplementary Table 7. Catalytic data of various catalyst in LT-FTO.[a]

Catalyst	CO Conv. (%)	CO_2 Sel. (%)	Selectivity (%)						o/a
			CH_4	C_{2-4}^{0}	$C_{2-4}^{=}$	C_{5-10}^{0}	$C_{5-10}^{=}$	C_{10}^{+} & C-oxy	
Na-FeC_x/*n*-S1	42.7	46.7	5.0	3.2	13.8	8.0	45.3	24.7	5.3
Na-FeC_x/*n*-S1-OH	57.5	47.1	6.4	10.0	16.2	20.4	19.9	27.1	1.2
Na-FeC_x/*s*-S1	50.2	43.8	2.6	4.1	21.4	2.9	24.7	44.3	6.6
Na-FeC_x/*s*-S1-OH	66.5	47.7	3.0	7.6	32.5	8.3	36.2	12.4	4.3

[a] Reaction conditions: weight ratio of zeolite/Na-FeC_x at 3, CO/H_2/Ar at 45/45/10, 2400 mL·h^{-1}·g^{-1}, 2.0 MPa, 260 °C, and granule mixture of zeolite and Na-FeC_x.

Supplementary Table 8. Catalytic data of the Na-FeC_x/*s*-S1-OH catalyst with H_2/CO ratios of 1, 2, and 3.[a]

H_2/CO	CO Conv. (%)	CO_2 Sel. (%)	Selectivity (%)						o/a
			CH_4	C_{2-4}^{0}	$C_{2-4}^{=}$	C_{5-10}^{0}	$C_{5-10}^{=}$	C_{10}^{+} & C-oxy	
1	66.5	47.7	3.0	7.6	32.5	8.3	36.2	12.4	4.3
2	81.3	45.2	4.3	8.8	36.8	8.6	32.6	8.9	4.0
3	82.6	37.1	6.0	9.2	39.3	9.2	30.0	6.3	3.8

[a] Reaction conditions: weight ratio of *s*-S1-OH/Na-FeC_x at 3, 2400 $mL \cdot h^{-1} \cdot g^{-1}$, 2.0 MPa, 260 °C, granule mixture of *s*-S1-OH and Na-FeC_x.

Supplementary Table 9. Catalytic data of the Na-FeC_x/*s*-ZSM-5 catalyst with different mixture manners of Na-FeC_x and *s*-ZSM-5.[a]

Mixture manner	CO Conv. (%)	CO_2 Sel. (%)	Selectivity (%)						o/a
			CH_4	C_{2-4}^0	$C_{2-4}^=$	C_{5-10}^0	$C_{5-10}^=$	C_{10}^+ & C-oxy	
Powder-mixing	21.5	20.2	18.8	17.2	23.8	18.3	21.9	<1	1.3
Dual-bed	13.8	47.6	7.0	13.9	24.6	14.8	30.7	9.0	1.9
Granule-mixing	82.5	46.6	3.0	8.9	29.1	10.9	39.9	8.2	3.5

[a] Reaction conditions: weight ratio of *s*-ZSM-5/Na-FeC_x at 3, CO/H_2/Ar at 45/45/10, 2400 $mL \cdot h^{-1} \cdot g^{-1}$, 2.0 MPa, and 260 °C.

Supplementary Table 10. Catalytic data of the Na-FeC_x/*s*-ZSM-5 catalysts with different proximities between the Na-FeC_x and *s*-ZSM-5 zeolite in the LT-FTO.[a]

Na-FeC_x/*s*-ZSM-5 proximity	CO Conv. (%)	CO_2 Sel. (%)	Selectivity (%)						o/a
			CH_4	C_{2-4}^0	$C_{2-4}^=$	C_{5-10}^0	$C_{5-10}^=$	C_{10}^+ & C-oxy	
10-20 mesh	77.1	49.4	4.4	8.6	27.8	9.2	42.6	7.4	4.0
20-40 mesh	82.5	46.6	3.0	8.9	29.1	10.9	39.9	8.2	3.5
40-60 mesh	88.3	45.8	7.1	11.4	38.0	7.3	31.4	4.8	3.7
80-100 mesh	69.8	42.8	11.0	10.0	36.7	6.4	31.0	4.9	4.1
Powder-mixing	21.5	20.2	18.8	17.2	23.8	18.3	21.9	<1	1.3

[a] Reaction conditions: weight ratio of *s*-ZSM-5/Na-FeC_x at 3, CO/H_2/Ar at 45/45/10, 2400 $mL \cdot h^{-1} \cdot g^{-1}$, 2.0 MPa, 260 °C.

Supplementary Table 11. Catalytic data of the Na-FeC_x/*s*-ZSM-5 catalyst with GHSV of 2400, 4800, and 9600 mL·h^{-1}·g^{-1}.[a]

GHSV (mL·h^{-1}·g^{-1})	CO Conv. (%)	CO_2 Sel. (%)	Selectivity (%)						o/a
			CH_4	C_{2-4}^0	$C_{2-4}^=$	C_{5-10}^0	$C_{5-10}^=$	C_{10}^+ & C-oxy	
2400	82.5	46.6	3.0	8.9	29.1	10.9	39.9	8.2	3.5
4800	77.2	48.2	2.8	8.7	36.7	9.9	37.1	4.8	4.0
9600	43.6	48.1	3.2	7.7	41.3	8.1	36.8	2.9	4.9

[a] Reaction conditions: weight ratio of *s*-ZSM-5/Na-FeC_x at 3, CO/H_2/Ar at 45/45/10, 2.0 MPa, 260 °C, and granule mixture of *s*-ZSM-5 and Na-FeC_x.

Supplementary Table 12. Catalytic data of the Na-FeC$_x$/*s*-S1-OH catalyst with GHSV of 2400, 3600, and 4800 mL·h^{-1}·g^{-1}.[a]

GHSV (mL·h^{-1}·g^{-1})	CO Conv. (%)	CO_2 Sel. (%)	Selectivity (%)						o/a
			CH_4	C_{2-4}^{0}	$C_{2-4}^{=}$	C_{5-10}^{0}	$C_{5-10}^{=}$	C_{10}^{+} & C-oxy	
2400	66.5	47.7	3.0	7.6	32.5	8.3	36.2	12.4	4.3
3600	40.7	45.6	3.4	7.2	30.4	8.6	37.0	13.4	4.2
4800	29.9	42.3	3.7	6.9	28.3	8.7	37.8	14.6	4.2

[a] Reaction conditions: weight ratio of *s*-S1-OH/Na-FeC$_x$ at 3, CO/H_2/Ar at 45/45/10, 2.0 MPa, 260 °C, and granule mixture of *s*-S1-OH and Na-FeC$_x$.

Supplementary Table 13. Intensities of band in IR spectra of C_2H_4 and the residue percentage in 0-20 min over various catalysts.

	Time (min)		0	4	8	12	16	20
s-ZSM-5	V_s=CH_2	Strength[a]	0.1206	0.0226	0.0134	0.0041	0.0013	0.0000
	2986 cm^{-1}	Percentage (%)[b]	100	6.8	4.0	1.2	0.4	0.0
	V_{as}=CH_2	Strength	0.1670	0.0329	0.0118	0.0004	0.0000	0.0000
	1445 cm^{-1}	Percentage (%)	100	9.0	3.2	0.1	0.0	0.0
Na-FeC_x	V_s=CH_2	Strength	0.3238	0.2974	0.1988	0.0941	0.0502	0.0226
	2986 cm^{-1}	Percentage (%)	100	91.8	61.4	29.1	15.5	7.0
	V_{as}=CH_2	Strength	0.3856	0.3486	0.1825	0.0586	0.0333	0.0243
	1445 cm^{-1}	Percentage (%)	100	90.4	47.3	15.2	8.6	6.3
Na-FeC_x/*n*-ZSM-5	V_s=CH_2	Strength	0.3638	0.3174	0.0986	0.038	0.0153	0.0053
	2986 cm^{-1}	Percentage (%)	100	89.6	61.4	30.6	15.1	8.7
	V_{as}=CH_2	Strength	0.3767	0.3174	0.0986	0.038	0.0153	0.0053
	1445 cm^{-1}	Percentage (%)	100	91.0	43.2	16.5	6.0	3.0
Na-FeC_x/*s*-ZSM-5	V_s=CH_2	Strength	0.3327	0.294	0.1493	0.0547	0.0281	0.0105
	2986 cm^{-1}	Percentage (%)	100	88.4	44.9	16.4	8.4	3.2
	V_{as}=CH_2	Strength	0.3638	0.3174	0.0986	0.038	0.0153	0.0053
	1445 cm^{-1}	Percentage (%)	100	87.3	27.1	10.4	4.2	1.5

[a] The value of absorbance intensity in the IR spectra.

[b] Calculated from the residue percentage of 2986 cm^{-1} and 1445 cm^{-1} bands in 0-20 min, respectively.

Supplementary Table 14. Intensities of band in IR spectra of C_3H_6 and the residue percentage in 0-20 min over various catalysts.

	Time (min)		0	4	8	12	16	20
s-ZSM-5	V_{as}=CH_3	Strength[a]	0.4280	0.033	0.0243	0.0192	0.0088	0.0096
	2952 cm^{-1}	Percentage (%)[b]	100.0	4.1	3.0	2.4	1.1	1.2
	δ=CH_2	Strength	0.4503	0.0237	0.0142	0.0020	0.0028	0.0026
	910 cm^{-1}	Percentage (%)	100.0	3.0	1.8	0.3	0.4	0.3
Na-FeC_x	V_{as}=CH_3	Strength	0.7632	0.6942	0.4001	0.1564	0.0589	0.0265
	2952 cm^{-1}	Percentage (%)	100.0	91.0	52.4	20.5	7.7	3.5
	δ=CH_2	Strength	1.0476	0.9437	0.5867	0.2647	0.1069	0.0520
	910 cm^{-1}	Percentage (%)	100.0	90.1	56.0	25.3	10.2	5.0
Na-FeC_x/*n*-ZSM-5	V_{as}=CH_3	Strength	0.8982	0.8351	0.5205	0.2532	0.1469	0.1115
	2952 cm^{-1}	Percentage (%)	100.0	93.0	57.9	28.2	16.4	12.4
	δ=CH_2	Strength	0.9723	0.9470	0.6060	0.2788	0.1117	0.0653
	910 cm^{-1}	Percentage (%)	100.0	97.4	62.3	28.7	11.5	6.7
Na-FeC_x/*s*-ZSM-5	V_{as}=CH_3	Strength	0.7971	0.7065	0.3369	0.0871	0.0299	0.0161
	2952 cm^{-1}	Percentage (%)	100.0	88.6	42.3	10.9	3.8	2.0
	δ=CH_2	Strength	0.7967	0.7365	0.4355	0.1443	0.0411	0.0275
	910 cm^{-1}	Percentage (%)	100.0	92.4	54.7	18.1	5.2	3.5
Na-FeC_x/*s*-S1-OH	V_{as}=CH_3	Strength	0.9182	0.4996	0.1742	0.0682	0.0304	0.0127
	2952 cm^{-1}	Percentage (%)	100.0	54.4	19.0	7.4	3.3	1.4
	δ=CH_2	Strength	1.0715	0.6487	0.2802	0.1181	0.0452	0.0257
	910 cm^{-1}	Percentage (%)	100.0	60.5	26.2	11.0	4.2	2.4

[a] The value of absorbance intensity in the IR spectra.

[b] Calculated from the residue percentage of 2952 cm^{-1} and 910 cm^{-1} bands in 0-20 min, respectively.

Supplementary Table 15. Catalytic performances of Na-FeC_x with different Na loading in the LT-FTO tests.[a]

Catalyst	Na loadings (wt%)	CO Conv. (%)	CO_2 Sel. (%)	Selectivity (%)						o/a
				CH_4	$C_{2\text{-}4}^{0}$	$C_{2\text{-}4}^{=}$	$C_{5\text{-}10}^{0}$	$C_{5\text{-}10}^{=}$	C_{10}^{+} & C-oxy	
5.35NaFeC_x	5.35	1.9	30.3	10.8	6.7	33.0	7.7	32.0	9.8	4.4
4.08NaFeC_x	4.08	9.1	38.3	8.7	7.7	31.4	6.5	31.9	13.8	4.4
4.08NaFeC_x/*s*-ZSM-5	4.08	77.7	46.4	2.9	8.5	33.5	11.3	37.8	6.0	3.6
1.30NaFeC_x	1.30	14.6	43.7	13.2	13.4	31.0	12.7	23.0	6.7	2.0
Na-free FeC_x	<0.04	17.3	38.7	18.6	12.5	28.9	12.2	20.8	7.0	1.8
Used-5.35NaFeC_x	5.19	6.0	34.2	9.4	6.8	30.0	7.3	33.5	13.0	4.5

[a] Reaction conditions: weight ratio of zeolite to Na-FeC_x at 3, H_2/CO at 1, 2400 $mL\cdot h^{-1}\cdot g^{-1}$, 2.0 MPa, 260 °C, granule-mixing of the zeolite and Na-FeC_x individuals.

Note: *Discussion to exclude the Na amount change on influencing catalytic activity.* The FeC_x catalysts have been extensively used for the classical Fischer-Tropsch synthesis reactions at low temperatures (e.g. 240-280 °C), giving high CO conversion but poor selectivity to olefins. In these cases, the reactions followed a classical Fischer-Tropsch route that converts syngas to alkanes[36-39]. With regard to the Fischer-Tropsch synthesis to olefins (FTO), more alkaline species were necessary to reduce the catalyst hydrogenation activity and weaken the olefin adsorption on Na-FeC_x surface, thus to hinder the deep hydrogenation and selectively obtain olefins[1,40-44]. These Na-FeC_x catalysts for FTO processes have to work at higher reaction temperatures to obtain desired CO conversions (e.g. higher than 320 °C) because of their reduced activity compared with the FeC_x catalysts for classical Fischer-Tropsch synthesis. In the previously tested catalysts for FTO, the alkaline modified FeC_x exhibited extremely low CO conversions at temperatures lower than 300 °C[41,43,45], in good agreement with the performance of Na-FeC_x catalyst in this work. This Na-FeC_x catalyst is synthesized following the classical method described in the previous reference[1,46,47], and showed similar performances to that in literatures under their described reaction conditions (e.g. at 340 °C)[1].

We rationally synthesized the Na-FeC_x catalyst with Na loading at 4.08%, denoted as 4.08Na-FeC_x, which exhibited a higher CO conversion (9.1%, Supplementary Table 15) than the Na-FeC_x catalyst (CO conversion of 1.9%), but still far to meet the level using zeolite promoter. After mixing with the *s*-ZSM-5 promoter (4.08Na-FeC_x/*s*-ZSM-5), the CO conversion could reach as high as 77.7% with olefins as a dominant product. The Na-FeC$_x$ catalyst with Na loading at 1.3% (1.30Na-FeC$_x$) showed CO conversion at 14.6%. Even for the Na-free FeC$_x$ catalyst (Na loading <0.04%), the CO conversion was 17.3% with abundant alkane products, in good agreement with the phenomenon in classical Fischer-Tropsch synthesis. These results confirm that the sodium loading to influence the catalytic activity, but still much lower than that using zeolite promoters.

We also evaluated the used Na-FeC$_x$ catalyst (after reaction, the zeolite individual of used Na-FeC$_x$/*s*-ZSM-5 catalyst was removed to obtain the used Na-FeC$_x$, and reused in the next run) in the LT-FTO reaction process, showing the CO conversion at 6.0%.

Supplementary Table 16. Parameter of the force fields for zeolites and ethene in theoretical simulation.

Atom type	ε (kcal/mol)	σ (Å)	q (e)
Oa [Si-Oa-Al][11]	0.115	3.600	-0.840
Oh [Si-Oh-H][12]	0.122	3.090	-0.675
Os [Si-Os-Si][10]	0.105	3.300	-0.750
Ha [Al-O-Ha][11]	0.000	0.000	0.500
Hs [Si-O-Hs][12]	0.015	0.970	0.400
Sh [Si-O-H][12]	0.093	3.700	1.400[a]
Si [O-Si-O][10]	0.044	2.300	1.500
Al [O-Al-O][11]	0.044	2.300	1.360
CH_2 [CH_2-CH_2][9]	0.169	3.675	0.000

[a] The charge has been adjusted to keep the total change equal 0.

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